

A Study on Union and Intersection of Subspaces and its Application to Quantum Communication

A dissertation submitted by

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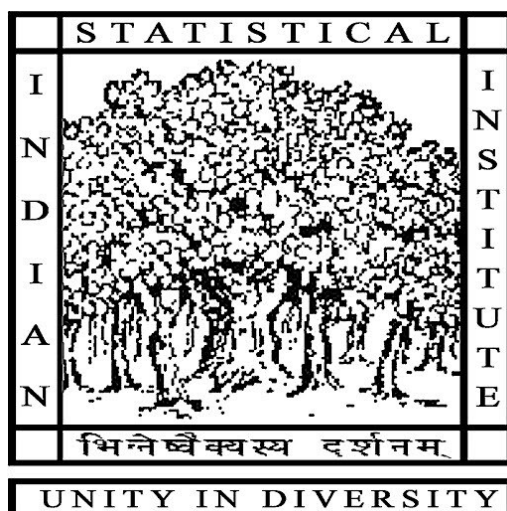
in partial fulfillment of the requirements

for the award of the degree of

Master of Technology

in

Computer Science



INDIAN STATISTICAL INSTITUTE, KOLKATA

THESIS CERTIFICATE

This is to certify that the thesis titled **A Study on Union and Intersection of Subspaces and its Application to Quantum Communication**, submitted by **Ayanava Dasgupta (CS2103)**, to the Indian Statistical Institute, Kolkata for the award of the degree of **Master of Technology in Computer Science**, is a *bona fide* record of the research work done by him under my supervision. The contents of this thesis, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

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ACKNOWLEDGEMENTS

I am deeply grateful to my thesis advisor, Prof. Naqeeb Ahmad Warsi, for his unwavering support, guidance, and insightful discussions that have been instrumental in the successful completion of this thesis. Our lengthy conversations about various ideas were truly enjoyable, and I am thankful for his invaluable contributions. During my time with Professor Warsi, I not only gained a solid foundation in the essential prerequisites but also had the privilege of being inspired by his insightful and motivational teachings.

I extend my deepest thanks to all the esteemed professors at ISI who have imparted their vast knowledge and expertise in computer science. Their teachings will undoubtedly serve as the foundation for my future endeavors. Their invaluable suggestions and engaging discussions have added a significant dimension to my research work and enhanced the quality of my findings.

I would like to express my heartfelt appreciation to my parents for their resolute faith in me and their encouragement to pursue my dreams, which led me to this esteemed institution. I will be forever grateful to ISI.

Lastly, I would also like to express my gratitude to all my classmates and friends, for their unwavering support and encouragement throughout this MTech journey. Their presence has consistently served as a source of motivation. I also extend my thanks to those individuals whom I may have unintentionally omitted from this acknowledgment.

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Date: [June 17, 2023](#)

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Abstract

In classical information theory, union and intersection of typical sets are often used to analyze variety of problems. Analogous to this, in quantum information theory one desires to construct a POVM (Positive Operator Valued Measurement) that functions as a measurement on the intersection/union of typical subspaces. However, unlike the classical case this is highly non-trivial in the quantum setting.

In this thesis, we study and explore this fundamental issue in quantum information. In particular, we study the techniques invented by Sen in his recent work [25] for analysing this fundamental issue for the case of classical-quantum states wherein he studies a very long standing open problem of simultaneous decoding for communication over classical-quantum multiple access channel. Our study finds that his techniques cannot be extended for arbitrary quantum states and therefore his results are not general enough to solve many other important long standing open problems in quantum information.

Further, we make some progress for the above mentioned problem by designing a binary POVM which behaves as measurement on the intersection subspace. Our construction utilizes Jordan's lemma and has practical applications, such as proving a converse for the asymmetric hypothesis testing problem. Our proof technique can be viewed as a quantum generalization of the proof presented in Thomas and Cover's book for the Chernoff Stein's lemma.

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Introduction

A fundamental bottleneck which often arises in quantum information theory is about calculating the probability of intersection of two events. Understanding this bottleneck will help us understand many quantum network information theory problems in the one-shot setting, e.g., entanglement assisted multiple access channel [17], entanglement broadcast channel with common message in the one-shot setting [3, 25, 1, 5]. Recently, the the broadcast channel with common message was studied for the case of classical-quantum channel in [24]. For details on one-shot setting see for example, [21], [19], [18], and references therein.

This bottleneck was first alluded by Fawzi et al in their paper [11]. Recently, Sen studied this problem from the point of view of communication over classical quantum multiple access channel [25]. In his work, he was able to resolve the intersection issue with respect to classical-quantum states and their respective marginals . In this work, we make some progress towards understanding the intersection bottle neck for general quantum states and as an application we give a proof for the converse of Quantum Chernoff Chernoff Stein's lemma. Quantum Chernoff Stein's lemma is a well studied problem, see for example, [22, 6], [7].

In the classical setting, when we calculate the probability of events under a particular probability distribution, it translates to measurement of quantum states in quantum setting. These measurements are in general POVM. However, because of the Gelfand-Neymark's lemma [13] it is enough to consider only projective measurements. Therefore, in this dissertation we will use projective measurements and POVMs interchangeably. In the following section we'll discuss about the motivation of this dissertation by going through two case studies to show why, exactly we need a well-formed notion of intersection (or union) of subspaces.

1.1 Backgrounds and Preliminaries

Abbreviations

CQ	Classical Quantum
MAC	Multiple Scses Channel
POVM	Positive Operator Valued Measure
AEP	Asymptotic Equi-partition Property
CQ-MAC	Classical Quantum Multiple Scses Channel

Notations

We use $\mathcal{H}$ to denote a finite dimensional Hilbert space and $\mathcal{D}(\mathcal{H})$ to represent the set of all quantum states acting on $\mathcal{H}$. For any vector we use ket-operator $|\cdot\rangle$ to denote it. For any quantum state ρ we define $\text{supp}(\rho) := \{|i\rangle : \lambda_i > 0\}$, where λ_i represents the non-zero eigenvalue of ρ . Similarly, for any distribution P defined over a set $\mathcal{X}$ we will use the notation $\text{supp}(P) := \{x \in \mathcal{X} : P(x) > 0\}$. Given two distribution P, Q over the set $\mathcal{X}$ such that $\text{supp}(P) \subseteq \text{supp}(Q)$, the relative entropy between P and Q is defined as,

$$D(P\|Q) := \sum_{x \in \mathcal{X}} P(x) \log \left(\frac{P(x)}{Q(x)} \right)$$

Given two quantum states $\rho, \sigma \in \mathcal{D}(\mathcal{H})$ such that $\text{supp}(\rho) \subseteq \text{supp}(\sigma)$, the quantum relative entropy (also known to be *quantum KL-divergence*) between ρ and σ is defined as,

$$D(\rho\|\sigma) := \text{Tr}[\rho(\log(\rho) - \log(\sigma))].$$

For two normal operators A and B we will use the notation $\Pi := \{A \succeq B\}$ to denote a projection operator on the positive eigen-space of the operator $(A - B)$. We use $\rho^{\otimes n}$ over $\mathbb{H}^{\otimes n}$, to represent n independent copies of ρ .

Facts

Fact 1. [26, Theorem 11.8.1] Let $X_1, X_2, \dots, X_n$ be a sequence of random variables drawn i.i.d. according to a probability distribution $P_1(x)$ and let $P_2(x)$ be any other probability distribution on sample space $\mathcal{X}$. Then

$$\frac{1}{n} \log \frac{P_1(X_1, X_2, \dots, X_n)}{P_2(X_1, X_2, \dots, X_n)} \rightarrow D(P_1\|P_2)$$

Definition 1.1.1. For a fixed n and $\epsilon > 0$, a sequence $(x_1, x_2, \dots, x_n) \in \mathcal{X}^n$ is said to be *relative*

entropy typical iff

$$D(P_1 \| P_2) - \epsilon \leq \frac{1}{n} \log \frac{P_1(x_1, x_2, \dots, x_n)}{P_2(x_1, x_2, \dots, x_n)} \rightarrow \leq D(P_1 \| P_2) + \epsilon$$

The set of *relative entropy typical sequences* is called *relative entropy typical set* $A_\epsilon^{(n)}(P_1 \| P_2)$. Using relative entropy asymptotic equi-partition property (AEP), we can show that the relative entropy typical set holds the following properties:

1. For $(x_1, x_2, \dots, x_n) \in A_\epsilon^{(n)}(P_1 \| P_2)$,

$$P_2(x_1, x_2, \dots, x_n) \in [P_1(x_1, x_2, \dots, x_n)2^{-n(D(P_1 \| P_2)+\epsilon)}, P_1(x_1, x_2, \dots, x_n)2^{-n(D(P_1 \| P_2)-\epsilon)}]. \quad (1.1.1)$$

2. $P_1(A_\epsilon^{(n)}(P_1 \| P_2)) > 1 - \epsilon$, for n sufficiently large.

3. $P_2(A_\epsilon^{(n)}(P_1 \| P_2)) < 2^{-n(D(P_1 \| P_2)-\epsilon)}$.

4. $P_2(A_\epsilon^{(n)}(P_1 \| P_2)) > (1 - \epsilon)2^{-n(D(P_1 \| P_2)+\epsilon)}$, for n sufficiently large.

To prove our main result in 6.1.1 we will use Jordan's lemma. This lemma is about simultaneous block diagonalisation of two non commuting projection operators. We mention this lemma below.

Fact 2. (Sum of subspaces) Suppose $S_1, S_2, \dots, S_n$ be n subspaces with in a vector space V over F . The sum of $S_1, S_2, \dots, S_n$ denoted as $\bigoplus_{i=1}^n S_i$ can be defined as follows:

$$\bigoplus_{i=1}^n S_i := \left\{ \sum_{i=1}^n s_i \mid s_i \in S_i (1 \leq i \leq n) \right\}$$

For this dissertation we'll use the phrase **span of subspaces** to refer to sum of subspaces.

Fact 3. (Direct Sum of vector spaces) Suppose we have n distinguishable vector spaces $V_1, V_2, \dots, V_n$ over the same field F . Then, n -fold direct sum (or, direct sum) of the vector spaces $V_1, V_2, \dots, V_n$ is defined to be the vector space $\bigoplus_{i=1}^n V_i$ where,

$$\bigoplus_{i=1}^n V_i = \times_{i=1}^n V_i = \{(v_1, v_2, \dots, v_n) \mid v_i \in V_i (1 \leq i \leq n)\}$$

and,

$$\dim \left\{ \bigoplus_{i=1}^n V_i \right\} = \sum_{i=1}^n \dim \{V_i\}$$

where $\times_{i=1}^n$ is the n -fold Cartesian product. Lets try to understand the concept with an example.

Say we have two vector spaces $V_1 = \mathbb{R}^2$ and $V_2 = \mathbb{R}^2$ over $\mathbb{R}$. Then $V_1 \oplus V_2$ is a 4 dimensional vector space over $\mathbb{R}$. Suppose we have a vector $|a\rangle := \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$ in V_1 and another vector $|b\rangle := \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$ in

V_2 , then $|a\rangle$ in the 4-dimensional vector space $V_1 \oplus V_2$ can be defined with padding $|\dim\{V_2\}|$ zeros

at the below of the vector $|a\rangle$, making $|a\rangle = \begin{bmatrix} a_1 \\ a_2 \\ 0 \\ 0 \end{bmatrix}$ and similarly $|b\rangle$ in $V_1 \oplus V_2$ would be $\begin{bmatrix} 0 \\ 0 \\ b_1 \\ b_2 \end{bmatrix}$. So

$$|a\rangle + |b\rangle = \begin{bmatrix} a_1 \\ a_2 \\ b_1 \\ b_2 \end{bmatrix}.$$

Fact 4. (Direct Sum of vector subspaces) Suppose $S_1, S_2, \dots, S_n$ be n subspaces with in a vector space V over F . $S := \bigoplus_{i=1}^n S_i$ is said to be the direct span of $S_1, S_2, \dots, S_n$ if for ever vector $s \in S$ can uniquely defined by set of $\{s_i \in S_i\}_{i \in [1, n]}$ vectors as

$$s = \sum_{i=1}^n s_i$$

Simply put, if $\forall i S_i \leq S$ then $S := \bigoplus_{i=1}^n S_i$ if and only if

1.

$$S_i \cap \bigoplus_{j \neq i} S_j = \{0\}$$

, where $\{0\}$ is the zero vector space containing the zero vector only.

2.

$$S = \bigoplus_{i=1}^n S_i$$

Fact 5. (Jordan's lemma)[20] For any two projectors Π_1, Π_2 there exists an orthogonal basis in which Π_1 and Π_2 simultaneously get block diagonalised and each of these blocks are either 2×2 or 1×1 . Further, for each 2×2 block both Π_1 and Π_2 behave as a one-dimensional projector.

Fact 6. (Markov's Inequality)[15] Let X be any non-negative random variable. Then, for any $\lambda > 0$,

$$\Pr\{X \geq \lambda\} \leq \frac{\mathbb{E}[X]}{\lambda}.$$

Fact 7. ([10], [27]) Let $\rho^{\otimes n}, \sigma^{\otimes n} \in \mathcal{D}(\mathcal{H}^n)$. Consider $\Pi_n := \{\sigma^{\otimes n} \succeq 2^{-n(D(\rho\|\sigma)+\delta)} \rho^{\otimes n}\}$, where $\delta \in (0, 1)$ Then, $\lim_{n \rightarrow \infty} \text{Tr}[\Pi_n \rho^{\otimes n}] \rightarrow 1$.

Fact 8. (Gao's union bound) [12] Let $\Pi_1, \Pi_2, \dots, \Pi_n$ be n projectors. Then,

$$\text{Tr}(\Pi_n \cdots \Pi_2 \Pi_1 \rho \Pi_1 \Pi_2 \cdots \Pi_n) \geq 1 - 4 \sum_{i=1}^n \text{Tr}[\Pi_i^c \rho],$$

where $\Pi_i^c = \mathbb{I} - \Pi_i$

Fact 9. (Schatten l_p norms or Operator norms) Given a field F , $F^{m \times n}$ represents set of all matrices with m rows and n columns. If the vector l_p norm (we are assuming that we already have prerequisite knowledge about vector l_p norms) is compatible to both F^n and F^m , then for a matrix $M \in F^{m \times n}$, for $1 \leq p \leq \infty$ Schatten l_p norm of M , denoted as $\|M\|_p$ can be defined as

$$\|M\|_p := \sup_{\|v\|_p \neq 0} \frac{\|M|v\rangle\|_p}{\|v\|_p}$$

We have two special cases of Schatten l_p norm:

1. for $p = 1$, $\|M\|_1$ also known as **trace norm** is nothing but summation of all singular values of M .
2. for $p = \infty$, $\|M\|_\infty$ also known as **operator norm** is nothing but largest singular value of M .

Fact 10. Given a Projective Measurement Π and two states $\rho_1, \rho_2 \in \mathcal{D}(\mathcal{H})$ then,

$$\text{Tr}[\Pi(\rho_1 - \rho_2)] \leq \|\rho_1 - \rho_2\|_1$$

Fact 11. (Operator norm and trace norm relationship) Given two operators A and B on $\mathcal{H}$, we have the following inequality

$$\|AB\|_1 \leq \min\{\|A\|_1\|B\|_\infty, \|A\|_\infty\|B\|_1\}.$$

Fact 12. (Gelfand-Neymark's Lemma)[13] Let Π be a POVM over a Hilbert space $\mathcal{H}$. For any arbitrary integer $d \geq 2$, there exists a normal projection Π' in $\mathcal{H} \otimes \mathbb{C}^d$ such that for all linear operators A acting on $\mathcal{H}$, we have

$$\text{Tr}[\Pi A] = \text{Tr}[\Pi'(A \otimes |0\rangle\langle 0|^{\mathbb{C}^d})]$$

Above, $|0\rangle$ is a fixed quantum state-vector which is not dependent of Π and A , in $\mathbb{C}^d$.

1.2 Motivation

There are quite a lot of classical problems which can be generalised in quantum scenario with the help of a 'notion' of intersection of subspaces. In this section we'll explore two of them and in the upcoming chapters we'll show certain construction of 'intersection' of subspaces.

1.2.1 Case Study A : Hypothesis Testing and Chernoff Stein's Lemma

Hypothesis Testing

A fundamental problem in information theory is asymmetric hypothesis testing. In classical setting, if we are provided with two probability distributions P and Q on a common sample space $\mathcal{A}$. Then a classical test on $\mathcal{A}$ can be defined as a function $T : \mathcal{A} \rightarrow [0, 1]$. For a sample instance $\mathcal{A}$, $T(a)$ denotes the probability of acceptance by the test. Let $0 < \epsilon < 1$. Now, we give the definition of the *classical hypothesis testing relative entropy* $D_H^\epsilon(P||Q)$ as following:

$$D_H^\epsilon(P||Q) := \max_{T: \sum_a T(a)P(a) \geq 1-\epsilon} -\log \sum_a T(a)Q(a) \quad (1.2.1)$$

Notice that, the above definition of hypothesis testing is asymmetric because, we are minimising over all tests on $\mathcal{A}$ accepting(i.e. accepting the true distribution which $\mathcal{A}$ follows) the distribution P with very high probability($1 - \epsilon$) or, very low (ϵ) probabilities of error to get the optimal test which minimises the acceptance probability of the distribution Q .

We can give an analogous definition in quantum scenario. In this problem we are given many independent and identically-distributed (i.i.d.) copies of a quantum system described either by ρ or σ . The aim is to minimize the probability of wrongly identifying ρ in place of σ under the constraint that we correctly identify the state ρ with high probability. Formally, the problem is defined as follows:

$$D_H^\epsilon(\rho^{\otimes n}||\sigma^{\otimes n}) := \min_{\Pi_n: \text{Tr}[\Pi_n \rho^{\otimes n}] \geq 1-\epsilon} \text{Tr}[\Pi_n \sigma^{\otimes n}]. \quad (1.2.2)$$

If we are provided with only one copy of a quantum system described either by ρ or σ , then *quantum hypothesis testing relative entropy* $D_H^\epsilon(\rho||\sigma)$ can be defined as following:

$$D_H^\epsilon(\rho||\sigma) := \min_{\Pi: \text{Tr}[\Pi \rho] \geq 1-\epsilon} \text{Tr}[\Pi \sigma]. \quad (1.2.3)$$

Classical Chernoff Stein's Lemma

In Chernoff Stein's lemma, we consider the problem of testing two hypotheses, P_1 vs P_2 . We fix one of the probabilities of the error and try to minimize the other probability of error. We can show

that the relative entropy is the best exponent in probability of error. Before stating Chernoff Stein's lemma let's give the following lemma which helps proving the converse of Chernoff Stein's lemma.

Lemma 1.2.1. [26, Lemma 11.8.1] *Let $B_n \subseteq \mathcal{X}^n$ be any set of sequences $x_1, x_2, \dots, x_n$ such that $P_1(B_n) > 1 - \epsilon$. Let P_2 be any other distribution such that $D(P_1 \| P_2) < \infty$. Then, $P_2(B_n) > (1 - 2\epsilon)2^{-n(D(P_1 \| P_2) + \epsilon)}$.*

Now we will state the classical Chernoff Stein's lemma

Lemma 1.2.2. [26, Theorem 11.8.1] *Let $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} Q$. Consider the hypothesis test between two alternatives. $H_1 : Q = P_1$ and $H_2 : Q = P_2$, where $D(P_1 \| P_2) < \infty$. Let $A_N \subseteq \mathcal{X}^n$ be an acceptance region for hypothesis H_1 . Let the probabilities of error be*

$$\alpha_n := P_1^n(A_n^C), \quad \beta_n := P_2^n(A_n). \quad (1.2.4)$$

and for $0 < \epsilon < \frac{1}{2}$, define

$$\beta_n^\epsilon := \min_{\substack{A_N \subseteq \mathcal{X}^n \\ \alpha_n < \epsilon}} \beta_n \quad (1.2.5)$$

Then,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \beta_n^\epsilon = -D(P_1 \| P_2) \quad (1.2.6)$$

We will derive a lemma analogous to Theorem 1.2.1 in quantum scenario but in Theorem 1.2.1, we used the concepts of union and intersection of two sequence of sets. So that's why we need a generalised notion of intersection and union of measurement operators. In chapter 6 we will further discuss about this and give a construction of intersection which will help us to prove the lemma which in turn will give the weak converse for Quantum Chernoff Stein's Lemma which is similar in flavor with Theorem 1.2.2.

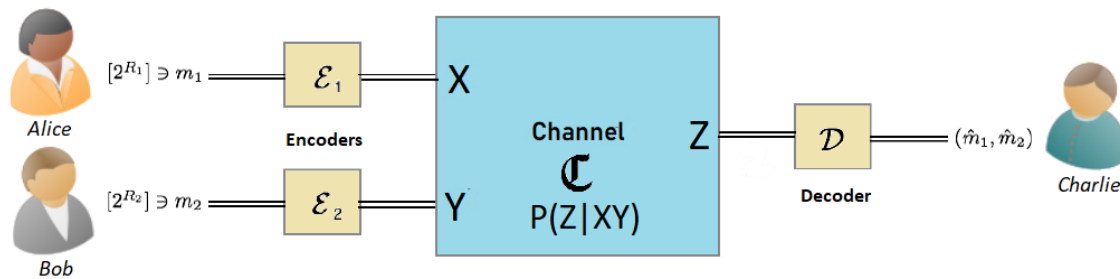


Figure 1.1: A classical MAC. Alice, Bob input their consecutive message-pair (m_1, m_2) to the encoder and then feed the encoded code-words to the channel. Charlie use the decoder to get his guess $(\hat{m}_1, \hat{m}_2)$ of the sent messages. X, Y are the random input alphabets and Z is the output alphabet of the channel.

1.2.2 Case Study B : Classical Multiple Access Channel (MAC)

In this particular case study we'll show the need of an intersection argument by demonstrating the working of a one-shot(i.e. the channel can be used once for transmission of the data) classical multiple access channel(MAC). For now let's assume two transmitters(or, senders).

Say, Alice and Bob are the two systems who are willing to transmit their corresponding messages to a common receiver Charlie with rates R_1 and R_2 bits/symbol. Let's say m_1, m_2 be the messages for Alice and Bob respectively. Then $m_1 \in [2^{R_1}]$ and $m_2 \in [2^{R_2}]$. Now a classical one-shot two-sender one-reciever MAC $\mathfrak{C}$ connected to Alice and Bob at the input terminals respectively with the corresponding input alphabets $\mathcal{X}$ and $\mathcal{Y}$ and to Charlie at the output terminal with an output alphabet $\mathcal{Z}$.

Upon getting the message m_1 , Alice encodes it to a letter $x(m_1) \in \mathcal{X}$ and similarly Bob, encodes it's message m_2 to a letter $y(m_2) \in \mathcal{Y}$. The channel has a well-defined probability distribution $p(z|x, y)$ for each $x \in \mathcal{X}, y \in \mathcal{Y}, z \in \mathcal{Z}$. So let's assume channel outputs a $z \in \mathcal{Z}$ according to the distribution $p(z|x(m_1), y(m_2))$. Let $0 < \epsilon < 1$. Charlie now has to devise some decoding algorithm to guess the messages m_1 and m_2 from the output with a condition on the average decoding error at most $O(\epsilon)$. To get a better understanding of it refer to Figure 1.1 which is depicting an image of classical MAC.

We construct the code-book C in a following randomised approach. Initially we fix the probability distribution $p(x)$ and $p(y)$ of the input alphabets $\mathcal{X}$ and $\mathcal{Y}$ respectively. Then for each $m_1 \in [2^{R_1}]$ we choose its corresponding encoding in $\mathcal{X}$ as $x(m_1)$ independently following the distribution $p(x)$ and for each $m_2 \in [2^{R_2}]$ we choose its corresponding encoding in $\mathcal{Y}$ as $x(m_1)$ independently following the distribution $p(y)$.

Generally the outcome z depends both input letters x and y . So z must follow the original channel probability distribution $p(z|x, y)$. So the true probability distribution on $\mathcal{X}\mathcal{Y}\mathcal{Z}$ is as following:

$$p^{\mathcal{X}\mathcal{Y}\mathcal{Z}}(x, y, z) := p(x)p(y)p(z|x, y) \quad (1.2.7)$$

The decoding strategy devised by Charlie must capture the true distribution only, but due to noise or some other reasons there can be room for certain errors and which can be of three possible types:

- There might be a possibility that the outcome z only depends on x , i.e. unlike the true distribution of z , the captured distribution is $p(z|x)$. Then the distribution on $\mathcal{X}\mathcal{Y}\mathcal{Z}$ can be given as :

$$(p^{\mathcal{X}\mathcal{Z}} \times p^{\mathcal{Y}})(x, y, z) := p(x)p(y)p(z|x) \quad (1.2.8)$$

where, $p(z|x) = \sum_y p(y)p(z|x, y)$

- Another possibility is that the outcome z only depends on y i.e. here the captured distribution is $p(z|y)$. Then the distribution on $\mathcal{X}\mathcal{Y}\mathcal{Z}$ can be given as :

$$(p^{\mathcal{Y}\mathcal{Z}} \times p^{\mathcal{X}})(x, y, z) := p(x)p(y)p(z|y) \quad (1.2.9)$$

where, $p(z|y) = \sum_x p(x)p(z|x, y)$

- the last one is that the outcome z depends neither on x nor y . So, in this case, the captured distribution is $p(z)$. Then the distribution on $\mathcal{X}\mathcal{Y}\mathcal{Z}$ can be given as :

$$(p^{\mathcal{X}} \times p^{\mathcal{Y}} \times p^{\mathcal{Z}})(x, y, z) := p(x)p(y)p(z) \quad (1.2.10)$$

where, $p(z) = \sum_{x,y} p(x)p(y)p(z|x, y)$

Now we consider the tests $T^{\mathcal{X}}, T^{\mathcal{Y}}$ and T which are the optimal tests coming from the classical hypothesis testing relative entropy definitions $D_H^\epsilon(p^{\mathcal{X}\mathcal{Y}\mathcal{Z}} || p^{\mathcal{X}\mathcal{Z}} \times p^{\mathcal{Y}})$, $D_H^\epsilon(p^{\mathcal{X}\mathcal{Y}\mathcal{Z}} || p^{\mathcal{Y}\mathcal{Z}} \times p^{\mathcal{X}})$ and $D_H^\epsilon(p^{\mathcal{X}\mathcal{Y}\mathcal{Z}} || p^{\mathcal{X}} \times p^{\mathcal{Y}} \times p^{\mathcal{Z}})$ respectively. For the sake of compactness, we will use hypothesis testing mutual information quantities $I_H^\epsilon(\mathcal{Y} : \mathcal{X}\mathcal{Z})$, $I_H^\epsilon(\mathcal{X} : \mathcal{Y}\mathcal{Z})$ and $I_H^\epsilon(\mathcal{X}\mathcal{Y} : \mathcal{Z})$ to represent the classical hypothesis testing relative entropy quantities $D_H^\epsilon(p^{\mathcal{X}\mathcal{Y}\mathcal{Z}} || p^{\mathcal{X}\mathcal{Z}} \times p^{\mathcal{Y}})$, $D_H^\epsilon(p^{\mathcal{X}\mathcal{Y}\mathcal{Z}} || p^{\mathcal{Y}\mathcal{Z}} \times p^{\mathcal{X}})$ and $D_H^\epsilon(p^{\mathcal{X}\mathcal{Y}\mathcal{Z}} || p^{\mathcal{X}} \times p^{\mathcal{Y}} \times p^{\mathcal{Z}})$ respectively. From the definition of classical hypothesis testing relative entropy 1.2.1 we have following properties for $T^{\mathcal{X}}, T^{\mathcal{Y}}$ and T , which are following:

$$\begin{aligned} \sum_{x,y,z} p(x, y, z) T^{\mathcal{X}}(x, y, z) &\geq 1 - \epsilon \\ \sum_{x,y,z} p(x, z) p(y) T^{\mathcal{X}}(x, y, z) &\leq 2^{-I_H^\epsilon(\mathcal{Y} : \mathcal{X}\mathcal{Z})}, \\ \sum_{x,y,z} p(x, y, z) T^{\mathcal{Y}}(x, y, z) &\geq 1 - \epsilon \\ \sum_{x,y,z} p(x) p(y, z) T^{\mathcal{Y}}(x, y, z) &\leq 2^{-I_H^\epsilon(\mathcal{X} : \mathcal{Y}\mathcal{Z})}, \end{aligned}$$

$$\sum_{x,y,z} p(x,y,z)T(x,y,z) \geq 1 - \epsilon$$

$$\sum_{x,y,z} p(x)p(y)p(z)T^{\mathcal{Y}}(x,y,z) \leq 2^{-I_H^{\epsilon}(\mathcal{X}\mathcal{Y}:\mathcal{Z})},$$

Notice that, each of $T^{\mathcal{X}}, T^{\mathcal{Y}}$ and T aims to keep the acceptance probability of the true distribution very high ($1 - \epsilon$) while making the acceptance probability of the erroneous distribution exponentially small.

Finally we will try to give the decoding strategy which will be followed by Charlie to guess the message pair (m_1, m_2) sent by Alice and Bob respectively. From the above discussion we have observed that each of these tests $T^{\mathcal{X}}, T^{\mathcal{Y}}$ and T are designed to solve particular types of errors we discussed earlier. But we need to design a test which can itself solve all the types of errors. So we need an intersection test $\hat{T} := T^{\mathcal{X}} \cap T^{\mathcal{Y}} \cap T$. If the channel output is z then Charlie uses following randomised algorithm for decoding the message pairs, which we have referred from [25].

Algorithm 1 Randomised Decoding Strategy [25]

```

for  $\hat{m}_1 = 1$  to  $2^{R_1}$  do
  for  $\hat{m}_2 = 1$  to  $2^{R_2}$  do
    Toss a coin with probability of HEAD
      being  $\hat{T}(x(\hat{m}_1), y(\hat{m}_2), z)$ ..
    if the coin comes up HEAD then
      declare  $(\hat{m}_1, \hat{m}_2)$  as Charlie's guess and HALT.
    end if
    if the coin comes up TAIL then
      go to next iteration.
    end if
  Declare FAIL if Charlie did not declare any guess above.

```

Now we will try to analyse the performance of the decoding strategy by finding its expected decoding error probability under a choice of random code-book C . Let's assume the input to the channel was the message pair (m_1, m_2) . A decoding error happens if Charlie gets **HEAD** for a message pair $(\hat{m}_1, \hat{m}_2) \neq (m_1, m_2)$ or if Charlie gets **TAIL** for the actual message pair (m_1, m_2) . Then the expected decoding error probability under a choice of random code-book C given the inputted message pair (m_1, m_2) is given by:

$$\mathbb{E}_C[p_e(C; m_1, m_2)] \leq 2^{R_1+R_2} 2^{-I_H^{\epsilon}(\mathcal{X}\mathcal{Y}:\mathcal{Z})} + 2^{R_2} 2^{-I_H^{\epsilon}(\mathcal{Y}:\mathcal{X}\mathcal{Z})} + 2^{R_1} 2^{-I_H^{\epsilon}(\mathcal{X}:\mathcal{Y}\mathcal{Z})} + 3\epsilon \quad (1.2.11)$$

Now the probability of average decoding error under a choice of random code-book C can be given as:

$$p_e(C) := \frac{1}{2^{(R_1+R_2)}} \sum_{(m_1, m_2)} p_e(C; m_1, m_2)$$

Now the expectation of the average decoding error probability under a choice of random code-book C can be given as:

$$\mathbf{E}_C[p_e(C)] \leq 2^{R_1+R_2} 2^{-I_H^\epsilon(\mathcal{X}\mathcal{Y}:\mathcal{Z})} + 2^{R_2} 2^{-I_H^\epsilon(\mathcal{Y}:\mathcal{X}\mathcal{Z})} + 2^{R_1} 2^{-I_H^\epsilon(\mathcal{X}:\mathcal{Y}\mathcal{Z})} + 3\epsilon \quad (1.2.12)$$

So, for any pair (R_1, R_2) lying in the following region:

$$\begin{aligned} R_1 &\leq I_H^\epsilon(\mathcal{X} : \mathcal{Y}\mathcal{Z}) - \log 1/\epsilon, \\ R_2 &\leq I_H^\epsilon(\mathcal{Y} : \mathcal{X}\mathcal{Z}) - \log 1/\epsilon. \\ R_1 + R_2 &\leq I_H^\epsilon(\mathcal{X}\mathcal{Y} : \mathcal{Z}) - \log 1/\epsilon \end{aligned}$$

there exists a code-book C with average decoding probability less than 6ϵ .

Now if we try to get something similar bounds in quantum scenario we can think of a quantum version of one-shot MAC $\mathfrak{C}$ with two input taking classical alphabets with underlying random probability distribution and one shared quantum output. Alice and Bob are connected to input terminal and Charlie is connected to the output terminal. We call this type of MAC as a classical-quantum MAC or CQ-MAC. Similar to that of classical case let's say $m_1 \in [2^{R_1}]$ and $m_2 \in [2^{R_2}]$ be two messages sent by Alice and Bob respectively. The two input alphabets of C are $\mathcal{X}$ and $\mathcal{Y}$ and the output is in the Hilbert space $\mathcal{Z}$. If $x \in \mathcal{X}$ and $y \in \mathcal{Y}$ are inputted to the channel then it outputs a quantum state with density matrix given as $\rho_{x,y} \in \mathcal{Z}$.

Upon getting the message m_1 , Alice encodes it to a letter $x(m_1) \in \mathcal{X}$ and similarly Bob, encodes it's message m_2 to a letter $y(m_2) \in \mathcal{Y}$. With $x(m_1)$ and $y(m_2)$ being inputted to The channel it outputs a state $\rho_{x(m_1),y(m_2)}$. Let $0 < \epsilon < 1$. Charlie now has to devise some decoding algorithm to guess the messages m_1 and m_2 from the output state with a condition on the average decoding error at most $O(\epsilon)$.

We basically can represent the channel with all of these systems $\mathcal{X}, \mathcal{Y}$ and $\mathcal{Z}$ with a classical-quantum state given as following:

$$\rho^{\mathcal{X}\mathcal{Y}\mathcal{Z}} := \sum_{x,y} p(x) |x\rangle\langle x|^{\mathcal{X}} \otimes p(y) |y\rangle\langle y|^{\mathcal{Y}} \otimes \rho_{x,y}^{\mathcal{Z}}$$

where $|x\rangle\langle x|^{\mathcal{X}}$'s and $|y\rangle\langle y|^{\mathcal{Y}}$'s are orthogonal states and interpreted only as classical systems because they are well-distinguishable.

We construct the code-book C in a following randomised approach, which is quite similar to what we have done previously in classical case. So, the tensor products of the marginal states of the original state, $\rho^{\mathcal{X}\mathcal{Z}} \otimes \rho^{\mathcal{Y}}$, $\rho^{\mathcal{Y}\mathcal{Z}} \otimes \rho^{\mathcal{X}}$ and $\rho^{\mathcal{X}} \otimes \rho^{\mathcal{Y}} \otimes \rho^{\mathcal{Z}}$ can occur erroneously. To resolve these types of error we need the following hypothesis testing mutual information terms or quantum

hypothesis testing relative entropy (which we have already defined in 1.2.3) terms:

$$\begin{aligned} I_H^\epsilon(\mathcal{Y} : \mathcal{XZ}) &:= D_H^\epsilon(\rho^{\mathcal{XYZ}} \| \rho^{\mathcal{XZ}} \otimes \rho^{\mathcal{Y}}), \\ I_H^\epsilon(\mathcal{X} : \mathcal{YZ}) &:= D_H^\epsilon(\rho^{\mathcal{XYZ}} \| \rho^{\mathcal{YZ}} \otimes \rho^{\mathcal{X}}), \\ I_H^\epsilon(\mathcal{XY} : \mathcal{Z}) &:= D_H^\epsilon(\rho^{\mathcal{XYZ}} \| \rho^{\mathcal{X}} \otimes \rho^{\mathcal{Y}} \otimes \rho^{\mathcal{Z}}) \end{aligned} \quad (1.2.13)$$

Notice that in classical case we constructed a test $\hat{T}$ by performing the intersection three different test elements $T^{\mathcal{X}}, T^{\mathcal{Y}}$ and T which are the optimal tests coming from the classical hypothesis testing relative entropy definitions $I_H^\epsilon(\mathcal{Y} : \mathcal{XZ}), I_H^\epsilon(\mathcal{X} : \mathcal{YZ})$ and $I_H^\epsilon(\mathcal{XY} : \mathcal{Z})$ respectively. Similarly, suppose $\Pi^{\mathcal{X}}, \Pi^{\mathcal{Y}}$ and Π be the optimal POVMs coming from the definition of quantum hypothesis testing relative entropy definitions 1.2.13 $I_H^\epsilon(\mathcal{Y} : \mathcal{XZ}), I_H^\epsilon(\mathcal{X} : \mathcal{YZ})$ and $I_H^\epsilon(\mathcal{XY} : \mathcal{Z})$ respectively. let $0 < \epsilon < 1$. We need such an intersection POVM $\hat{\Pi}$ of the above three POVMs such that it holds the following properties.

$$\text{Tr}[\hat{\Pi} \rho^{\mathcal{XYZ}}] \geq 1 - \mathcal{P}(\epsilon), \quad (1.2.14)$$

$$\text{Tr}[\hat{\Pi}(\rho^{\mathcal{XZ}} \otimes \rho^{\mathcal{Y}})] \leq 2^{-I_H^\epsilon(\mathcal{Y} : \mathcal{XZ})}, \quad (1.2.15)$$

$$\text{Tr}[\hat{\Pi}(\rho^{\mathcal{YZ}} \otimes \rho^{\mathcal{X}})] \leq 2^{-I_H^\epsilon(\mathcal{X} : \mathcal{YZ})}, \quad (1.2.16)$$

$$\text{Tr}[\hat{\Pi}(\rho^{\mathcal{X}} \otimes \rho^{\mathcal{Y}} \otimes \rho^{\mathcal{Z}})] \leq 2^{-I_H^\epsilon(\mathcal{XY} : \mathcal{Z})}, \quad (1.2.17)$$

where $\mathcal{P}(\epsilon)$ is a polynomial function of ϵ .

If such a POVM $\hat{\Pi}$ exists over the Hilbert space $\mathcal{XYZ}$, then we can achieve the following rate pair region

$$\begin{aligned} R_1 &\leq I_H^\epsilon(\mathcal{X} : \mathcal{YZ}) - \log 1/\epsilon, \\ R_2 &\leq I_H^\epsilon(\mathcal{Y} : \mathcal{XZ}) - \log 1/\epsilon, \\ R_1 + R_2 &\leq I_H^\epsilon(\mathcal{XY} : \mathcal{Z}) - \log 1/\epsilon \end{aligned} \quad (1.2.18)$$

with an average decoding error probability atmost $O(\epsilon)$.

1.3 Contribution of this dissertation

- We have examined the research conducted by Sen [25] on constructing POVMs on the intersection subspace for classical-quantum states using two non-trivial concepts: tilting and augmentation. Additionally, we have explored its potential application in proving error upper-bounds for a classical-quantum multiple access channel (MAC) under one-shot conditions.
- We conducted a study on the input space distribution of a classical-quantum multiple access channel and found that they are generally mutually independent. However, we also discovered that even if they are dependent, we can still utilize Sen's work [25] on the joint distribution of input spaces to obtain a comparable upper-bound for error probabilities. This approach may prove useful in addressing other quantum information theory problems in the future.
- We attempted to apply Sen's approach from [25] to assess its effectiveness on general quantum states. Establishing an error upper bound for general quantum states would enable us to determine an error upper bound for Entanglement assisted MAC, which differs from vanilla MAC in that it operates on general quantum states. Regrettably, Sen's ideas are only partially applicable to general quantum states. We conducted an analysis to determine the cause of this issue.
- Our work presents a new approach to defining the intersection of two measurement operators and establishes a lower bound for measurement probabilities. We conducted research on asymmetric hypothesis testing and the Chernoff Stein's lemma, a significant breakthrough in information theory. Using our approach, we were able to prove a weak converse of the Quantum Chernoff Stein's lemma.

In the next chapter, we'll try to show that the notion of intersection of two or more measurement operators is highly non trivial and will see some approaches to create a robust notion of "Intersection of Measurement Operators" gradually.

1.4 Organization

- In the first chapter Introduction we discussed about the main issue that our dissertation focuses followed by providing some needful preliminaries to for understanding the dissertation better and then giving the motivation behind working on this issue.
- The chapter 2 starts with demsonstrating the highly-nontriviality of Intersection notion of certain POVMs and then discussing about the concept of geometrically tilting[25] a state.
- Chapter 3 discuss about the techniqe of 'smoothing and augmentation'[25].
- In chapter 4 we investigate an one-shot probability innerbound for a CQ-MAC[278] and then provide a more generalised version of the proof.
- Chapter 5 provides an attempt to map the strategy discussed in chapter 4 to prove similar kind of upper-bounds for general quantum states.
- In chapter 6 we give a novel construction of intersection of two measurement operators and provide a lower-bound of probability for general qunatum states.
- Finally in chapter 7 we draw our conclision of this thesis work and discuss some further open ideas which we will be interesting to work on.

Chapter 2

Tilted Span

2.1 Non-triviality of Intersection of measurements

Before going into the main construction we will first try to show that the notion of intersection of two or more measurement operators is highly non trivial. But at first we need to set some presumptions. Let Π_1, Π_2 and Π_3 be three non-commutative POVMs(i.e. they might not have the same support set). Using Naimark's dilation theorem we can have an embedding to an extended Hilbert space such that we will have an isometric orthogonal projective measurements $\hat{\Pi}_1, \hat{\Pi}_2$ and $\hat{\Pi}_3$ corresponding to each POVMs mentioned above. Now we can try taking the projector $\hat{\Pi}$ over the intersection of the support set of $\hat{\Pi}_1, \hat{\Pi}_2$ and $\hat{\Pi}_3$. Now, let's see some attempts to create a robust notion of "Intersection of Measurement Operators".

2.1.1 Intersection of supports

If $\hat{\Pi}_1, \hat{\Pi}_2$ and $\hat{\Pi}_3$ are three projective measurement(i.e. they might not have the same support set) Now we can try taking the projector $\hat{\Pi}$ over the intersection of the support set of $\hat{\Pi}_1, \hat{\Pi}_2$ and $\hat{\Pi}_3$

The above approach of taking intersection suffers an issue which is due to the reason that the intersection can be a zero subspace(since the projectors can be non-commutative sometimes).

2.1.2 Simultaneous Product of Measurements

Another simple approach can be taking the product of the projectors $\Pi := \hat{\Pi}_1 \hat{\Pi}_2 \hat{\Pi}_3 \hat{\Pi}_3 \hat{\Pi}_2 \hat{\Pi}_1$. But this approach could fail since the projectors are non-commutative in general , sandwiching one projector by the other will depend on the ordering of the projectors appearing in Π .

Let's try to analyse it with taking Case Study 1.2.2 as a reference.

We had POVMs Π^x, Π^y and Π as defined in Case Study 1.2.2 and we need a $\hat{\Pi}$ such that it satisfies Properties [1.2.14-1.2.17]. If we define $\hat{\Pi} := \Pi^x \Pi^y \Pi \Pi \Pi^y \Pi^x$, Property 1.2.14 can be satisfied with a lower-bound of $1 - o(\epsilon)$ using Fact8. But if for the other three properties it's quite unclear that whether they can be satisfied synchronously.

For an example, if we try to analyse the below expression,

$$\begin{aligned} & \text{Tr}[\hat{\Pi}(\rho^{\mathcal{Y}\mathcal{Z}} \otimes \rho^{\mathcal{X}})] \\ &= \text{Tr}[\Pi^{\mathcal{X}} \Pi^{\mathcal{Y}} \Pi \Pi^{\mathcal{Y}} \Pi^{\mathcal{X}} (\rho^{\mathcal{Y}\mathcal{Z}} \otimes \rho^{\mathcal{X}})] \end{aligned}$$

If someone tries to prove Property 1.2.16, we know $\text{Tr}[\hat{\Pi}(\rho^{\mathcal{Y}\mathcal{Z}} \otimes \rho^{\mathcal{X}})] \leq 2^{-I_H^{\epsilon}(\mathcal{X}:\mathcal{Y}\mathcal{Z})}$

2.1.3 Span of Orthogonal Complements

In classical scenario, we know intersection of a family of events(classical) is nothing but the complement of the union of the individual complement of the events. *Using similar intuitions we can take the span of the supports of the projectors as a notion of 'union' of a family of projectors. Now complement of a POVM Π in a Hilbert space $\mathcal{H}$ is defined as $\mathbf{I}_{\mathcal{H}} - \Pi$, which is a projection(for the sake of simplicity let's assume POVMs to be projectors) on the orthogonal complement of the subspace Π is projecting onto.*

But this method fails again due to the fact that the span of the support can take up the whole Hilbert space the projectors act on, making the complement of the span a **zero subspace**.

The concept of **"tilted span of subspaces"** aims to enhance our understanding of the union of POVMs, or more specifically, the union of subspaces that the POVMs project onto. The next section will delve into the notion of tilted span subspaces.

2.2 Tilted span of subspaces

Let $\mathcal{H}$ be our original Hilbert space. Let's consider Hilbert spaces $\mathcal{H}_j, j \in [l]$, each of the same dimension as of $\mathcal{H}$ i.e $\dim \mathcal{H}$. Now define the conjugated and extended Hilbert space $\hat{\mathcal{H}} := \mathcal{H} \oplus \{\oplus_{j \in [l]} \mathcal{H}_j\}$. Consider $\mathcal{E}_j, j \in [l]$ be linear maps mapping $\mathcal{H}$ isometrically onto $\mathcal{H}_j$, for $j \in [l]$, let $0 < a_j < 1$, we design our a_j -tilting map $\mathcal{E}_{j,a_j} : \mathcal{H}_j \mapsto \hat{\mathcal{H}}$ as follows:

$$\mathcal{E}_{j,a_j} := \sqrt{1 - a_j} \mathcal{I}_{\mathcal{H}} + \sqrt{a_j} \mathcal{E}_j \quad (2.2.1)$$

where $\mathcal{E}_j$ is an isometric embedding of $\mathcal{H}$ to $\mathcal{H}_j$ and $\mathcal{I}_{\mathcal{H}}$ is the identity embedding of $\mathcal{H}$ to $\hat{\mathcal{H}}$. We will now give the formal definition of tilted span of subspaces. $\mathcal{E}_{j,a_j}$

Definition 2.2.1 (Tilted Span). *Let $0 < a_j < 1, \forall j \in [l]$. The subspace $\mathcal{E}_{j,a_j}(W_j) \leq \hat{\mathcal{H}}$ will be called the a_j -tilt of W_j along the j -th direction. The subspace $\mathbf{W}_{(a_1, \dots, a_l)}$ of $\hat{\mathcal{H}}$ defined by*

$$\mathbf{W}_{(a_1, \dots, a_l)} := \bigoplus_{j=1}^l \mathcal{E}_{j,a_j}(W_j)$$

is called the $(a_1, \dots, a_l)$ -tilted span of $W_1, \dots, W_l$.

2.2.1 Illustration of Tilted Span

Now let's try to understand how the above concept is working with an example. Let's assume our original Hilbert space is $\mathbb{C}^2$. Now let $W_1 := \text{span}\{|0\rangle\}$ and $W_2 := \text{span}\{\sqrt{1-\epsilon}|0\rangle + \sqrt{\epsilon}|1\rangle\}$. Consider a quantum state $\rho := |1\rangle\langle 1|$. Now the probability of ρ being accepted by W_1, W_2 is 0 and ϵ respectively, but the probability of ρ being accepted by $\text{span}\{W_1 + W_2\}$ is 1. The reason behind such behaviour is due to a small angle between W_1, W_2 . Now using tilting of subspaces we can actually push(tilt) W_1 and W_2 away in a mutually orthogonal direction while embedding them isometrically in an extended Hilbert space. Let's say the original Hilbert space $\mathcal{A} \equiv \mathbb{C}^2$. Now $\forall j \in [2]$, we define $\mathcal{A}_j \equiv \mathbb{C}^2$ and the extended Hilbert space is given as $\hat{\mathcal{A}} := \mathcal{A} \oplus \mathcal{A}_1 \oplus \mathcal{A}_2$. Now, using the above mentioned concept of tilted span of subspace, for a given $0 < \alpha < 1$ we have the tilted subspace W'_1 and W'_2 as follows:

$$\begin{aligned} W'_1 &:= \text{span}\{\sqrt{1-\alpha}|0\rangle^{\mathcal{A}} + \sqrt{\alpha}|0\rangle^{\mathcal{A}_1}\} \\ W'_2 &:= \text{span}\{\sqrt{(1-\alpha)(1-\epsilon)}|0\rangle^{\mathcal{A}} + \sqrt{\alpha(1-\epsilon)}|0\rangle^{\mathcal{A}_2} + \sqrt{(1-\alpha)\epsilon}|1\rangle^{\mathcal{A}} + \sqrt{\alpha\epsilon}|1\rangle^{\mathcal{A}_2}\} \end{aligned}$$

Now we define the α -tilted span $W_\alpha := \text{span}\{W'_1, W'_2\}$ or $W_\alpha := \text{span}\{W'_1 + W'_2\}$.

Let's define the basis of tilted subspace W'_1 and W'_2 as $|u\rangle := \sqrt{1-\alpha}|0\rangle^{\mathcal{A}} + \sqrt{\alpha}|0\rangle^{\mathcal{A}_1}$ and $|v\rangle := \sqrt{(1-\alpha)(1-\epsilon)}|0\rangle^{\mathcal{A}} + \sqrt{\alpha(1-\epsilon)}|0\rangle^{\mathcal{A}_2} + \sqrt{(1-\alpha)\epsilon}|1\rangle^{\mathcal{A}} + \sqrt{\alpha\epsilon}|1\rangle^{\mathcal{A}_2}$ respectively. But they are

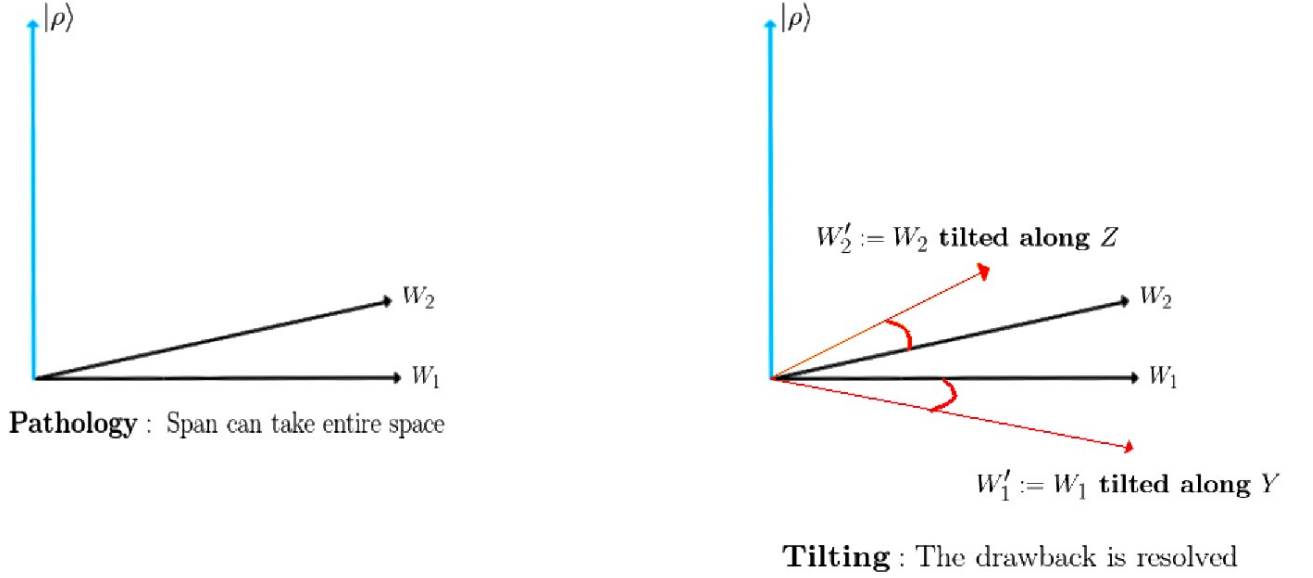


Figure 2.1: Span versus tilted span : W_1 and W_2 are two subspaces in the same plane as the paper is having two orthogonal co-ordinate basis X and Y . We have a quantum state $|\rho\rangle$ also in the same plane. The left figure shows $|\rho\rangle$ has small projections on the two subspaces but very large projection on their span $W_1 + W_2$. In this particular scenario $|\rho\rangle$ is lying completely on the span $W_1 + W_2$. On the contrary in the right figure, we can see that W_1 is tilted along Y and W_2 is tilted along a orthogonal direction Z which is assumed to be perpendicular to the plane of paper. The quantum state $|\rho\rangle$ now has small projection onto the span $W'_1 + W'_2$ of the the two tilted subspaces W'_1 and W'_2 .

not orthogonal since $\langle u|v\rangle = (1 - \alpha)\sqrt{1 - \epsilon}$. Notice that $\|u\|_{2^2=1}$, we need to find a orthogonal projection $|p\rangle$ of $|v\rangle$ on $|u\rangle$. So , $|p\rangle = (|u\rangle\langle u|) \bullet |v\rangle = (1 - \alpha)\sqrt{(1 - \alpha)(1 - \epsilon)}|0\rangle^{\mathcal{A}} + (1 - \alpha)\sqrt{\alpha(1 - \epsilon)}|0\rangle^{\mathcal{A}_1}$. So an orthonormal basis $|v'\rangle$ along $|v\rangle - |p\rangle$ is as following :

$$|v'\rangle = \frac{1}{\sqrt{\epsilon + \alpha(1 - \epsilon)(2 - \alpha)}} \left(\alpha\sqrt{(1 - \alpha)(1 - \epsilon)}|0\rangle^{\mathcal{A}} + \sqrt{\alpha(1 - \epsilon)}|0\rangle^{\mathcal{A}_2} + \sqrt{(1 - \alpha)\epsilon}|1\rangle^{\mathcal{A}} + \sqrt{\alpha\epsilon}|1\rangle^{\mathcal{A}_2} - (1 - \alpha)\sqrt{\alpha(1 - \epsilon)}|0\rangle^{\mathcal{A}_1} \right)$$

So,

$$W'_\alpha = \text{span} \left\{ \sqrt{1 - \alpha}|0\rangle^{\mathcal{A}} + \sqrt{\alpha}|0\rangle^{\mathcal{A}_1}, \frac{1}{\sqrt{\epsilon + \alpha(1 - \epsilon)(2 - \alpha)}} \left(\alpha\sqrt{(1 - \alpha)(1 - \epsilon)}|0\rangle^{\mathcal{A}} + \sqrt{\alpha(1 - \epsilon)}|0\rangle^{\mathcal{A}_2} + \sqrt{(1 - \alpha)\epsilon}|1\rangle^{\mathcal{A}} + \sqrt{\alpha\epsilon}|1\rangle^{\mathcal{A}_2} - (1 - \alpha)\sqrt{\alpha(1 - \epsilon)}|0\rangle^{\mathcal{A}_1} \right) \right\}$$

Now we can see the projection of $|\rho\rangle$ onto W'_α is of length

$$\frac{\sqrt{(1-\alpha)\epsilon}}{\sqrt{\epsilon + \alpha(1-\epsilon)(2-\alpha)}} < \frac{\sqrt{(1-\alpha)\epsilon}}{\sqrt{\alpha}} \quad (2.2.2)$$

$$< \frac{\sqrt{\epsilon}}{\sqrt{\alpha}} \quad (2.2.3)$$

Now here 2.2.2 is due to the fact that $\alpha \in (0, 1)$ and 2.2.3 is due to the fact that $\epsilon \in (0, 1)$. Setting $\alpha = \sqrt{\epsilon}$ we get the length less than $\epsilon^{\frac{1}{4}}$ and the probability of $|\rho\rangle$ getting accepted by W'_α is less than $\sqrt{\epsilon}$ and the l_2 -norm of the difference between the original subspaces and the tilted subspaces are in $O(2^{\frac{1}{2}}, \epsilon^{\frac{1}{4}})$.

In the next section we will work out an example to get a better view of this notion of "tilted span".

Before going into that let's prove a proposition which will later help us in chapter 4 while proving error upper-bounds for a classical quantum multiple access channel using the concept we discussed in chapter 2 and chapter 3.

Proposition 2.2.2. [25, Proposition 2] Let $0 < a < 1$. Let $|h\rangle \in \mathcal{H}$ be a unit vector. Let $W_1, \dots, W_l$ be l subspaces over $\mathcal{H}$. $\forall j \in [l]$, define $\epsilon_j := \|\Pi_{W_j}|h\rangle\|_2^2$ and let $0 < a_j < 1$. Define $a := \min_{j \in [l]} a_j$. Then,

$$\max_{j \in [l]} (1 - a_j) \epsilon_j \leq \left\| \Pi_{\mathbf{W}_{(a_1, \dots, a_l)}} |h\rangle \right\|_2^2 \leq \frac{1 - a}{a} \sum_{j=1}^l \epsilon_j$$

Proof. At first, we will try to prove the lower bound.

$$\left\| \Pi_{\mathbf{W}_{(a_1, \dots, a_l)}} |h\rangle \right\|_2^2 \geq \sum_{j=1}^l \left\| \Pi_{\mathcal{E}_{j, a_j}(W_j)} |h\rangle \right\|_2^2 \quad (2.2.4)$$

$$\geq \sum_{j=1}^l \left\| \sqrt{1 - a_j} \Pi_{W_j} |h\rangle \right\|_2^2 \quad (2.2.5)$$

$$= \sum_{j=1}^l (1 - a_j) \left\| \Pi_{W_j} |h\rangle \right\|_2^2$$

$$= \sum_{j=1}^l (1 - a_j) \epsilon_j$$

$$\geq \max_{j \in [l]} (1 - a_j) \epsilon_j \quad (2.2.6)$$

Here the inequality in 2.2.4 is due to the fact that the length of the projection of a vector in a subspace should be greater than equal to the sum of the square of the length of the projection of that particular vector on the individual basis and 2.2.5 follows from the definition in 2.2.1. Now we have to prove the upper bound. Let's define a vector $|\hat{h}\rangle = \Pi_{\mathbf{W}_{(a_1, \dots, a_l)}} |h\rangle$. So as $|\hat{h}\rangle \in \mathbf{W}_{(a_1, \dots, a_l)}$, let think of $|\hat{h}\rangle = \sum_{j=1}^l \lambda_j |u_j\rangle$, where $\forall j \in [l], \lambda_j \in \mathbb{C}$, and $|u_j\rangle$ is an arbitrary unit vector in $\mathcal{E}_{j, a_j}(W_j)$. Let $|u_j\rangle = \mathcal{E}_{j, a_j}(|v_j\rangle)$, where $|v_j\rangle$ is a unit vector in W_j and can uniquely determined by $|u_j\rangle$, since for any $j \in [l]$, $\mathcal{E}_{j, a_j}$ is an isometric embedding. So now we can write,

$$\begin{aligned} \left\| |\hat{h}\rangle \right\|_2^2 &= \left\| \sum_{j=1}^l \sqrt{1 - a_j} \mu_j |v_j\rangle + \sum_{j=1}^l \sqrt{a_j} \lambda_j \mathcal{E}_j(|v_j\rangle) \right\|_2^2 \\ &\geq \left\| \sum_{j=1}^l \sqrt{a_j} \lambda_j \mathcal{E}_j(|v_j\rangle) \right\|_2^2 \\ &\geq \alpha \sum_{j=1}^l |\lambda_j|^2. \end{aligned} \quad (2.2.7)$$

In the other way,

$$\begin{aligned} \left\| |\hat{h}\rangle \right\|_2^2 &= \langle \hat{h} | \hat{h} \rangle \\ &= \left| \langle h | \Pi_{\mathbf{W}_{(a_1, \dots, a_l)}} |h\rangle \right| \\ &\stackrel{a}{\leq} \left| \langle h | \hat{h} \rangle \right| = \left| \sum_{j=1}^l \lambda_j \langle h | u_j \rangle \right| \end{aligned}$$

$$\stackrel{b}{\leq} \sqrt{\sum_{j=1}^l |\lambda_j|^2} \cdot \sqrt{\sum_{j=1}^l |\langle h|u_j\rangle|^2}. \quad (2.2.8)$$

Here a is due to the fact *forgot the theorem* and b follows from Cauchy-Schwartz inequality. Now from the above inequalities 2.2.7 and 2.2.8 we get,

$$\begin{aligned} \alpha \sqrt{\sum_{j=1}^l |\lambda_j|^2} &\leq \sqrt{\sum_{j=1}^l |\langle h|u_j\rangle|^2} \\ &= \sqrt{\sum_{j=1}^l (1 - a_j) |\langle h|v_j\rangle|^2} \end{aligned} \quad (2.2.9)$$

$$= \sqrt{\sum_{j=1}^l (1 - a_j) |\langle h|\Pi_{W_j}|v_j\rangle|^2} \quad (2.2.10)$$

$$\begin{aligned} &= \sqrt{\sum_{j=1}^l (1 - a_j) \langle h|\Pi_{W_j}|v_j\rangle \langle v_j|\Pi_{W_j}|h\rangle} \\ &\leq \sqrt{\sum_{j=1}^l (1 - a_j) \langle h|\Pi_{W_j}|\Pi_{W_j}|h\rangle} \end{aligned} \quad (2.2.11)$$

$$\begin{aligned} &= \sqrt{\sum_{j=1}^l (1 - a_j) \|\Pi_{W_j}|h\rangle\|_2^2} \\ &\leq \sqrt{1 - \alpha} \sqrt{\sum_{j=1}^l \epsilon_j} \end{aligned} \quad (2.2.12)$$

$$\Rightarrow \sqrt{\sum_{j=1}^l |\lambda_j|^2} \leq \frac{\sqrt{1 - \alpha}}{\alpha} \sqrt{\sum_{j=1}^l \epsilon_j}. \quad (2.2.13)$$

From the above working we also have,

$$\sqrt{\sum_{j=1}^l |\langle h|u_j\rangle|^2} \leq \sqrt{1 - \alpha} \sqrt{\sum_{j=1}^l \epsilon_j}. \quad (2.2.14)$$

Here 2.2.9 follows from the definition of $|u_j\rangle$, and notice that $\mathcal{E}_j(|v_j\rangle)$ is in $\mathcal{H}_j$, while $|h\rangle$ belongs to $\mathcal{H}$, so they are mutually orthogonal since they reside in mutually orthogonal Hilbert subspaces in the extended Hilbert space i.e. $\hat{\mathcal{H}}$. So, $\langle h|\mathcal{E}_j(|v_j\rangle) = 0$ and we are only left with $\langle h|v_j\rangle$. For 2.2.10 since $|v_j\rangle \in W_j$ projecting it in W_j will left us with $|v_j\rangle$ only i.e $\Pi_{W_j}|v_j\rangle = |v_j\rangle$. 2.2.11 follows from the fact that $|v_j\rangle\langle v_j| \leq I_{\mathcal{H}}$. Now if we put the RHS of 2.2.13 and 2.2.14 in 2.2.8 we get,

$$\begin{aligned}
\left\| |\hat{h}\rangle \right\|_2^2 &\leq \sqrt{\sum_{j=1}^l |\lambda_j|^2} \cdot \sqrt{\sum_{j=1}^l |\langle h|u_j\rangle|^2} \\
&\leq \frac{\sqrt{1-\alpha}}{\alpha} \sqrt{\sum_{j=1}^l \epsilon_j} \cdot \sqrt{1-\alpha} \sqrt{\sum_{j=1}^l \epsilon_j} = \frac{1-\alpha}{\alpha} \sum_{j=1}^l \epsilon_j
\end{aligned} \tag{2.2.15}$$

□

So we have successfully proved our proposition. As a corollary we now give a small extension to Propostion 1.

Corollary 2.2.3. [25, Corollary 1] Let $|h\rangle \in \mathcal{H}$ be a unit vector. Let $W_0, W_1, \dots, W_l$ be subspaces of $\mathcal{H}$. $\forall j \in [l] \cup 0$, define $\epsilon_j := \|\Pi_{W_j}|h\rangle\|_{2^2}$ and let $0 < \alpha < \frac{1}{3}$. Define the subspace $\mathbf{W}$ of $\hat{\mathcal{H}}$ as $\mathbf{W} := W_0 + \mathbf{W}_\alpha$. Define $\epsilon := \frac{1-\alpha}{\alpha} \sum_{j=1}^l \epsilon_j$. Then,

$$\max\{\epsilon_0, (1-\alpha)(\max_{j:j \in [l]} \epsilon_j)\} \leq \|\Pi_{\mathbf{W}}|h\rangle\|_{2^2} \leq \frac{3l}{\alpha}(\epsilon_0 + \epsilon)$$

2.3 An Instance to Exemplify

Now let's try to apply this result and see how our approach works in an example. Let's say we have three POVMs Π_{W_1} , Π_{W_2} and Π_{W_3} projects (for the sake of generalization) on the subspaces W_1 , W_2 and W_3 each belongs to $\mathcal{H}$ and we have a state vector $|\rho\rangle$ and $\rho := |\rho\rangle\langle\rho|$. Now $\forall j \in [l]$, for $\epsilon \in (0, 1)$ $\text{tr}[\Pi_{W_j}\rho] \geq 1 - \epsilon$ and for each $j \in [l]$, $\Pi_{W_j^\perp} := (\mathbf{I}_{\mathcal{H}} - \Pi_{W_j})$ is a POVM projecting on $W_j^\perp$. Now let's define the α -tilted span $W^\alpha \in \mathcal{H}$ as follows:

$$W_\alpha^\perp = \bigoplus_{j=1}^3 \mathcal{E}_{j,\alpha}(W_j^\perp) \text{ (where } 0 < \alpha < 1 \text{)}$$

So our required POVM is $(\mathbf{I}_{\hat{\mathcal{H}}} - \Pi_{W_\alpha^\perp})$ where $\hat{\mathcal{H}} = \mathcal{H} \oplus \mathcal{H}_1 \oplus \mathcal{H}_2 \oplus \mathcal{H}_3$, where $\forall j \in [3]$, $\mathcal{H}_j$ is a Hilbert space with $\dim(\mathcal{H})$. Thus,

$$\begin{aligned} \text{tr}[(\mathbf{I}_{\hat{\mathcal{H}}} - \Pi_{W_\alpha^\perp})\rho] &= 1 - \text{tr}[\Pi_{W_\alpha^\perp}\rho] \\ &= 1 - \text{tr}[\Pi_{W_\alpha^\perp}|\rho\rangle\langle\rho|] \\ &= 1 - \text{tr}[\langle\rho|\Pi_{W_\alpha^\perp}|\rho\rangle] \\ &= 1 - \langle\rho|\Pi_{W_\alpha^\perp}|\Pi_{W_\alpha^\perp}|\rho\rangle \\ &= 1 - \|\Pi_{W_\alpha^\perp}|\rho\rangle\|_{2^2}^2 \end{aligned} \tag{2.3.1}$$

Now Similarly $\forall j \in [3]$,

$$\begin{aligned} \text{tr}[\Pi_{W_j^\perp}\rho] &= \text{tr}[(\mathbf{I}_{\mathcal{H}} - \Pi_{W_j})\rho] \\ &= 1 - \text{tr}[\langle\rho|\Pi_{W_j}|\rho\rangle] \\ &= 1 - \text{tr}[\langle\rho|\Pi_{W_j}|\Pi_{W_j}|\rho\rangle] \\ &= 1 - \langle\rho|\Pi_{W_j}|\Pi_{W_j}|\rho\rangle \\ &= 1 - \|\Pi_{W_j}|\rho\rangle\|_{2^2}^2 \leq \epsilon \\ \Rightarrow \|\Pi_{W_j^\perp}|\rho\rangle\|_{2^2}^2 &= 1 - \|\Pi_{W_j}|\rho\rangle\|_{2^2}^2 \leq \epsilon \end{aligned} \tag{2.3.2}$$

Continuing from 2.3.1, using the result proved in 2.2.15 we can write,

$$\begin{aligned} \text{tr}[(\mathbf{I}_{\hat{\mathcal{H}}} - \Pi_{W_\alpha^\perp})\rho] &= 1 - \|\Pi_{W_\alpha^\perp}|\rho\rangle\|_{2^2}^2 \\ &\geq 1 - 3 \left(\frac{1 - \alpha}{\alpha} \right) \epsilon \\ &\geq 1 - 3 \left(\frac{\epsilon}{\alpha} \right) \end{aligned}$$

Taking $\alpha = \sqrt{\epsilon}$, we have the bound $\text{tr}[(\mathbf{I}_{\hat{\mathcal{H}}} - \Pi_{W_{\alpha}^{\perp}})\rho] \geq 1 - 3\sqrt{\epsilon}$. Continuing from 2.3.1, using the result proved in 2.2.6 we can write,

$$\begin{aligned} \text{tr}[(\mathbf{I}_{\hat{\mathcal{H}}} - \Pi_{W_{\alpha}^{\perp}})\rho] &= 1 - \|\Pi_{W_{\alpha}^{\perp}}|\rho\rangle\|_{2^2} \\ &\leq 1 - (1 - \alpha)\epsilon \end{aligned}$$

Taking $\alpha = \sqrt{\epsilon}$, we have the bound $\text{tr}[(\mathbf{I}_{\hat{\mathcal{H}}} - \Pi_{W_{\alpha}^{\perp}})\rho] \leq 1 - \sqrt{\epsilon}$.

Chapter 3

Smoothing and Augmentation

Let's try to build a more robust notion of intersection of subspaces. Consider we have a pure state $\Psi^{\mathcal{X}\mathcal{Y}}$ on a bipartite hilbert space $\mathcal{X} \otimes \mathcal{Y}$. The state can be expressed in state-vector form as :

$$|\Psi\rangle^{\mathcal{AB}} = \sum_{a,b} \mu_{a,b} |a\rangle^{\mathcal{A}} \otimes |b\rangle^{\mathcal{B}},$$

where $|a\rangle^{\mathcal{A}}, |b\rangle^{\mathcal{B}}$ are computational basis of the Hilbert space $\mathcal{A}$ and $\mathcal{B}$ respectively.

3.1 Smoothing

If we tilt $|\Psi\rangle^{\mathcal{AB}}$ isometrically in a particular way to get a state $|\Psi\rangle^{\mathcal{A}'\mathcal{B}'}$ over a larger Hilbert space $\mathcal{A}' \otimes \mathcal{B}'$ which can be expressed as following:

$$|\Psi'\rangle^{\mathcal{A}'\mathcal{B}'} := \sum_{a,b} \mu_{a,b} (\sqrt{1-2\epsilon}|a\rangle^{\mathcal{A}} + \sqrt{\epsilon}|a\rangle^{\mathcal{A}_a} + \sqrt{\epsilon}|a\rangle^{\mathcal{A}_b}) \otimes (\sqrt{1-2\epsilon}|b\rangle^{\mathcal{B}} + \sqrt{\epsilon}|b\rangle^{\mathcal{B}_a} + \sqrt{\epsilon}|b\rangle^{\mathcal{B}_b}),$$

where a and b are the indices for corresponding computational basis $|a\rangle$ and $|b\rangle$ and for each (a, b) we assume $\mathcal{A}'$ and $\mathcal{B}'$ are symmetric in nature and given as:

$$\begin{aligned} \mathcal{A}' &:= \mathcal{A} \oplus \left(\bigoplus_a \mathcal{A}_a \right) \oplus \left(\bigoplus_b \mathcal{A}_b \right), \\ \mathcal{B}' &:= \mathcal{B} \oplus \left(\bigoplus_a \mathcal{B}_a \right) \oplus \left(\bigoplus_b \mathcal{B}_b \right) \end{aligned}$$

Here, $\mathcal{A}_a$ and $\mathcal{A}_b$ are orthogonal Hilbert spaces of equal dimension as that of $\mathcal{A}$ and $\mathcal{B}_a$ and $\mathcal{B}_b$ are orthogonal Hilbert spaces of equal dimension as that of $\mathcal{B}$. Now if we try to represent $|\Psi'\rangle^{\mathcal{A}'\mathcal{B}'}$ as a density matrix $\Psi'^{\mathcal{A}'\mathcal{B}'}$, which can be expressed as following:

$$\begin{aligned} &(\Psi')^{\mathcal{A}'\mathcal{B}'} \\ &= \sum_{a,a'} \sum_{b,b'} \mu_{a,b} \mu_{a',b'}^* (\sqrt{1-2\epsilon}|a\rangle^{\mathcal{A}} + \sqrt{\epsilon}|a\rangle^{\mathcal{A}_a} + \sqrt{\epsilon}|a\rangle^{\mathcal{A}_b}) (\sqrt{1-2\epsilon}\langle a'|^{\mathcal{A}} + \sqrt{\epsilon}\langle a'|^{\mathcal{A}_{a'}} + \sqrt{\epsilon}\langle a'|^{\mathcal{A}_{b'}}) \\ &\quad \otimes (\sqrt{1-2\epsilon}|b\rangle^{\mathcal{B}} + \sqrt{\epsilon}|b\rangle^{\mathcal{B}_a} + \sqrt{\epsilon}|b\rangle^{\mathcal{B}_b}) (\sqrt{1-2\epsilon}\langle b'|^{\mathcal{B}} + \sqrt{\epsilon}\langle b'|^{\mathcal{B}_{a'}} + \sqrt{\epsilon}\langle b'|^{\mathcal{B}_{b'}}) \end{aligned}$$

Then, we can write,

$$\begin{aligned}
& (\Psi')^{\mathcal{A}'} \\
&= \text{Tr}_{\mathcal{B}'} \left[\Psi'^{\mathcal{A}'\mathcal{B}'} \right] \\
&= \text{Tr}_{\mathcal{B}'} \left[\sum_{a,a'} \sum_b \mu_{a,b} \mu_{a',b}^* (\sqrt{1-2\epsilon}|a\rangle^{\mathcal{A}} + \sqrt{\epsilon}|a\rangle^{\mathcal{A}_a} + \sqrt{\epsilon}|a\rangle^{\mathcal{A}_b}) (\sqrt{1-2\epsilon}\langle a'|^{\mathcal{A}} + \sqrt{\epsilon}\langle a'|^{\mathcal{A}_{a'}} + \sqrt{\epsilon}\langle a'|^{\mathcal{A}_{b'}}) \right. \\
&\quad \left. \otimes (\sqrt{1-2\epsilon}|b\rangle^{\mathcal{B}} + \sqrt{\epsilon}|b\rangle^{\mathcal{B}_a} + \sqrt{\epsilon}|b\rangle^{\mathcal{B}_b}) (\sqrt{1-2\epsilon}\langle b'|^{\mathcal{B}} + \sqrt{\epsilon}\langle b'|^{\mathcal{B}_{a'}} + \sqrt{\epsilon}\langle b'|^{\mathcal{B}_{b'}}) \right] \\
&= \sum_{a,a'} \sum_b \mu_{a,b} \mu_{a',b}^* (\sqrt{1-2\epsilon}|a\rangle^{\mathcal{A}} + \sqrt{\epsilon}|a\rangle^{\mathcal{A}_a} + \sqrt{\epsilon}|a\rangle^{\mathcal{A}_b}) (\sqrt{1-2\epsilon}\langle a'|^{\mathcal{A}} + \sqrt{\epsilon}\langle a'|^{\mathcal{A}_a} + \sqrt{\epsilon}\langle a'|^{\mathcal{A}_b}) \\
&= (\Psi'')^{\mathcal{A}'} + \Phi^{\mathcal{A}'}
\end{aligned}$$

where

$$(\Psi'')^{\mathcal{A}'} = \sum_{a,a'} \left(\sum_b \mu_{a,b} \mu_{a',b}^* \right) (\sqrt{1-2\epsilon}|a\rangle^{\mathcal{A}} + \sqrt{\epsilon}|a\rangle^{\mathcal{A}_a}) (\sqrt{1-2\epsilon}\langle a'|^{\mathcal{A}} + \sqrt{\epsilon}\langle a'|^{\mathcal{A}_a})$$

and

$$\begin{aligned}
& \Phi^{\mathcal{A}'} \\
&= \sum_{a,a'} \sum_b \mu_{a,b} \mu_{a',b}^* ((\sqrt{1-2\epsilon}|a\rangle^{\mathcal{A}} + \sqrt{\epsilon}|a\rangle^{\mathcal{A}_a}) (\sqrt{\epsilon}\langle a'|^{\mathcal{A}_b}) + (\sqrt{\epsilon}|a\rangle^{\mathcal{A}_b}) (\sqrt{\epsilon}\langle a'|^{\mathcal{A}_a}) + \\
&\quad (\sqrt{\epsilon}|a\rangle^{\mathcal{A}_b}) (\sqrt{1-2\epsilon}\langle a'|^{\mathcal{A}} \sqrt{\epsilon}\langle a'|^{\mathcal{A}_a}))
\end{aligned}$$

If we define the following two normalised isometric embeddings $\mathcal{E}_{\mathcal{A}}$ and $\mathcal{E}_b$ from $\mathcal{A}$ to $\mathcal{A}'$:

$$\begin{aligned}
\mathcal{E}_{\mathcal{A}}(|a\rangle^{\mathcal{A}}) &:= \sqrt{1-2\epsilon}|a\rangle^{\mathcal{A}} + \sqrt{\epsilon}|a\rangle^{\mathcal{A}_a}, \\
\mathcal{E}_b(|a\rangle^{\mathcal{A}}) &:= \sqrt{\epsilon}|a\rangle^{\mathcal{A}_b}
\end{aligned}$$

It's trivial to see that $\|\mathcal{E}_{\mathcal{A}}\|_{\infty} \leq 1$ and $\|\mathcal{E}_b\|_{\infty} = 1$ (since it's an identity embedding) and for every b , $\|\mathcal{E}_b\|_{\infty}$ are equal in magnitude. Now, consider the following mathematical expression: $(\Psi'')^{\mathcal{A}'}$, which is an isometric tilt of $(\Psi)^{\mathcal{A}}$. Now, assume a state $|\Psi_b\rangle$ with the following form:

$$|\Psi_b\rangle := \sum_a \mu_{a,b} |a\rangle^{\mathcal{A}}$$

If we substitute $|\Psi_b\rangle$ in the expression $\Phi^{\mathcal{A}'}$ we get,

$$\Phi^{\mathcal{A}'} = \sqrt{\epsilon} \sum_b \mathcal{E}_{\mathcal{A}} |\Psi_b\rangle^{\mathcal{A}} \langle \Psi_b | \mathcal{E}_b^{\dagger} + \sqrt{\epsilon} \sum_b \mathcal{E}_b |\Psi_b\rangle^{\mathcal{A}} \langle \Psi_b | \mathcal{E}_{\mathcal{A}}^{\dagger} + \epsilon \sum_b \mathcal{E}_b |\Psi_b\rangle^{\mathcal{A}} \langle \Psi_b | \mathcal{E}_b^{\dagger}$$

To prove that the expression $\Phi^{\mathcal{A}'}$ is small in the operator (l_∞) -norm defined in ??, we need to show...

$$\begin{aligned}
\|\Phi^{\mathcal{A}'}\|_\infty &= \left\| \sqrt{\epsilon} \sum_b \mathcal{E}_{\mathcal{A}} |\Psi_b\rangle \langle \Psi_b|^{\mathcal{A}} \mathcal{E}_b^\dagger + \sqrt{\epsilon} \sum_b \mathcal{E}_b |\Psi_b\rangle \langle \Psi_b|^{\mathcal{A}} \mathcal{E}_{\mathcal{A}}^\dagger + \epsilon \sum_b \mathcal{E}_b |\Psi_b\rangle \langle \Psi_b|^{\mathcal{A}} \mathcal{E}_b^\dagger \right\|_\infty \\
&\leq \left\| \sqrt{\epsilon} \sum_b \mathcal{E}_{\mathcal{A}} |\Psi_b\rangle \langle \Psi_b|^{\mathcal{A}} \mathcal{E}_b^\dagger \right\|_\infty + \left\| \sqrt{\epsilon} \sum_b \mathcal{E}_b |\Psi_b\rangle \langle \Psi_b|^{\mathcal{A}} \mathcal{E}_{\mathcal{A}}^\dagger \right\|_\infty + \left\| \epsilon \sum_b \mathcal{E}_b |\Psi_b\rangle \langle \Psi_b|^{\mathcal{A}} \mathcal{E}_b^\dagger \right\|_\infty \\
&\leq \frac{2}{a} \sqrt{\epsilon} \left\| \sum_b \mathcal{E}_{\mathcal{A}} |\Psi_b\rangle \langle \Psi_b|^{\mathcal{A}} \mathcal{E}_b^\dagger \right\|_\infty + \epsilon \left\| \sum_b \mathcal{E}_b |\Psi_b\rangle \langle \Psi_b|^{\mathcal{A}} \mathcal{E}_b^\dagger \right\|_\infty \\
&\leq \frac{2}{b} \sqrt{\epsilon} \sum_b \left\| \mathcal{E}_{\mathcal{A}} |\Psi_b\rangle \langle \Psi_b|^{\mathcal{A}} \mathcal{E}_b^\dagger \right\|_\infty + \epsilon \sum_b \left\| \mathcal{E}_b |\Psi_b\rangle \langle \Psi_b|^{\mathcal{A}} \mathcal{E}_b^\dagger \right\|_\infty \\
&\leq \frac{2}{c} \sqrt{\epsilon} |\mathcal{B}| \left(\max_b \left\| \mathcal{E}_{\mathcal{A}} |\Psi_b\rangle \langle \Psi_b|^{\mathcal{A}} \mathcal{E}_b^\dagger \right\|_\infty \right) + \epsilon |\mathcal{B}| \left(\max_b \left\| \mathcal{E}_b |\Psi_b\rangle \langle \Psi_b|^{\mathcal{A}} \mathcal{E}_b^\dagger \right\|_\infty \right) \\
&\leq \frac{2}{d} \sqrt{\epsilon} |\mathcal{B}| \left(\max_b \left\| \mathcal{E}_{\mathcal{A}} |\Psi_b\rangle \langle \Psi_b|^{\mathcal{A}} \right\|_\infty \cdot \left\| \mathcal{E}_b^\dagger \right\|_\infty \right) + \epsilon |\mathcal{B}| \left(\max_b \left\| \mathcal{E}_b \right\|_\infty \left\| |\Psi_b\rangle \langle \Psi_b|^{\mathcal{A}} \right\|_\infty \left\| \mathcal{E}_b^\dagger \right\|_\infty \right) \\
&= \frac{2}{e} \sqrt{\epsilon} |\mathcal{B}| \left(\max_b \left\| \mathcal{E}_{\mathcal{A}} |\Psi_b\rangle \langle \Psi_b|^{\mathcal{A}} \right\|_\infty \right) + \epsilon |\mathcal{B}| \left(\max_b \left\| |\Psi_b\rangle \langle \Psi_b|^{\mathcal{A}} \right\|_\infty \right) \\
&= \frac{2}{f} \sqrt{\epsilon} |\mathcal{B}| \left(\left\| \mathcal{E}_{\mathcal{A}} \right\|_\infty \max_b \left\| |\Psi_b\rangle \langle \Psi_b|^{\mathcal{A}} \right\|_\infty \right) + \epsilon |\mathcal{B}| \left(\max_b \left\| |\Psi_b\rangle \langle \Psi_b|^{\mathcal{A}} \right\|_\infty \right) \\
&\leq \frac{2}{g} \sqrt{\epsilon} |\mathcal{B}| \left(\max_b \left\| |\Psi_b\rangle \langle \Psi_b|^{\mathcal{A}} \right\|_\infty \right) + \epsilon |\mathcal{B}| \left(\max_b \left\| |\Psi_b\rangle \langle \Psi_b|^{\mathcal{A}} \right\|_\infty \right) \\
&\leq \frac{2}{h} \sqrt{\epsilon} |\mathcal{B}| \left(\max_b \left\| |\Psi_b\rangle \langle \Psi_b|^{\mathcal{A}} \right\|_1 \right) + \epsilon |\mathcal{B}| \left(\max_b \left\| |\Psi_b\rangle \langle \Psi_b|^{\mathcal{A}} \right\|_1 \right) \\
&\leq 3\sqrt{\epsilon} |\mathcal{B}| \left(\max_b \left\| |\Psi_b\rangle \langle \Psi_b|^{\mathcal{A}} \right\|_1 \right) \\
&\leq \frac{3}{i} \sqrt{\epsilon} |\mathcal{B}| \left(\max_b \left\| |\Psi_b\rangle^{\mathcal{A}} \right\|_2^2 \right)
\end{aligned}$$

where **a** follows from symmetry and If we can recall, $\left\| |\Psi_b\rangle^{\mathcal{A}} \right\|_2^2$ is the measurement probability of getting the outcome b . Hence, if the probability 'distribution of measurement outcomes of $\mathcal{B}$ has low (l_∞) -norm i.e. it is 'smooth', then we can say $\|\Phi^{\mathcal{A}'}\|_\infty = \|(\Psi')^{\mathcal{A}'} - (\Psi'')^{\mathcal{A}'}\|_\infty$ is small.

Lastly, with all these analysis, what we are trying to say is, although $(\Psi')^{\mathcal{A}'\mathcal{B}'}$ is tilted along both 'a' and 'b' directions(by 'a' direction we mean tilting along the $\{\mathcal{A}_a, \mathcal{B}_a\}_a$ family of Hilbert spaces and by 'b' direction we mean tilting along the $\{\mathcal{A}_b, \mathcal{B}_b\}_b$ family of Hilbert spaces), after tracing it out over $\mathcal{B}'$, $(\Psi')^{\mathcal{A}'}$ is predominantly tilted along 'a' because it's very close to $(\Psi'')^{\mathcal{A}'}$ in operator (l_∞) -norm. Thus we can conclude that although if a bipartite quantum state (i.e. with two different subsystems), is tilted along two mutually orthogonal directions, its reduced (traced-out) state is primarily tilted along only one of those directions. This phenomenon is called **smoothing** a traced-out quantum state over a subsystem. In the above discussion we actually smoothed $(\Psi')^{\mathcal{A}'\mathcal{B}'}$ with respect to $\text{Tr}_{\mathcal{B}'}[\cdot]$.

3.2 Augmentation

In the above section we learnt '**Smoothing**' of a quantum state. However, notice that smoothing $(\Psi')^{\mathcal{A}'\mathcal{B}'}$ with respect to $\text{Tr}_{\mathcal{B}'}[\cdot]$ in the above-mentioned example is only possible if the measurement probability distribution of measurement outcomes of $\mathcal{B}$ is 'smooth' (simply put, the probability distribution must not have any peak). But for an arbitrary bipartite quantum state $\Psi^{\mathcal{A}\mathcal{B}}$ this might not always be true. So to ensure 'smoothing' we need to augment the state $\Psi^{\mathcal{A}\mathcal{B}}$.

Observe that the state $\Psi^{\mathcal{A}\mathcal{B}}$ is basically defined over two Hilbert spaces $\mathcal{A}$ and $\mathcal{B}$. To ensure smoothness of the measurement probability distribution of a reduced state over any of the two Hilbert spaces, we have to enlarge the Hilbert space and uniformly spread the distribution over the larger Hilbert space.

We define two larger Hilbert spaces $\hat{\mathcal{A}}$ and $\hat{\mathcal{B}}$ as:

$$\begin{aligned}\hat{\mathcal{A}} &:= \mathcal{A} \otimes \mathcal{L}^{\mathcal{A}} \\ \hat{\mathcal{B}} &:= \mathcal{B} \otimes \mathcal{L}^{\mathcal{B}}\end{aligned}$$

where, $\mathcal{L}^{\mathcal{A}}$ and $\mathcal{L}^{\mathcal{B}}$ are new augmented Hilbert spaces of dimension $|\mathcal{L}|$. We can now define our augmented state $\hat{\Psi}^{\hat{\mathcal{A}}\hat{\mathcal{B}}}$ over $\hat{\mathcal{A}}$ and $\hat{\mathcal{B}}$ as follows:

$$\hat{\Psi}^{\hat{\mathcal{A}}\hat{\mathcal{B}}} := |\mathcal{L}|^{-2} \Psi^{\mathcal{A}\mathcal{B}} \otimes \mathbb{1}^{\mathcal{L}^{\mathcal{A}}} \otimes \mathbb{1}^{\mathcal{L}^{\mathcal{B}}}$$

Now we can tilt $\hat{\Psi}^{\hat{\mathcal{A}}\hat{\mathcal{B}}}$ to get a state $(\hat{\Psi}')^{\hat{\mathcal{A}}'\hat{\mathcal{B}'}}$ in the similar manner we have done in the previous section. Now if we have a mixed state (ensemble of certain pure states) we can tilt it by tilting all the pure states components of it. So for $(\hat{\Psi}')^{\hat{\mathcal{A}}'\hat{\mathcal{B}'}}$ it's trivial to see that

$$\left\| (\hat{\Psi}')^{\hat{\mathcal{A}}'} - (\hat{\Psi}'')^{\hat{\mathcal{A}}'} \right\|_{\infty} \leq 3 \frac{\sqrt{\epsilon} |\mathcal{B}|}{|\mathcal{L}|}$$

If we set the dimension of augmenting system $|\mathcal{L}|$ is large enough we can certainly achieve smoothing with respect to $\text{Tr}_{\mathcal{B}'}[\cdot]$.

In the next chapter we'll try to apply the proposed techniques discussed in **chapter 2** and **chapter 3** in a two party classical-quantum multiple access channel (cq-MAC) to check whether we can get our desired properties defined in 1.2.18.

Chapter 4

One-shot inner bound for a CQ-MAC

4.1 Structure of a CQ-MAC

We have already discussed about classical two-party MACs in subsection 1.2.2 and showed the optimal error bound and rate region for the two senders. Here in this chapter, we are considering a two-party multiple access classical-quantum channel (CQ-MAC).

Now let's say Alice and Bob are the two senders who would send classical messages $m_1 \in [2^{R_1}](\mathcal{M}_1)$ and $m_2 \in [2^{R_2}](\mathcal{M}_2)$, where R_1 and R_2 are the rate of transmission of the sender Alice and Bob respectively. There is a MAC $\mathfrak{C}$ taking two alphabet as classical input and output a quantum state to Charlie. Alice and Bob has the input random alphabet(can be treated as random variable) $\mathcal{X}, \mathcal{Y}$ respectively and the channel outputs in Hilbert space $\mathcal{Z}$. Alice and Bob has the encoders $\mathcal{E}_1$ and $\mathcal{E}_2$, where $\mathcal{E}_1$ takes the input m_1 and encode it to a random variable realization $x(m_1) \in \mathcal{X}$ and $\mathcal{E}_2$ takes the input m_2 and encode it to a random variable realization $y(m_2) \in \mathcal{Y}$ and these are feed-ed in the channel and get a quantum state $\rho_{x(m_1), y(m_2)}^{\mathcal{Z}}$ in $\mathcal{Z}$ as output. Now from this quantum state Charlie has to guess the correct message pair (m_1, m_2) . Our aim is to bound the average probability of decoding error at Charlie's end by ϵ .

$$\bar{p}_e \equiv \frac{1}{|\mathcal{M}_1||\mathcal{M}_2|} \sum_{m_1, m_2} p_e(m_1, m_2) \leq \epsilon$$

Where $p_e(m_1, m_2)$ refers to the decoding error in Charlie's end given input messages from Alice and Bob is m_1 and m_2 respectively.

4.1.1 CQ State to describe a CQ-MAC

[We can actually elucidate a two sender CQ-MAC with a classical quantum state For input alphabets $\mathcal{X}$ and $\mathcal{Y}$. if we have fixed probability densities $\{p(x)\}_{x \in \mathcal{X}}$ and $\{p(y)\}_{y \in \mathcal{Y}}$ then we can represent a

classical quantum state as follows:

$$\rho^{\mathcal{X}\mathcal{Y}\mathcal{Z}} := \sum_{x,y} p(x)|x\rangle\langle x|^{\mathcal{X}} \otimes p(y)|y\rangle\langle y|^{\mathcal{Y}} \otimes \rho_{x,y}^{\mathcal{Z}}$$

Similarly we have the following marginal states by performing partial traces:

$$\rho^{\mathcal{X}\mathcal{Z}} = \sum_x p(x)|x\rangle\langle x|^{\mathcal{X}} \otimes \rho_x^{\mathcal{Z}} \quad (4.1.1)$$

$$\text{where, } \rho_x^{\mathcal{Z}} = \sum_y p(y)\rho_{x,y}^{\mathcal{Z}}$$

$$\rho^{\mathcal{Y}\mathcal{Z}} = \sum_y p(y)|y\rangle\langle y|^{\mathcal{Y}} \otimes \rho_y^{\mathcal{Z}} \quad (4.1.2)$$

$$\text{where, } \rho_y^{\mathcal{Z}} = \sum_x p(x)\rho_{x,y}^{\mathcal{Z}}$$

$$\rho^{\mathcal{Z}} = \sum_{x,y} p(x)p(y)\rho_{x,y}^{\mathcal{Z}} \quad (4.1.3)$$

Here in 4.1 the output state is dependent only on X and similarly in 4.2 the output state is dependent only on Y and while in 4.3 the output state is independent of both the inputs.

4.1.2 Augmentation of the CQ state

We now augment input alphabet with a Hilbert space $\mathcal{L}$, $\mathcal{X}' := \mathcal{X} \otimes \mathcal{L}$ and $\mathcal{Y}' := \mathcal{Y} \otimes \mathcal{L}$ and the extended output Hilbert space $\mathcal{Z}' := (\mathcal{Z} \otimes \mathbb{C}^2) \oplus (\mathcal{Z} \otimes \mathbb{C}^2 \otimes \mathcal{L}^{\mathcal{X}}) \oplus (\mathcal{Z} \otimes \mathbb{C}^2 \otimes \mathcal{L}^{\mathcal{Y}})$ where $\{\mathcal{L}^x\}_{x \in \mathcal{X}}$, $\{\mathcal{L}^y\}_{y \in \mathcal{Y}}$ are distinct orthogonal Hilbert spaces labelled with superscripts x and y depending on the input from Alice and Bob respectively and they are isomorphic to $\mathcal{L}$. Let $\{l_x\}$'s are the eigenbasis of $\mathcal{L}^x$ and $\{l_y\}$'s are the eigenbasis of $\mathcal{L}^y$

$$\begin{aligned} \sum_{l_x} |l_x\rangle\langle l_x| &= \mathcal{I}^{\mathcal{L}^x} \equiv \mathcal{I}^{\mathcal{L}} \\ \sum_{l_y} |l_y\rangle\langle l_y| &= \mathcal{I}^{\mathcal{L}^y} \equiv \mathcal{I}^{\mathcal{L}} \end{aligned}$$

So now let's define an augmented version of the existing cq-mac channel $\mathfrak{C}'$ with input alphabet $\mathcal{X}'$ and $\mathcal{Y}'$ and output Hilbert spaces $\mathcal{Z}'$. Unlike the previous channel here we input $(x, l_x), (y, l_y)$ and as an output we get the augmented state $\rho_{x,y}^{\mathcal{Z}} \otimes |0\rangle\langle 0|^{\mathbb{C}^2}$. So the classical-quantum state for the channel $\mathfrak{C}'$ is:

$$\rho^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'} = |\mathcal{L}|^{-2} \sum_{x,l_x,y,l_y} p(x)p(y)|x.l_x\rangle\langle x.l_x|^{\mathcal{X}'} \otimes |y.l_y\rangle\langle y.l_y|^{\mathcal{Y}'} \otimes (\rho_{x,y}^{\mathcal{Z}} \otimes |0\rangle\langle 0|^{\mathbb{C}^2})$$

$$\begin{aligned}
&= |\mathcal{L}|^{-2} \sum_{x,y} p(x)p(y) \left\{ |x\rangle\langle x| \otimes \sum_{l_x} |l_x\rangle\langle l_x| \otimes |y\rangle\langle y| \otimes \sum_{l_y} |l_y\rangle\langle l_y| \otimes \right. \\
&\quad \left. (\rho_{x,y}^{\mathcal{Z}} \otimes |0\rangle\langle 0|) \right\} \\
&= |\mathcal{L}|^{-2} \sum_{x,y} p(x)p(y) |x, y\rangle\langle x, y| \otimes \rho_{x,y}^{\mathcal{Z}} \otimes \mathcal{I}^{\mathcal{L}^x} \otimes \mathcal{I}^{\mathcal{L}^y} \otimes |0\rangle\langle 0| \\
&= |\mathcal{L}|^{-2} \rho^{\mathcal{X}\mathcal{Y}\mathcal{Z}} \otimes \mathcal{I}^{\mathcal{L}^x} \otimes \mathcal{I}^{\mathcal{L}^y} \otimes |0\rangle\langle 0| \\
&= |\mathcal{L}|^{-2} \rho^{\mathcal{X}\mathcal{Y}\mathcal{Z}} \otimes |0\rangle\langle 0| \otimes \mathcal{I}^{\mathcal{L}^{\otimes 2}}
\end{aligned}$$

Now the reason we need this orthogonal eigenbasis $\{|x\rangle\}$'s and $\{|y\rangle\}$'s, is that we need to tilt the output quantum state $(\rho_{x,y}^{\mathcal{Z}} \otimes |0\rangle\langle 0|)$ in $\mathcal{X}'$ direction, $\mathcal{Y}'$ direction or both $\mathcal{X}'\mathcal{Y}'$ direction.

4.1.3 Tilting of the CQ state

Now when we say we are tilting in $\mathcal{X}'$ direction we mean given a vector $l_x \in \mathcal{L}^x$ the tilting map project a vector $|x\rangle \in \hat{\mathcal{Z}}$ in a orthogonal larger isomorphic augmented Hilbert space $\hat{\mathcal{Z}} \otimes \mathcal{L}^x$ by the help of l_x , and then we get the final output by the tilted sum(span if we use a subspace instead of a vector) of the existing vector $|x\rangle$ and it's orthogonal isomorphic embedding. So we finally can define the tilting map for some $\hat{\mathcal{Z}} := \mathcal{Z} \otimes \mathbb{C}^2$ and $0 \leq \delta \leq 1/10$, $\mathcal{T}_{\mathcal{X};l_x,\delta} : \hat{\mathcal{Z}} \mapsto \hat{\mathcal{Z}} \oplus (\hat{\mathcal{Z}} \otimes \mathcal{L}^x)$ as follows:

$$\mathcal{T}_{\mathcal{X};l_x,\delta} : |x\rangle \mapsto \frac{1}{\sqrt{1+\delta^2}}(|x\rangle + \delta|x\rangle|l_x\rangle)$$

Similarly we have tilting map tilting in $\mathcal{Y}$ direction only $\mathcal{T}_{\mathcal{Y};l_y,\delta} : \hat{\mathcal{Z}} \mapsto \hat{\mathcal{Z}} \oplus (\hat{\mathcal{Z}} \otimes \mathcal{L}^y)$ as follows :

$$\mathcal{T}_{\mathcal{Y};l_y,\delta} : |y\rangle \mapsto \frac{1}{\sqrt{1+\delta^2}}(|y\rangle + \delta|y\rangle|l_y\rangle)$$

For the tilting map tilting in both $\mathcal{X}$ and $\mathcal{Y}$ direction $\mathcal{T}_{\mathcal{X}\mathcal{Y};l_x,l_y,\delta} : \hat{\mathcal{Z}} \mapsto \mathcal{Z}'$ is given as:

$$\mathcal{T}_{\mathcal{X}\mathcal{Y};l_x,l_y,\delta} : |h\rangle \mapsto \frac{1}{\sqrt{1+2\delta^2}}(|h\rangle + \delta|h\rangle|l_x\rangle + \delta|h\rangle|l_y\rangle)$$

Now we have a 'perturbed' or 'tilted' variant of channel $\mathfrak{C}''$ of the existing augmented channel, which have the same input alphabets but a tilted version of output quantum state given as

$$(\rho')_{(x,l_x),(y,l_y),\delta}^{\mathcal{Z}'} := \mathcal{T}_{\mathcal{X}\mathcal{Y};l_x,l_y,\delta}(\rho_{x,y}^{\mathcal{Z}} \otimes |0\rangle\langle 0|),$$

where, $\mathcal{T}_{\mathcal{X}\mathcal{Y};l_x,l_y,\delta}$ will act on a mixed state ensemble by acting on each pure state in the ensemble individually. It's easy to see that,

$$\left\| (\rho')_{(x,l_x),(y,l_y),\delta}^{\mathcal{Z}'} - \rho_{x,y}^{\mathcal{Z}} \otimes |0\rangle\langle 0|^{\mathbb{C}^2} \right\|_1 \leq 4\delta^2 \quad (4.1.4)$$

So finally the augmented and tilted classical-quantum state is given as

$$(\rho')^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'} := |\mathcal{L}|^{-2} \sum_{x,l_x,y,l_y} p(x)p(y) |x.l_x\rangle^{\mathcal{X}'} \langle x.l_x| \otimes |y.l_y\rangle^{\mathcal{Y}'} \langle y.l_y| \otimes (\rho')_{(x,l_x),(y,l_y),\delta}^{\mathcal{Z}'}$$

Similarly, we can show that

$$\begin{aligned} & \left\| (\rho')^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'} - \rho^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'} \right\|_1 \\ &= \left\| |\mathcal{L}|^{-2} \sum_{x,l_x,y,l_y} p(x)p(y) |x.l_x\rangle^{\mathcal{X}'} \langle x.l_x| \otimes |y.l_y\rangle^{\mathcal{Y}'} \langle y.l_y| \otimes \left\{ (\rho')_{(x,l_x),(y,l_y),\delta}^{\mathcal{Z}'} - \rho_{x,y}^{\mathcal{Z}} \otimes |0\rangle\langle 0|^{\mathbb{C}^2} \right\} \right\|_1 \\ &\leq |\mathcal{L}|^{-2} \sum_{x,l_x,y,l_y} p(x)p(y) \left\| |x.l_x\rangle^{\mathcal{X}'} \langle x.l_x| \otimes |y.l_y\rangle^{\mathcal{Y}'} \langle y.l_y| \otimes \left\{ (\rho')_{(x,l_x),(y,l_y),\delta}^{\mathcal{Z}'} - \rho_{x,y}^{\mathcal{Z}} \otimes |0\rangle\langle 0|^{\mathbb{C}^2} \right\} \right\|_1 \\ &= |\mathcal{L}|^{-2} \sum_{x,l_x,y,l_y} p(x)p(y) \left\| (\rho')_{(x,l_x),(y,l_y),\delta}^{\mathcal{Z}'} - \rho_{x,y}^{\mathcal{Z}} \otimes |0\rangle\langle 0|^{\mathbb{C}^2} \right\|_1 \\ &\leq |\mathcal{L}|^{-2} \sum_{x,l_x,y,l_y} p(x)p(y) (4\delta^2) \\ &= 4\delta^2 \end{aligned} \quad (4.1.5)$$

4.2 Intersection POVM

In this particular section we'll go through selection of appropriate POVMs for the original CQ state $\rho^{\mathcal{X}\mathcal{Y}\mathcal{Z}}$ of MAC $\mathfrak{C}$ and then applying suitable modification to them in augmented space and then defining the intersection POVM for the modified state $(\rho')^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'}$ of the modified MAC $\mathfrak{C}''$. Let's first discuss about the selection of the POVMs with satisfiability for some particular properties we discussed in 1.2.2

4.2.1 Appropriate selection of POVMs for the original CQ state

Now, let's define the POVM element $(\Pi'')_{\mathcal{X}}^{\mathcal{X}\mathcal{Y}\mathcal{Z}}, (\Pi'')_{\mathcal{Y}}^{\mathcal{X}\mathcal{Y}\mathcal{Z}}$ and $(\Pi'')^{\mathcal{X}\mathcal{Y}\mathcal{Z}}$ inside Hilbert space $\mathcal{X} \otimes \mathcal{Y} \otimes \mathcal{Z}$ in the following way:

$$\begin{aligned}
(\Pi'')^{\mathcal{X}\mathcal{Y}\mathcal{Z}} &:= \sum_{x,y} |x\rangle^{\mathcal{X}}\langle x| \otimes |y\rangle^{\mathcal{Y}}\langle y| \otimes (\Pi'')^{\mathcal{Z}}_{\mathcal{X},x,y} \\
(\Pi'')^{\mathcal{X}\mathcal{Y}\mathcal{Z}} &:= \sum_{x,y} |x\rangle^{\mathcal{X}}\langle x| \otimes |y\rangle^{\mathcal{Y}}\langle y| \otimes (\Pi'')^{\mathcal{Z}}_{\mathcal{Y},x,y} \\
(\Pi'')^{\mathcal{X}\mathcal{Y}\mathcal{Z}} &:= \sum_{x,y} |x\rangle^{\mathcal{X}}\langle x| \otimes |y\rangle^{\mathcal{Y}}\langle y| \otimes (\Pi'')^{\mathcal{Z}}_{x,y}
\end{aligned}$$

For each $(x, y) \in \mathcal{X} \otimes \mathcal{Y}$ and $1 > \epsilon > 0$, POVM elements $(\Pi'')^{\mathcal{Z}}_{\mathcal{X},x,y}$, $(\Pi'')^{\mathcal{Z}}_{\mathcal{Y},x,y}$ and $(\Pi'')^{\mathcal{Z}}_{x,y}$ are defined in a particular way such that following three properties are satisfied:

$$\begin{aligned}
\text{Tr}[(\Pi'')^{\mathcal{X}\mathcal{Y}\mathcal{Z}} \rho^{\mathcal{X}\mathcal{Y}\mathcal{Z}}] &\geq 1 - \epsilon, \\
\text{Tr}[(\Pi'')^{\mathcal{X}\mathcal{Y}\mathcal{Z}} (\rho^{\mathcal{Y}} \otimes \rho^{\mathcal{X}\mathcal{Z}})] &\leq 2^{-I_H^{\epsilon}(\mathcal{Y}:\mathcal{X}\mathcal{Z})} \\
\text{Tr}[(\Pi'')^{\mathcal{X}\mathcal{Y}\mathcal{Z}} \rho^{\mathcal{X}\mathcal{Y}\mathcal{Z}}] &\geq 1 - \epsilon, \\
\text{Tr}[(\Pi'')^{\mathcal{X}\mathcal{Y}\mathcal{Z}} (\rho^{\mathcal{X}} \otimes \rho^{\mathcal{Y}\mathcal{Z}})] &\leq 2^{-I_H^{\epsilon}(\mathcal{X}:\mathcal{Y}\mathcal{Z})} \\
\text{Tr}[(\Pi'')^{\mathcal{X}\mathcal{Y}\mathcal{Z}} \rho^{\mathcal{X}\mathcal{Y}\mathcal{Z}}] &\geq 1 - \epsilon, \\
\text{Tr}[(\Pi'')^{\mathcal{X}\mathcal{Y}\mathcal{Z}} (\rho^{\mathcal{X}\mathcal{Y}} \otimes \rho^{\mathcal{Z}})] &\leq 2^{-I_H^{\epsilon}(\mathcal{X}\mathcal{Y}:\mathcal{Z})}
\end{aligned} \tag{4.2.1}$$

4.2.2 Modification of the selected POVMs for finding the Intersection POVM

By Fact 12, there are orthogonal projections $(\Pi)_{\mathcal{X},x,y}^{\hat{\mathcal{Z}}}$, $(\Pi)_{\mathcal{Y},x,y}^{\hat{\mathcal{Z}}}$ and $(\Pi)_{x,y}^{\hat{\mathcal{Z}}}$ in $\hat{\mathcal{Z}}$ that give equal measurement probability on states $(\sigma^{\mathcal{Z}} \otimes |0\rangle\langle 0|^{\mathbb{C}^d})$ that POVM elements $(\Pi'')^{\mathcal{Z}}_{\mathcal{X},x,y}$, $(\Pi'')^{\mathcal{Z}}_{\mathcal{Y},x,y}$ and $(\Pi'')^{\mathcal{Z}}_{x,y}$ gives on states $\sigma^{\mathcal{Z}}$. Let $W_{\mathcal{X},x,y}$ denotes the orthogonal complement of the support of $(\Pi)_{\mathcal{X},x,y}^{\hat{\mathcal{Z}}}$ in $\hat{\mathcal{Z}}$. Define $W_{\mathcal{Y},x,y}$, $W_{\mathcal{X},x,y}$ analogously. Define the tilted spaces

$$\begin{aligned}
W'_{\mathcal{X};(x,l_x),(y,l_y),\delta} &:= \mathcal{T}_{\mathcal{X};l_x,\delta}(W_{\mathcal{X},x,y}), \\
W'_{\mathcal{Y};(x,l_x),(y,l_y),\delta} &:= \mathcal{T}_{\mathcal{Y};l_x,\delta}(W_{\mathcal{Y},x,y}),
\end{aligned}$$

residing in the Hilbert space $\mathcal{Z}'$. Define the subs ace

$$W'_{(x,l_x),(y,l_y),\delta} := W'_{\mathcal{X};(x,l_x),(y,l_y),\delta} + W'_{\mathcal{Y};(x,l_x),(y,l_y),\delta} + W_{\mathcal{X},x,y},$$

and $(\Pi')_{W'_{(x,l_x),(y,l_y),\delta}}^{\mathcal{Z}'}$ to be the orthogonal projection in $\mathcal{Z}'$ onto $W'_{(x,l_x),(y,l_y),\delta}$. Let $\Pi_{\hat{\mathcal{Z}}}^{\mathcal{Z}'}$ be the orthogonal projection in $\mathcal{Z}'$ onto $\hat{\mathcal{Z}}$. Then define the POVM element $(\Pi')_{(x,l_x),(y,l_y),\delta}^{\mathcal{Z}'}$ in $\mathcal{Z}'$ to be

$$(\Pi')_{(x,l_x),(y,l_y),\delta}^{\mathcal{Z}'} := (\mathbb{1}^{\mathcal{Z}'} - (\Pi')_{W'_{(x,l_x),(y,l_y),\delta}}^{\mathcal{Z}'}) \Pi_{\hat{\mathcal{Z}}}^{\mathcal{Z}'} (\mathbb{1}^{\mathcal{Z}'} - (\Pi')_{W'_{(x,l_x),(y,l_y),\delta}}^{\mathcal{Z}'})$$

Define POVM element

$$(\Pi')^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'} := \sum_{x,y,l_x,l_y} |x.l_x\rangle^{\mathcal{X}'}\langle x.l_x| \otimes |y.l_y\rangle^{\mathcal{Y}'}\langle y.l_y| \otimes (\Pi')_{(x,l_x),(y,l_y),\delta}^{\mathcal{Z}'} \tag{4.2.2}$$

So we have our Intersection POVM $(\Pi')^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'}$ in the larger augmented Hilbert space $\mathcal{X}'\mathcal{Y}'\mathcal{Z}'$. In the next section we will try to check whether we can satisfy all the properties discussed in (1.2.14)-(1.2.17) and the optimal rate regions given in (1.2.18)

4.3 Performance Analysis

Let's first start with the upper-bound of error probability of not getting the actual state i.e. $(\rho')^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'}$. By using equations 4.1.4, 4.1.5, 4.2.1, Fact 8 and corollary 2.2.3, we can show

$$\begin{aligned}
& \text{Tr}[(\mathbb{1}^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'} - (\Pi')^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'})(\rho')^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'}] \\
&= \text{Tr}(\rho')^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'} - \text{Tr}[(\Pi')^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'}(\rho')^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'}] \\
&= |\mathcal{L}|^{-2} \sum_{x,l_x,y,l_y} p(x)p(y) \left(\text{Tr}[(\rho')_{(x,l_x),(y,l_y),\delta}^{\mathcal{Z}'}] - \text{Tr}[(\Pi')_{(x,l_x),(y,l_y),\delta}^{\mathcal{Z}'}(\rho')_{(x,l_x),(y,l_y),\delta}^{\mathcal{Z}'}] \right) \\
&= |\mathcal{L}|^{-2} \sum_{x,l_x,y,l_y} p(x)p(y) \left(\text{Tr}[(\rho')_{(x,l_x),(y,l_y),\delta}^{\mathcal{Z}'}] - \text{Tr}[\Pi_{\hat{\mathcal{Z}}}^{\mathcal{Z}'}(\mathbb{1}^{\mathcal{Z}'} - (\Pi')_{W'_{(x,l_x),(y,l_y),\delta}}^{\mathcal{Z}'})(\rho')_{(x,l_x),(y,l_y),\delta}^{\mathcal{Z}'}] \right. \\
&\quad \left. (\mathbb{1}^{\mathcal{Z}'} - (\Pi')_{W'_{(x,l_x),(y,l_y),\delta}}^{\mathcal{Z}'})\Pi_{\hat{\mathcal{Z}}}^{\mathcal{Z}'}] \right) \\
&\leq_a 4|\mathcal{L}|^{-2} \sum_{x,l_x,y,l_y} p(x)p(y) \left(\text{Tr}[(\mathbb{1}^{\mathcal{Z}'} - \Pi_{\hat{\mathcal{Z}}}^{\mathcal{Z}'})(\rho')_{(x,l_x),(y,l_y),\delta}^{\mathcal{Z}'}] + \text{Tr}[(\Pi')_{W'_{(x,l_x),(y,l_y),\delta}}^{\mathcal{Z}'}(\rho')_{(x,l_x),(y,l_y),\delta}^{\mathcal{Z}'}] \right) \\
&\leq_b 4\delta^2 + 4|\mathcal{L}|^{-2} \sum_{x,l_x,y,l_y} p(x)p(y) \left(\text{Tr}[(\mathbb{1}^{\mathcal{Z}'} - \Pi_{\hat{\mathcal{Z}}}^{\mathcal{Z}'})(\rho_{x,y}^{\mathcal{Z}} \otimes |0\rangle\langle 0|)]^{\mathbb{C}^2} \right. \\
&\quad \left. + \text{Tr}[(\Pi')_{W'_{(x,l_x),(y,l_y),\delta}}^{\mathcal{Z}'}(\rho_{x,y}^{\mathcal{Z}} \otimes |0\rangle\langle 0|)]^{\mathbb{C}^2} \right) \\
&\leq_c 4\delta^2 + \frac{4.3.2|\mathcal{L}|^{-2}}{\delta^2} \sum_{x,l_x,y,l_y} p(x)p(y) \left(\{ (1 - \text{Tr}[\Pi_{\hat{\mathcal{Z}}}^{\mathcal{Z}'}(\rho_{x,y}^{\mathcal{Z}} \otimes |0\rangle\langle 0|)]^{\mathbb{C}^2}) \right. \\
&\quad \left. + (1 - \text{Tr}[\Pi_{Y,x,y}^{\hat{\mathcal{Z}}}(\rho_{x,y}^{\mathcal{Z}} \otimes |0\rangle\langle 0|)]^{\mathbb{C}^2}) \right. \\
&\quad \left. + (1 - \text{Tr}[\Pi_{x,y}^{\hat{\mathcal{Z}}}(\rho_{x,y}^{\mathcal{Z}} \otimes |0\rangle\langle 0|)]^{\mathbb{C}^2}) \right) \\
&= 4\delta^2 + \frac{24|\mathcal{L}|^{-2}}{\delta^2} \sum_{x,l_x,y,l_y} p(x)p(y) \left((1 - \text{Tr}[(\Pi'')_{X,x,y}^{\mathcal{Z}}(\rho_{x,y}^{\mathcal{Z}})]) + (1 - \text{Tr}[(\Pi'')_{Y,x,y}^{\mathcal{Z}}(\rho_{x,y}^{\mathcal{Z}})]) \right. \\
&\quad \left. + (1 - \text{Tr}[(\Pi'')_{x,y}^{\mathcal{Z}}(\rho_{x,y}^{\mathcal{Z}})]) \right) \\
&= 4\delta^2 + \frac{24}{\delta^2} \left((1 - \text{Tr}[(\Pi'')_{\mathcal{X}}^{\mathcal{X}\mathcal{Y}\mathcal{Z}}\rho^{\mathcal{X}\mathcal{Y}\mathcal{Z}}]) + (1 - \text{Tr}[(\Pi'')_{\mathcal{Y}}^{\mathcal{X}\mathcal{Y}\mathcal{Z}}\rho^{\mathcal{X}\mathcal{Y}\mathcal{Z}}]) \right. \\
&\quad \left. + (1 - \text{Tr}[(\Pi'')^{\mathcal{X}\mathcal{Y}\mathcal{Z}}\rho^{\mathcal{X}\mathcal{Y}\mathcal{Z}}]) \right) \\
&\leq 4\delta^2 + \frac{72\epsilon}{\delta^2}
\end{aligned}$$

where, inequality **a** follows from Gao's union bound 8, **b** follows 4.1.4 and **c** follows from 2.2.3.

We choose $\delta := \epsilon^{1/4}$, then

$$\text{Tr}[(\mathbb{1}^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'} - (\Pi')^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'})(\rho')^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'}] \leq 76\epsilon^{1/2} \quad (4.3.1)$$

4.3.1 Marginal states of $(\rho')^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'}$

Notice the following construction of different marginal states of $(\Pi')^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'}$ using partial trace:

$$\begin{aligned} (\rho')^{\mathcal{X}'\mathcal{Z}'} &= |\mathcal{L}|^{-1} \sum_{x, l_x} p(x) |x.l_x\rangle\langle x.l_x| \otimes \sum_{y, l_y} \frac{p(y)}{|\mathcal{L}|} (\rho')_{(x, l_x), (y, l_y)}^{\mathcal{Z}'} \\ (\rho')^{\mathcal{Y}'\mathcal{Z}'} &= |\mathcal{L}|^{-1} \sum_{y, l_y} p(y) |y.l_y\rangle\langle y.l_y| \otimes \sum_{x, l_x} \frac{p(x)}{|\mathcal{L}|} (\rho')_{(x, l_x), (y, l_y)}^{\mathcal{Z}'} \\ (\rho')^{\mathcal{Z}'} &= \sum_{x, y, l_x, l_y} \frac{p(x)p(y)}{|\mathcal{L}|^2} (\rho')_{(x, l_x), (y, l_y)}^{\mathcal{Z}'} \end{aligned}$$

Similarly,

$$\begin{aligned} (\rho')^{\mathcal{X}'} &= |\mathcal{L}|^{-1} \sum_{x, l_x} p(x) \left\{ \sum_{y, l_y} \frac{p(y)}{|\mathcal{L}|} \right\} |x.l_x\rangle\langle x.l_x| = |\mathcal{L}|^{-1} \sum_{x, l_x} p(x) |x.l_x\rangle\langle x.l_x| \\ (\rho')^{\mathcal{Y}'} &= |\mathcal{L}|^{-1} \sum_{y, l_y} p(y) \left\{ \sum_{x, l_x} \frac{p(x)}{|\mathcal{L}|} \right\} |y.l_y\rangle\langle y.l_y| = |\mathcal{L}|^{-1} \sum_{y, l_y} p(y) |y.l_y\rangle\langle y.l_y| \\ (\rho')^{\mathcal{X}'\mathcal{Y}'} &= |\mathcal{L}|^{-2} \sum_{x, l_x, y, l_y} p(x)p(y) |x.l_x\rangle\langle x.l_x| \otimes |y.l_y\rangle\langle y.l_y| \end{aligned}$$

So the marginal states are following

$$\begin{aligned} (\rho')^{\mathcal{X}'} \otimes (\rho')^{\mathcal{Y}'\mathcal{Z}'} &= |\mathcal{L}|^{-2} \sum_{x, l_x, y, l_y} p(x) |x.l_x\rangle\langle x.l_x| \otimes p(y) |y.l_y\rangle\langle y.l_y| \otimes \left\{ \sum_{x, l_x} \frac{p(x)}{|\mathcal{L}|} (\rho')_{(x, l_x), (y, l_y)}^{\mathcal{Z}'} \right\} \\ (\rho')^{\mathcal{Y}'} \otimes (\rho')^{\mathcal{X}'\mathcal{Z}'} &= |\mathcal{L}|^{-2} \sum_{x, l_x, y, l_y} p(y) |y.l_y\rangle\langle y.l_y| \otimes p(x) |x.l_x\rangle\langle x.l_x| \otimes \left\{ \sum_{y, l_y} \frac{p(y)}{|\mathcal{L}|} (\rho')_{(x, l_x), (y, l_y)}^{\mathcal{Z}'} \right\} \\ (\rho')^{\mathcal{X}'\mathcal{Y}'} \otimes (\rho')^{\mathcal{Z}'} &= |\mathcal{L}|^{-2} \sum_{x, l_x, y, l_y} p(x)p(y) \left(|x.l_x\rangle\langle x.l_x| \otimes |y.l_y\rangle\langle y.l_y| \right. \\ &\quad \left. \otimes \left\{ \sum_{x, y, l_x, l_y} \frac{p(x)p(y)}{|\mathcal{L}|^2} (\rho')_{(x, l_x), (y, l_y)}^{\mathcal{Z}'} \right\} \right) \end{aligned}$$

4.3.2 Smoothness of the Marginal states

In chapter 3 we noticed that 'Smoothing' says that if a bipartite quantum state is tilted along two mutually orthogonal directions, its reduced (traced-out) state is primarily tilted along only one of those directions. Now observe that

$$\begin{aligned} |\mathcal{L}|^{-1} \sum_{l_y} \mathcal{T}_{\mathcal{X}; l_x, l_y, \delta}(|h\rangle\langle h|) &= \frac{|\mathcal{L}|^{-1}}{1 + 2\delta^2} \sum_{l_y} ((|h\rangle + \delta|h\rangle|l_x\rangle)(\langle h| + \delta\langle h|\langle l_x|) + \delta^2|h\rangle\langle h||l_y\rangle\langle l_y| \\ &\quad + \delta(|h\rangle + \delta|h\rangle|l_x\rangle)\langle h|\langle l_y| + \delta|h\rangle|l_y\rangle(\langle h| + \delta\langle h|\langle l_x|)) \\ &= \frac{1 + \delta^2}{1 + 2\delta^2} \mathcal{T}_{\mathcal{X}; l_x, \delta}(|h\rangle\langle h|) + \mathcal{N}_{\mathcal{X}; l_x, \delta}(|h\rangle\langle h|) \end{aligned}$$

where,

$$\mathcal{N}_{\mathcal{X}; l_x, \delta}(|h\rangle\langle h|) := \frac{|\mathcal{L}|^{-1}}{1 + 2\delta^2} \sum_{l_y} (\delta(|h\rangle + \delta|h\rangle|l_x\rangle)\langle h|\langle l_y| + \delta|h\rangle|l_y\rangle(\langle h| + \delta\langle h|\langle l_x|) + \delta^2|h\rangle\langle h||l_y\rangle\langle l_y|)$$

Now notice that,

$$\begin{aligned} \|\mathcal{N}_{\mathcal{X}; l_x, \delta}(|h\rangle\langle h|)\|_{\infty} &\leq \frac{|\mathcal{L}|^{-1}}{1 + 2\delta^2} \left(\left\| \sum_{l_y} \delta(|h\rangle + \delta|h\rangle|l_x\rangle)\langle h|\langle l_y| \right\|_{\infty} + \left\| \sum_{l_y} \delta|h\rangle|l_y\rangle(\langle h| + \delta\langle h|\langle l_x|) \right\|_{\infty} \right. \\ &\quad \left. + \left\| \sum_{l_y} \delta^2|h\rangle\langle h||l_y\rangle\langle l_y| \right\|_{\infty} \right) \\ &= \frac{|\mathcal{L}|^{-1}}{1 + 2\delta^2} \left(2\delta \| |h\rangle + \delta|h\rangle|l_x\rangle \|_2 \left\| \sum_{l_y} |l_y\rangle \right\|_2 + \delta^2 \left\| \sum_{l_y} |l_y\rangle \right\|_2 \right) \\ &\leq \frac{|\mathcal{L}|^{-1}}{1 + 2\delta^2} \left(3\delta \| |h\rangle + \delta|h\rangle|l_x\rangle \|_2 \left\| \sum_{l_y} |l_y\rangle \right\|_2 \right) \\ &\leq |\mathcal{L}|^{-1} \left(3\delta \| |h\rangle + \delta|h\rangle|l_x\rangle \|_2^2 \sqrt{\sum_{l_y} \| |l_y\rangle \|_2^2} \right) \\ &= |\mathcal{L}|^{-1} \left(3\delta \| |h\rangle + \delta|h\rangle|l_x\rangle \|_2 \sqrt{|\mathcal{L}|} \right) \leq \frac{3\delta}{\sqrt{|\mathcal{L}|}} \end{aligned}$$

Using the similar deductions, observe:

$$\begin{aligned} |\mathcal{L}|^{-1} \sum_{l_x} \mathcal{T}_{\mathcal{Y}; l_x, l_y, \delta}(|h\rangle\langle h|) &= \frac{1 + \delta^2}{1 + 2\delta^2} \mathcal{T}_{\mathcal{Y}; l_y, \delta}(|h\rangle\langle h|) + \mathcal{N}_{\mathcal{Y}; l_y, \delta}(|h\rangle\langle h|) \\ |\mathcal{L}|^{-2} \sum_{l_x, l_y} \mathcal{T}_{\mathcal{X}; l_x, l_y, \delta}(|h\rangle\langle h|) &= \frac{1 + \delta^2}{1 + 2\delta^2} (|h\rangle\langle h|) + \mathcal{N}_{\delta}(|h\rangle\langle h|) \end{aligned}$$

where,

$$\begin{aligned}\mathcal{N}_{\mathcal{Y};l_y,\delta}(|h\rangle\langle h|) &:= \frac{|\mathcal{L}|^{-1}}{1+2\delta^2} \sum_{l_x} (\delta(|h\rangle + \delta|h\rangle|l_y\rangle)\langle h|\langle l_x| + \delta|h\rangle|l_x\rangle(\langle h| + \delta\langle h|\langle l_y|) + \delta^2|h\rangle\langle h||l_x\rangle\langle l_x|) \\ \mathcal{N}_\delta(|h\rangle\langle h|) &:= \frac{|\mathcal{L}|^{-2}}{1+2\delta^2} \sum_{l_x, l_y} (\delta(|h\rangle + \delta|h\rangle|l_x\rangle)\langle h|\langle l_y| + \delta|h\rangle|l_y\rangle(\langle h| + \delta\langle h|\langle l_x|) + \\ &\quad \delta^2|h\rangle\langle h|(|l_x\rangle + |l_y\rangle)(\langle l_x| + \langle l_y|))\end{aligned}$$

and also it can be shown that,

$$\begin{aligned}\|\mathcal{N}_{\mathcal{Y};l_y,\delta}(|h\rangle\langle h|)\|_\infty &\leq \frac{3\delta}{\sqrt{|\mathcal{L}|}} \\ \|\mathcal{N}_\delta(|h\rangle\langle h|)\|_\infty &\leq \frac{3\delta}{\sqrt{|\mathcal{L}|}}\end{aligned}$$

So from the above discussion we can easily see that,

$$\begin{aligned}\sum_{x, l_x} \frac{p(x)}{|\mathcal{L}|} (\rho')_{(x, l_x), (y, l_y)}^{\mathcal{Z}'} &= \sum_x \frac{p(x)}{|\mathcal{L}|} \sum_{l_x} (\rho')_{(x, l_x), (y, l_y)}^{\mathcal{Z}'} \\ &= \sum_x p(x) \left\{ \frac{1+\delta^2}{1+2\delta^2} \mathcal{T}_{\mathcal{Y};l_y,\delta}(\rho_{x,y}^{\mathcal{Z}} \otimes |0\rangle\langle 0|)^{\mathbb{C}^2} + \mathcal{N}_{\mathcal{Y};l_y,\delta}(\rho_{x,y}^{\mathcal{Z}} \otimes |0\rangle\langle 0|)^{\mathbb{C}^2} \right\} \\ &\stackrel{a}{=} \frac{1+\delta^2}{1+2\delta^2} \mathcal{T}_{\mathcal{Y};l_y,\delta} \left(\rho_y^{\mathcal{Z}} \otimes |0\rangle\langle 0| \right)^{\mathbb{C}^2} + \\ &\quad \mathcal{N}_{\mathcal{Y};l_y,\delta} \left(\rho_y^{\mathcal{Z}} \otimes |0\rangle\langle 0| \right)^{\mathbb{C}^2} \\ &\stackrel{b}{=} \frac{1+\delta^2}{1+2\delta^2} \mathcal{T}_{\mathcal{Y};l_y,\delta} \left(\rho_y^{\mathcal{Z}} \otimes |0\rangle\langle 0| \right)^{\mathbb{C}^2} + \\ &\quad \mathcal{N}_{\mathcal{Y};l_y,\delta} \left(\rho_y^{\mathcal{Z}} \otimes |0\rangle\langle 0| \right)^{\mathbb{C}^2}\end{aligned}$$

Here the equality **a** is due to the following two equalities. Let $|h_{x,y}\rangle\langle h_{x,y}| := (\rho_{x,y}^{\mathcal{Z}} \otimes |0\rangle\langle 0|)^{\mathbb{C}^2}$ and $|h_y\rangle\langle h_y| := \sum_x p(x) (\rho_{x,y}^{\mathcal{Z}} \otimes |0\rangle\langle 0|)^{\mathbb{C}^2} = \rho_y^{\mathcal{Z}} \otimes |0\rangle\langle 0|$. Then,

$$\begin{aligned}\sum_x p(x) \mathcal{T}_{\mathcal{Y};l_y,\delta}(|h_{x,y}\rangle\langle h_{x,y}|) &= \sum_x p(x) ((|h_{x,y}\rangle + \delta|h_{x,y}\rangle|l_y\rangle)(\langle h_{x,y}| + \delta\langle h_{x,y}|\langle l_y|)) \\ &= ((|h_y\rangle + \delta|h_y\rangle|l_y\rangle)(\langle h_y| + \delta\langle h_y|\langle l_y|)) \\ &= \mathcal{T}_{\mathcal{Y};l_y,\delta}(|h_y\rangle\langle h_y|) \\ &= \mathcal{T}_{\mathcal{Y};l_y,\delta} \left(\rho_y^{\mathcal{Z}} \otimes |0\rangle\langle 0| \right)^{\mathbb{C}^2}\end{aligned}$$

and,

$$\begin{aligned}
\sum_x p(x) \mathcal{N}_{\mathcal{Y}; l_y, \delta}(|h_{x,y}\rangle\langle h_{x,y}|) &= \frac{|\mathcal{L}|^{-1}}{1+2\delta^2} \sum_x p(x) \sum_{l_x} (\delta(|h_{x,y}\rangle + \delta|h_{x,y}\rangle|l_y\rangle) \langle h_{x,y}| \langle l_x| \\
&\quad + \delta|h_{x,y}\rangle|l_x\rangle (\langle h_{x,y}| + \delta\langle h_{x,y}| \langle l_y|) + \delta^2|h_{x,y}\rangle\langle h_{x,y}||l_x\rangle\langle l_x|) \\
&= \frac{|\mathcal{L}|^{-1}}{1+2\delta^2} \sum_{l_x} (\delta(|h_y\rangle + \delta|h_y\rangle|l_y\rangle) \langle h_y| \langle l_x| \\
&\quad + \delta|h_y\rangle|l_x\rangle (\langle h_y| + \delta\langle h_y| \langle l_y|) + \delta^2|h_y\rangle\langle h_y||l_x\rangle\langle l_x|) \\
&= \mathcal{N}_{\mathcal{Y}; l_y, \delta}(|h_y\rangle\langle h_y|) \\
&= \mathcal{N}_{\mathcal{Y}; l_y, \delta} \left(\rho_y^{\mathbb{Z}} \otimes |0\rangle\langle 0| \right)
\end{aligned}$$

and equality **b** follows from the definition of $\rho_y^{\mathbb{Z}}$ in equation (4.4.2). Similarly,

$$\begin{aligned}
\sum_{y, l_y} \frac{p(y)}{|\mathcal{L}|} (\rho')_{(x, l_x), (y, l_y)}^{\mathbb{Z}'} &= \frac{1+\delta^2}{1+2\delta^2} \mathcal{T}_{\mathcal{X}; l_x, \delta} \left(\rho_x^{\mathbb{Z}} \otimes |0\rangle\langle 0| \right) + \\
&\quad \mathcal{N}_{\mathcal{X}; l_x, \delta} \left(\rho_x^{\mathbb{Z}} \otimes |0\rangle\langle 0| \right) \\
\sum_{x, y, l_x, l_y} \frac{p(x)p(y)}{|\mathcal{L}|^2} (\rho')_{(x, l_x), (y, l_y)}^{\mathbb{Z}'} &= \frac{1+\delta^2}{1+2\delta^2} \left(\rho^{\mathbb{Z}} \otimes |0\rangle\langle 0| \right) + \\
&\quad \mathcal{N}_{\delta} \left(\rho^{\mathbb{Z}} \otimes |0\rangle\langle 0| \right)
\end{aligned}$$

with

$$\begin{aligned}
\left\| \mathcal{N}_{\mathcal{X}; l_x, \delta} \left(\rho_X^{\mathbb{Z}} \otimes |0\rangle\langle 0| \right) \right\|_{\infty} &\leq \frac{3\delta}{\sqrt{|\mathcal{L}|}} \\
\left\| \mathcal{N}_{\mathcal{Y}; l_y, \delta} \left(\rho_y^{\mathbb{Z}} \otimes |0\rangle\langle 0| \right) \right\|_{\infty} &\leq \frac{3\delta}{\sqrt{|\mathcal{L}|}} \\
\left\| \mathcal{N}_{\delta} \left(\rho^{\mathbb{Z}} \otimes |0\rangle\langle 0| \right) \right\|_{\infty} &\leq \frac{3\delta}{\sqrt{|\mathcal{L}|}}
\end{aligned}$$

4.3.3 Upper-bounds of Error Probabilities

Here we'll give an upper-bound of error probabilities we discussed in (1.2.14)-(1.2.17) in subsection 1.2.2 using the smoothness property we proved in subsection 4.3.2. Now observe that using

Fact 11 we can have the following inequality

$$\begin{aligned}
\left\| (\Pi')_{(x,l_x),(y,l_y),\delta}^{\mathcal{Z}'} \right\|_1 &= \left\| (\mathbb{1}^{\mathcal{Z}'} - (\Pi')_{W'_{(x,l_x),(y,l_y),\delta}}^{\mathcal{Z}'}) \Pi_{\hat{\mathcal{Z}}}^{\mathcal{Z}'} (\mathbb{1}^{\mathcal{Z}'} - (\Pi')_{W'_{(x,l_x),(y,l_y),\delta}}^{\mathcal{Z}'}) \right\|_1 \\
&\leq \left\| (\mathbb{1}^{\mathcal{Z}'} - (\Pi')_{W'_{(x,l_x),(y,l_y),\delta}}^{\mathcal{Z}'}) \Pi_{\hat{\mathcal{Z}}}^{\mathcal{Z}'} \right\|_1 \left\| (\mathbb{1}^{\mathcal{Z}'} - (\Pi')_{W'_{(x,l_x),(y,l_y),\delta}}^{\mathcal{Z}'}) \right\|_\infty \\
&\leq \left\| (\mathbb{1}^{\mathcal{Z}'} - (\Pi')_{W'_{(x,l_x),(y,l_y),\delta}}^{\mathcal{Z}'}) \right\|_\infty \left\| \Pi_{\hat{\mathcal{Z}}}^{\mathcal{Z}'} \right\|_1 \left\| (\mathbb{1}^{\mathcal{Z}'} - (\Pi')_{W'_{(x,l_x),(y,l_y),\delta}}^{\mathcal{Z}'}) \right\|_\infty \\
&= \left\| (\mathbb{1}^{\mathcal{Z}'} - (\Pi')_{W'_{(x,l_x),(y,l_y),\delta}}^{\mathcal{Z}'}) \right\|_\infty^2 \left\| \Pi_{\hat{\mathcal{Z}}}^{\mathcal{Z}'} \right\|_1 = 2|\mathcal{Z}|
\end{aligned}$$

Now we try to find the upper-bound,

$$\begin{aligned}
&\text{Tr} \left[(\Pi')_{(x,l_x),(y,l_y),\delta}^{\mathcal{Z}'} \left\{ \sum_{y,l_y} \frac{p(y)}{|\mathcal{L}|} (\rho')_{(x,l_x),(y,l_y)}^{\mathcal{Z}'} \right\} \right] \\
&\leq_a \text{Tr} \left[(\Pi')_{(x,l_x),(y,l_y),\delta}^{\mathcal{Z}'} \left(\mathcal{T}_{\mathcal{X};l_x,\delta} \left(\rho_x^{\mathcal{Z}} \otimes |0\rangle\langle 0| \right) \right) \right] + \left\| (\Pi')_{(x,l_x),(y,l_y),\delta}^{\mathcal{Z}'} \left(\mathcal{N}_{\mathcal{X};l_x,\delta} \left(\rho_x^{\mathcal{Z}} \otimes |0\rangle\langle 0| \right) \right) \right\|_1 \\
&\leq_b \text{Tr} \left[(\Pi')_{(x,l_x),(y,l_y),\delta}^{\mathcal{Z}'} \left(\mathcal{T}_{\mathcal{X};l_x,\delta} \left(\rho_x^{\mathcal{Z}} \otimes |0\rangle\langle 0| \right) \right) \right] + \left\| (\Pi')_{(x,l_x),(y,l_y),\delta}^{\mathcal{Z}'} \right\|_1 \cdot \left\| \mathcal{N}_{\mathcal{X};l_x,\delta} \left(\rho_x^{\mathcal{Z}} \otimes |0\rangle\langle 0| \right) \right\|_\infty \\
&\leq \text{Tr} \left[(\Pi')_{(x,l_x),(y,l_y),\delta}^{\mathcal{Z}'} \left(\mathcal{T}_{\mathcal{X};l_x,\delta} \left(\rho_x^{\mathcal{Z}} \otimes |0\rangle\langle 0| \right) \right) \right] + \frac{6\delta\mathcal{Z}}{\sqrt{|\mathcal{L}|}} \\
&= \text{Tr} \left[\Pi_{\hat{\mathcal{Z}}}^{\mathcal{Z}'} (\mathbb{1}^{\mathcal{Z}'} - (\Pi')_{W'_{(x,l_x),(y,l_y),\delta}}^{\mathcal{Z}'}) \left(\mathcal{T}_{\mathcal{X};l_x,\delta} \left(\rho_x^{\mathcal{Z}} \otimes |0\rangle\langle 0| \right) \right) (\mathbb{1}^{\mathcal{Z}'} - (\Pi')_{W'_{(x,l_x),(y,l_y),\delta}}^{\mathcal{Z}'}) \right] \\
&\quad + \frac{6\delta\mathcal{Z}}{\sqrt{|\mathcal{L}|}} \\
&\leq \text{Tr} \left[(\mathbb{1}^{\mathcal{Z}'} - (\Pi')_{W'_{(x,l_x),(y,l_y),\delta}}^{\mathcal{Z}'}) \left(\mathcal{T}_{\mathcal{X};l_x,\delta} \left(\rho_x^{\mathcal{Z}} \otimes |0\rangle\langle 0| \right) \right) (\mathbb{1}^{\mathcal{Z}'} - (\Pi')_{W'_{(x,l_x),(y,l_y),\delta}}^{\mathcal{Z}'}) \right] \\
&\quad + \frac{6\delta\mathcal{Z}}{\sqrt{|\mathcal{L}|}} \\
&\leq_c \text{Tr} \left[(\mathbb{1}^{\mathcal{Z}'} - (\Pi')_{W'_{\mathcal{X};(x,l_x),(y,l_y),\delta}}^{\mathcal{Z}'}) \left(\mathcal{T}_{\mathcal{X};l_x,\delta} \left(\rho_x^{\mathcal{Z}} \otimes |0\rangle\langle 0| \right) \right) (\mathbb{1}^{\mathcal{Z}'} - (\Pi')_{W'_{\mathcal{X};(x,l_x),(y,l_y),\delta}}^{\mathcal{Z}'}) \right] \\
&\quad + \frac{6\delta\mathcal{Z}}{\sqrt{|\mathcal{L}|}} \\
&= \text{Tr} \left[(\mathbb{1}^{\mathcal{Z}'} - (\Pi')_{W'_{\mathcal{X};(x,l_x),(y,l_y),\delta}}^{\mathcal{Z}'}) \left(\mathcal{T}_{\mathcal{X};l_x,\delta} \left(\rho_x^{\mathcal{Z}} \otimes |0\rangle\langle 0| \right) \right) \right] + \frac{6\delta\mathcal{Z}}{\sqrt{|\mathcal{L}|}}
\end{aligned}$$

$$\begin{aligned}
&= \text{Tr} \left[\left(\mathbb{1}^{\mathcal{Z}'} - (\Pi')_{W'_{\mathcal{X};(x,l_x),(y,l_y),\delta}}^{\mathcal{Z}'} \right) \Pi_{\mathcal{T}_{\mathcal{X};l_x,\delta}(\hat{\mathcal{Z}})}^{\mathcal{Z}'} \left(\mathcal{T}_{\mathcal{X};l_x,\delta} \left(\rho_x^{\mathcal{Z}} \otimes |0\rangle\langle 0|^{\mathbb{C}^2} \right) \right) \right] + \frac{6\delta\mathcal{Z}}{\sqrt{|\mathcal{L}|}} \\
&\stackrel{\mathbf{d}}{=} \text{Tr} \left[\left(\mathbb{1}_{\mathcal{T}_{\mathcal{X};l_x,\delta}(\hat{\mathcal{Z}})} - (\Pi')_{W'_{\mathcal{X};(x,l_x),(y,l_y),\delta}}^{\mathcal{T}_{\mathcal{X};l_x,\delta}(\hat{\mathcal{Z}})} \right) \left(\mathcal{T}_{\mathcal{X};l_x,\delta} \left(\rho_x^{\mathcal{Z}} \otimes |0\rangle\langle 0|^{\mathbb{C}^2} \right) \right) \right] + \frac{6\delta\mathcal{Z}}{\sqrt{|\mathcal{L}|}} \\
&= \text{Tr} \left[\left(\mathbb{1}_{\mathcal{T}_{\mathcal{X};l_x,\delta}(\hat{\mathcal{Z}})} - (\Pi')_{\mathcal{T}_{\mathcal{X};l_x,\delta}(W_{\mathcal{X};x,y})}^{\mathcal{T}_{\mathcal{X};l_x,\delta}(\hat{\mathcal{Z}})} \right) \left(\mathcal{T}_{\mathcal{X};l_x,\delta} \left(\rho_x^{\mathcal{Z}} \otimes |0\rangle\langle 0|^{\mathbb{C}^2} \right) \right) \right] + \frac{6\delta\mathcal{Z}}{\sqrt{|\mathcal{L}|}} \\
&\stackrel{\mathbf{e}}{=} \text{Tr} \left[\left(\mathbb{1}^{\hat{\mathcal{Z}}} - \Pi_{W_{\mathcal{X};x,y}}^{\hat{\mathcal{Z}}} \right) \left(\rho_x^{\mathcal{Z}} \otimes |0\rangle\langle 0|^{\mathbb{C}^2} \right) \right] + \frac{6\delta\mathcal{Z}}{\sqrt{|\mathcal{L}|}} \\
&= \text{Tr} \left[\Pi_{\mathcal{X};x,y}^{\hat{\mathcal{Z}}} \left(\rho_x^{\mathcal{Z}} \otimes |0\rangle\langle 0|^{\mathbb{C}^2} \right) \right] + \frac{6\delta\mathcal{Z}}{\sqrt{|\mathcal{L}|}} \\
&= \text{Tr} \left[(\Pi'')_{\mathcal{X};x,y}^{\mathcal{Z}} \rho_x^{\mathcal{Z}} \right] + \frac{6\delta\mathcal{Z}}{\sqrt{|\mathcal{L}|}}
\end{aligned}$$

Here, inequality **a** follows from Fact 10, **b** follows from Fact 11, **c** follows from the fact that

$(\Pi')_{W'_{\mathcal{X};(x,l_x),(y,l_y),\delta}}^{\mathcal{Z}'} \leq (\Pi')_{W'_{(x,l_x),(y,l_y),\delta}}^{\mathcal{Z}'}$, the equality **d** follows from that fact $\mathcal{T}_{\mathcal{X};l_x,\delta}(\hat{\mathcal{Z}}) \leq \mathcal{Z}'$, and equality **e** follows from the fact that $\mathcal{T}_{\mathcal{X};l_x,\delta}$ is an isometric map. Similarly we'll have the other two upper-bounds,

$$\begin{aligned}
&\text{Tr} \left[(\Pi')_{(x,l_x),(y,l_y),\delta}^{\mathcal{Z}'} \left\{ \sum_{x,l_x} \frac{p(x)}{|\mathcal{L}|} (\rho')_{(x,l_x),(y,l_y)}^{\mathcal{Z}'} \right\} \right] \leq \text{Tr} \left[(\Pi'')_{\mathcal{Y};x,y}^{\mathcal{Z}} \rho_y^{\mathcal{Z}} \right] + \frac{6\delta\mathcal{Z}}{\sqrt{|\mathcal{L}|}} \\
&\text{Tr} \left[(\Pi')_{(x,l_x),(y,l_y),\delta}^{\mathcal{Z}'} \left\{ \sum_{x,y,l_x,l_y} \frac{p(x)p(y)}{|\mathcal{L}|^2} (\rho')_{(x,l_x),(y,l_y)}^{\mathcal{Z}'} \right\} \right] \leq \text{Tr} \left[(\Pi'')_{x,y}^{\mathcal{Z}} \rho^{\mathcal{Z}} \right] + \frac{6\delta\mathcal{Z}}{\sqrt{|\mathcal{L}|}}
\end{aligned}$$

Now using equation (4.2.1) and the definition of $(\Pi')^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'}$ in (4.4.7), we can write

$$\begin{aligned}
&\text{Tr} \left[(\Pi')^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'} \left((\rho')^{\mathcal{Y}'} \otimes (\rho')^{\mathcal{X}'\mathcal{Z}'} \right) \right] \\
&= |\mathcal{L}|^{-2} \sum_{y,l_y} p(y) \sum_{x,l_x} p(x) \text{Tr} \left[(\Pi')_{(x,l_x),(y,l_y),\delta}^{\mathcal{Z}'} \left\{ \sum_{y,l_y} \frac{p(y)}{|\mathcal{L}|} (\rho')_{(x,l_x),(y,l_y)}^{\mathcal{Z}'} \right\} \right] \\
&\leq |\mathcal{L}|^{-2} \sum_{y,l_y} p(y) \sum_{x,l_x} p(x) \text{Tr} \left[(\Pi'')_{\mathcal{X};x,y}^{\mathcal{Z}} \rho_x^{\mathcal{Z}} \right] + \frac{6\delta\mathcal{Z}}{\sqrt{|\mathcal{L}|}} \\
&= \sum_{x,y} p(x)p(y) \text{Tr} \left[(\Pi'')_{\mathcal{X};x,y}^{\mathcal{Z}} \rho_x^{\mathcal{Z}} \right] + \frac{6\delta\mathcal{Z}}{\sqrt{|\mathcal{L}|}} \\
&= \text{Tr} \left[(\Pi'')_{\mathcal{X}}^{\mathcal{X}\mathcal{Y}\mathcal{Z}} (\rho^{\mathcal{Y}} \otimes \rho^{\mathcal{X}\mathcal{Z}}) \right] + \frac{6\delta\mathcal{Z}}{\sqrt{|\mathcal{L}|}} \\
&\stackrel{\mathbf{a}}{\leq} 2^{-I_H^{\epsilon}(\mathcal{Y}:\mathcal{X}\mathcal{Z})} + \frac{6\delta\mathcal{Z}}{\sqrt{|\mathcal{L}|}}
\end{aligned}$$

where inequality **a** follows from (4.3.1) and (4.3.2). Now we choose $|\mathcal{L}|$ large enough so the second term in the above inequality is less than $\min\{2^{-I_H^\epsilon(\mathcal{Y}:\mathcal{XZ})}, 2^{-I_H^\epsilon(\mathcal{X}:\mathcal{YZ})}, 2^{-I_H^\epsilon(\mathcal{XY}:\mathcal{Z})}\}$ such that we can have the following inequalities

$$\begin{aligned} \text{Tr} \left[(\Pi')^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'} \left((\rho')^{\mathcal{Y}'} \otimes (\rho')^{\mathcal{X}'\mathcal{Z}'} \right) \right] &\leq 2^{-I_H^\epsilon(\mathcal{Y}:\mathcal{XZ})+1} \\ \text{Tr} \left[(\Pi')^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'} \left((\rho')^{\mathcal{X}'} \otimes (\rho')^{\mathcal{Y}'\mathcal{Z}'} \right) \right] &\leq 2^{-I_H^\epsilon(\mathcal{X}:\mathcal{YZ})+1} \\ \text{Tr} \left[(\Pi')^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'} \left((\rho')^{\mathcal{X}'\mathcal{Y}'} \otimes (\rho')^{\mathcal{Z}'} \right) \right] &\leq 2^{-I_H^\epsilon(\mathcal{XY}:\mathcal{Z})+1} \end{aligned} \quad (4.3.2)$$

So we have given the error upper-bounds of the modified CQ-MAC $\mathfrak{C}''$ and Now we'll describe a decoding strategy Charlie needs to follow in order to guess the message pair sent by Alice and Bob respectively.

4.3.4 Decoding Strategy and Expected Average Decoding Error

For any of the message pair $(m_1, m_2) \in [2^{R_1}] \times [2^{R_2}]$ we have a corresponding POVM element $(\Pi')_{(x,l_x)(m_1),(y,l_y)(m_2),\delta}^{\mathcal{Z}'}$. Charlie uses *pretty good measurement* [4],[23],[16] using the POVM element set $\left\{ (\Pi')_{(x,l_x)(m_1),(y,l_y)(m_2),\delta}^{\mathcal{Z}'} \right\}_{(m_1,m_2) \in [2^{R_1}] \times [2^{R_2}]}$. We now analyse the expectation of the average decoding error probability of this particular strategy performed by Charlie under a choice of random code-book $\mathcal{C}$.

Let's assume the message pair are feed-ed in $\mathfrak{C}''$ by Alice and Bob is (m_1, m_2) , then the $\mathfrak{C}''$ should output a state $(\rho')_{(x,l_x)(m_1),(y,l_y)(m_2),\delta}^{\mathcal{Z}'}$. Let $\beta_{(\hat{m}_1, \hat{m}_2)}^{\mathcal{Z}'}$ be decoder POVM corresponding to decoded output $(\hat{m}_1, \hat{m}_2)$ coming from pretty good measurement which can be written as follows

$$\beta_{(\hat{m}_1, \hat{m}_2)}^{\mathcal{Z}'} := \left\{ \sum_{\substack{(\hat{m}_1, \hat{m}_2) \\ \in [2^{R_1}] \times [2^{R_2}]}} (\Pi')_{(x,l_x)(\hat{m}_1), (y,l_y)(\hat{m}_2), \delta}^{\mathcal{Z}'} \right\}^{-\frac{1}{2}} (\Pi')_{(x,l_x)(\hat{m}_1), (y,l_y)(\hat{m}_2), \delta}^{\mathcal{Z}'} \left\{ \sum_{\substack{(\hat{m}_1, \hat{m}_2) \\ \in [2^{R_1}] \times [2^{R_2}]}} (\Pi')_{(x,l_x)(\hat{m}_1), (y,l_y)(\hat{m}_2), \delta}^{\mathcal{Z}'} \right\}^{-\frac{1}{2}} \quad (4.3.3)$$

Now using Hayashi-Nagoka inequality [14], the decoding error for (m_1, m_2) can be upper bounded by

$$\begin{aligned} &\text{Tr} \left[\left(\mathbb{1}^{\mathcal{Z}'} - \beta_{(m_1, m_2)}^{\mathcal{Z}'} \right) (\rho')_{(x,l_x)(m_1),(y,l_y)(m_2),\delta}^{\mathcal{Z}'} \right] \\ &\leq 2 \text{Tr} \left[\left(\mathbb{1}^{\mathcal{Z}'} - (\Pi')_{(x,l_x)(m_1), (y,l_y)(m_2), \delta}^{\mathcal{Z}'} \right) (\rho')_{(x,l_x)(m_1), (y,l_y)(m_2), \delta}^{\mathcal{Z}'} \right] + 4 \sum_{\substack{(\hat{m}_1, \hat{m}_2) \\ \neq (m_1, m_2)}} \text{Tr} \left[(\Pi')_{(x,l_x)(\hat{m}_1), (y,l_y)(\hat{m}_2), \delta}^{\mathcal{Z}'} (\rho')_{(x,l_x)(m_1), (y,l_y)(m_2), \delta}^{\mathcal{Z}'} \right] \\ &= 2 \text{Tr} \left[\left(\mathbb{1}^{\mathcal{Z}'} - (\Pi')_{(x,l_x)(m_1), (y,l_y)(m_2), \delta}^{\mathcal{Z}'} \right) (\rho')_{(x,l_x)(m_1), (y,l_y)(m_2), \delta}^{\mathcal{Z}'} \right] + 4 \sum_{\hat{m}_1 \neq m_1} \text{Tr} \left[(\Pi')_{(x,l_x)(\hat{m}_1), (y,l_y)(m_2), \delta}^{\mathcal{Z}'} (\rho')_{(x,l_x)(m_1), (y,l_y)(m_2), \delta}^{\mathcal{Z}'} \right] \end{aligned}$$

$$+ 4 \sum_{\hat{m}_2 \neq m_2} \text{Tr} \left[\begin{pmatrix} (\Pi')_{(x,l_x)(m_1), (y,l_y)(\hat{m}_2), \delta}^{\mathcal{Z}'} & (\rho')_{(x,l_x)(m_1), (y,l_y)(m_2), \delta}^{\mathcal{Z}'} \end{pmatrix} \right] + 4 \sum_{\substack{\hat{m}_1 \neq m_1, \\ \hat{m}_2 \neq m_2}} \text{Tr} \left[\begin{pmatrix} (\Pi')_{(x,l_x)(\hat{m}_1), (y,l_y)(\hat{m}_2), \delta}^{\mathcal{Z}'} & (\rho')_{(x,l_x)(m_1), (y,l_y)(m_2), \delta}^{\mathcal{Z}'} \end{pmatrix} \right].$$

The expectation of the average decoding error probability of this particular strategy performed by Charlie under a choice of random code-book C can be upper bounded by

$$\begin{aligned} & \mathbf{E}_C \left[\text{Tr} \left[\left(\mathbb{1}^{\mathcal{Z}'} - \beta_{(m_1, m_2)}^{\mathcal{Z}'} \right) (\rho')_{(x,l_x)(m_1), (y,l_y)(m_2), \delta}^{\mathcal{Z}'} \right] \right] \\ & \leq 2 \mathbf{E}_C \left[\text{Tr} \left[\left(\mathbb{1}^{\mathcal{Z}'} - (\Pi')_{(x,l_x)(m_1), (y,l_y)(m_2), \delta}^{\mathcal{Z}'} \right) (\rho')_{(x,l_x)(m_1), (y,l_y)(m_2), \delta}^{\mathcal{Z}'} \right] \right] \\ & \quad + 4 \sum_{\hat{m}_1 \neq m_1} \mathbf{E}_C \left[\text{Tr} \left[\begin{pmatrix} (\Pi')_{(x,l_x)(\hat{m}_1), (y,l_y)(m_2), \delta}^{\mathcal{Z}'} & (\rho')_{(x,l_x)(m_1), (y,l_y)(m_2), \delta}^{\mathcal{Z}'} \end{pmatrix} \right] \right] + 4 \sum_{\hat{m}_2 \neq m_2} \mathbf{E}_C \left[\text{Tr} \left[\begin{pmatrix} (\Pi')_{(x,l_x)(m_1), (y,l_y)(\hat{m}_2), \delta}^{\mathcal{Z}'} & (\rho')_{(x,l_x)(m_1), (y,l_y)(m_2), \delta}^{\mathcal{Z}'} \end{pmatrix} \right] \right] \\ & \quad + 4 \sum_{\substack{\hat{m}_1 \neq m_1, \\ \hat{m}_2 \neq m_2}} \mathbf{E}_C \left[\text{Tr} \left[\begin{pmatrix} (\Pi')_{(x,l_x)(\hat{m}_1), (y,l_y)(\hat{m}_2), \delta}^{\mathcal{Z}'} & (\rho')_{(x,l_x)(m_1), (y,l_y)(m_2), \delta}^{\mathcal{Z}'} \end{pmatrix} \right] \right] \\ & = 2|\mathcal{L}|^{-2} \sum_{x, l_x, y, l_y} p(x)p(y) \left(\text{Tr} \left[\left(\mathbb{1}^{\mathcal{Z}'} - (\Pi')_{(x,l_x), (y,l_y), \delta}^{\mathcal{Z}'} \right) (\rho')_{(x,l_x), (y,l_y), \delta}^{\mathcal{Z}'} \right] \right) \\ & \quad + 4(2^{R_1} - 1)|\mathcal{L}|^{-3} \sum_{x, l_x, x', l'_x, y, l_y} p(x)p(x')p(y) \left(\text{Tr}[(\Pi')_{(x', l'_x), (y, l_y), \delta}^{\mathcal{Z}'} (\rho')_{(x, l_x), (y, l_y), \delta}^{\mathcal{Z}'}] \right) \\ & \quad + 4(2^{R_2} - 1)|\mathcal{L}|^{-3} \sum_{x, l_x, y, l_y, y', l'_y} p(x)p(y)p(y') \left(\text{Tr}[\text{Tr}[(\Pi')_{(x, l_x), (y', l'_y), \delta}^{\mathcal{Z}'} (\rho')_{(x, l_x), (y, l_y), \delta}^{\mathcal{Z}'}] \right) \\ & \quad + 4(2^{R_1} - 1)(2^{R_2} - 1)|\mathcal{L}|^{-4} \sum_{\substack{x, l_x, x', l'_x, \\ y, l_y, y', l'_y}} p(x)p(x')p(y)p(y') \left(\text{Tr}[\text{Tr}[(\Pi')_{(x', l'_x), (y', l'_y), \delta}^{\mathcal{Z}'} (\rho')_{(x, l_x), (y, l_y), \delta}^{\mathcal{Z}'}] \right) \\ & = 2\text{Tr}[(\mathbb{1}^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'} - (\Pi')^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'})(\rho')^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'}] + 4(2^{R_1} - 1)\text{Tr}[(\Pi')^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'} ((\rho')^{\mathcal{X}'} \otimes (\rho')^{\mathcal{Y}'\mathcal{Z}'})] \\ & \quad + 4(2^{R_2} - 1)\text{Tr}[(\Pi')^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'} ((\rho')^{\mathcal{Y}'} \otimes (\rho')^{\mathcal{X}'\mathcal{Z}'})] \\ & \quad + 4(2^{R_1} - 1)(2^{R_2} - 1)\text{Tr}[(\Pi')^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'} ((\rho')^{\mathcal{X}'\mathcal{Y}'} \otimes (\rho')^{\mathcal{Z}'})] \\ & \leq 152\epsilon^{1/2} + 4(2^{R_1} - 1)2^{-I_H^\epsilon(\mathcal{X}:\mathcal{Y}\mathcal{Z})+1} + 4(2^{R_2} - 1)2^{-I_H^\epsilon(\mathcal{Y}:\mathcal{X}\mathcal{Z})+1} + 4(2^{R_1} - 1)(2^{R_2} - 1)2^{-I_H^\epsilon(\mathcal{X}\mathcal{Y}:\mathcal{Z})+1} \\ & \leq 152\epsilon^{1/2} + 2^{R_1 - I_H^\epsilon(\mathcal{X}:\mathcal{Y}\mathcal{Z})+3} + 2^{R_2 - I_H^\epsilon(\mathcal{Y}:\mathcal{X}\mathcal{Z})+3} + 2^{R_1 + R_2 - I_H^\epsilon(\mathcal{X}\mathcal{Y}:\mathcal{Z})+3} \end{aligned}$$

where **a** follows from (4.2.1) and (4.3.1). Choosing a rate pair (R_1, R_2) satisfying the following constraints:

$$\begin{aligned} R_1 & \leq I_H^\epsilon(\mathcal{X} : \mathcal{Y}\mathcal{Z}) - 3 - \log 1/\epsilon, \\ R_2 & \leq I_H^\epsilon(\mathcal{Y} : \mathcal{X}\mathcal{Z}) - 3 - \log 1/\epsilon. \\ R_1 + R_2 & \leq I_H^\epsilon(\mathcal{X}\mathcal{Y} : \mathcal{Z}) - 3 - \log 1/\epsilon \end{aligned}$$

can guarantee the expected average decoding error for $\mathfrak{C}''$ to be at most $155\epsilon^{1/2}$. Now from (4.1.4) we can say for the original MAC $\mathfrak{C}$, the expected average decoding error should be at most $159\epsilon^{1/2}$. Hence, we have proved existence of at least one code-book C with probability of average decoding error is at most $155\epsilon^{1/2}$.

4.4 Generalised Approach

4.4.1 Structural Changes of CQ State

In this section we will show that the construction works even also if the input alphabets of a MAC are not mutually independent in probability. This will give us a more generalised form of the above approach of Sen's work [25].

Here instead of what we discussed in subsection 4.1.1, we say for the joint input alphabet space $\mathcal{X} \otimes \mathcal{Y}$, if we have a joint probability density $\{p(x, y)\}_{(x, y) \in \mathcal{X} \otimes \mathcal{Y}}$ then we can represent a classical quantum state as follows:

$$\rho^{\mathcal{X}\mathcal{Y}\mathcal{Z}} := \sum_{x, y} p(x, y) |x, y\rangle^{\mathcal{X}\mathcal{Y}} \langle x, y| \otimes \rho_{x, y}^{\mathcal{Z}}$$

Similarly we have the partial traces

$$\rho^{\mathcal{X}\mathcal{Z}} = \sum_x |x\rangle^{\mathcal{X}} \langle x| \otimes \left\{ \sum_y p(x, y) \rho_{x, y}^{\mathcal{Z}} \right\} \quad (4.4.1)$$

$$\rho^{\mathcal{Y}\mathcal{Z}} = \sum_y |y\rangle^{\mathcal{Y}} \langle y| \otimes \left\{ \sum_x p(x, y) \rho_{x, y}^{\mathcal{Z}} \right\} \quad (4.4.2)$$

$$\rho^{\mathcal{Z}} = \sum_{x, y} p(x, y) \rho_{x, y}^{\mathcal{Z}} \quad (4.4.3)$$

Similarly,

$$\rho^{\mathcal{X}} = \sum_x \left\{ \sum_y p(x, y) \right\} |x\rangle^{\mathcal{X}} \langle x| = \sum_x p(x) |x\rangle^{\mathcal{X}} \langle x| \quad (4.4.4)$$

$$\rho^{\mathcal{Y}} = \sum_y \left\{ \sum_x p(x, y) \right\} |y\rangle^{\mathcal{Y}} \langle y| = \sum_y p(y) |y\rangle^{\mathcal{Y}} \langle y| \quad (4.4.5)$$

$$\rho^{\mathcal{X}\mathcal{Y}} = \sum_{x, y} p(x, y) |x\rangle^{\mathcal{X}} \langle x| \otimes |y\rangle^{\mathcal{Y}} \langle y| \quad (4.4.6)$$

Using the same technique we discussed in subsection 4.1.3 and subsection 4.1.2, we can finally have the tilted, augmented and enlarged state $(\rho')^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'}$

$$(\rho')^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'} := |\mathcal{L}|^{-2} \sum_{x, l_x, y, l_y} p(x, y) |x.l_x\rangle^{\mathcal{X}'} \langle x.l_x| \otimes |y.l_y\rangle^{\mathcal{Y}'} \langle y.l_y| \otimes (\rho')_{(x, l_x), (y, l_y)}^{\mathcal{Z}'}$$

with the marginals,

$$(\rho')^{\mathcal{X}'} \otimes (\rho')^{\mathcal{Y}'\mathcal{Z}'} = |\mathcal{L}|^{-2} \sum_{x, l_x, y, l_y} p(x) |x.l_x\rangle^{\mathcal{X}'} \langle x.l_x| \otimes |y.l_y\rangle^{\mathcal{Y}'} \langle y.l_y| \otimes \left\{ \sum_{x, l_x} \frac{p(x, y)}{|\mathcal{L}|} (\rho')_{(x, l_x), (y, l_y)}^{\mathcal{Z}'} \right\}$$

$$(\rho')^{\mathcal{Y}'} \otimes (\rho')^{\mathcal{X}'\mathcal{Z}'} = |\mathcal{L}|^{-2} \sum_{x, l_x, y, l_y} p(y) |y.l_y\rangle^{\mathcal{Y}'} \langle y.l_y| \otimes |x.l_x\rangle^{\mathcal{X}'} \langle x.l_x| \otimes \left\{ \sum_{y, l_y} \frac{p(x, y)}{|\mathcal{L}|} (\rho')_{(x, l_x), (y, l_y)}^{\mathcal{Z}'} \right\}$$

$$(\rho')^{\mathcal{X}'\mathcal{Y}'} \otimes (\rho')^{\mathcal{Z}'} = |\mathcal{L}|^{-2} \sum_{x, l_x, y, l_y} p(x, y) |x, l_x\rangle^{\mathcal{X}'} \langle x, l_x| \otimes |y, l_y\rangle^{\mathcal{Y}'} \langle y, l_y| \otimes \left\{ \sum_{x, y, l_x, l_y} \frac{p(x, y)}{|\mathcal{L}|^2} (\rho')^{\mathcal{Z}'}_{(x, l_x), (y, l_y)} \right\}$$

Similar to (4.1.5) we can show,

$$\left\| (\rho')^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'} - \rho^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'} \right\|_1 \leq 4\delta^2$$

4.4.2 Intersection POVM Construction

Using the same strategy we used in section 4.2, we can have our Intersection POVM $(\Pi')^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'}$ in the larger augmented Hilbert space $\mathcal{X}'\mathcal{Y}'\mathcal{Z}'$. In the next section we will try to check whether we can satisfy all the properties discussed in (1.2.14)-(1.2.17) and the optimal rate regions given in (1.2.18), where,

$$(\Pi')^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'} := \sum_{x, y, l_x, l_y} |x, l_x\rangle^{\mathcal{X}'} \langle x, l_x| \otimes |y, l_y\rangle^{\mathcal{Y}'} \langle y, l_y| \otimes (\Pi')^{\mathcal{Z}'}_{(x, l_x), (y, l_y), \delta} \quad (4.4.7)$$

4.4.3 Performance Analysis

In section 4.3 We have already given all the desired properties of the intersection POVM in section 4.3. Using similar intuitions here also we'll have the following properties :

$$\begin{aligned} \text{Tr}[(\mathbb{1}^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'} - (\Pi')^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'})(\rho')^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'}] &\leq 76\epsilon^{1/2} \\ \text{Tr}[(\Pi')^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'} ((\rho')^{\mathcal{Y}'} \otimes (\rho')^{\mathcal{X}'\mathcal{Z}'})] &\leq 2^{-I_H^\epsilon(\mathcal{Y}:\mathcal{X}\mathcal{Z})+1} \\ \text{Tr}[(\Pi')^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'} ((\rho')^{\mathcal{X}'} \otimes (\rho')^{\mathcal{Y}'\mathcal{Z}'})] &\leq 2^{-I_H^\epsilon(\mathcal{X}:\mathcal{Y}\mathcal{Z})+1} \\ \text{Tr}[(\Pi')^{\mathcal{X}'\mathcal{Y}'\mathcal{Z}'} ((\rho')^{\mathcal{X}'\mathcal{Y}'} \otimes (\rho')^{\mathcal{Z}'})] &\leq 2^{-I_H^\epsilon(\mathcal{X}\mathcal{Y}:\mathcal{Z})+1} \end{aligned} \quad (4.4.8)$$

So we have shown a generalised version of Sen's strategy in [25]. Although this generalised construction may help us in Distributed Source Compression problem with quantum side information and making it a good application of Sen's work since Distributed Source Compression problem with classical side information shares a very similar idea as that of classical MAC with two mutually dependent inputs. However, further work is required to fully explore these connections.

Chapter 5

Extension of Construction

In this chapter we'll attempt to build an extension of the construction we discussed in chapter 4. We will try to find whether we'll be able to perform this approach quantum states instead of having any classical system appended with the quantum system.

5.1 Quantum State Description and Smoothness of marginal states

Let's say we have 3 systems Alice, Bob and Charlie and we represent them with Hilbert spaces $\mathcal{A}$, $\mathcal{B}$ and $\mathcal{C}$. Suppose we have a shared quantum state ρ_{ABC} . We have the marginals defined as,

$$\begin{aligned}\rho_{AC} &:= \text{Tr}_{\mathcal{B}}[\rho_{ABC}] \\ \rho_{BC} &:= \text{Tr}_{\mathcal{A}}[\rho_{ABC}] \\ \rho_{\mathcal{C}} &:= \text{Tr}_{\mathcal{A}\mathcal{B}}[\rho_{ABC}]\end{aligned}$$

Let's consider $\mathcal{C}_{\mathcal{A}}$ and $\mathcal{C}_{\mathcal{B}}$ be two Hilbert spaces where we assume $\dim(\mathcal{C}_{\mathcal{A}}) = \dim(\mathcal{C}_{\mathcal{B}}) \gg \dim(\mathcal{C})$. Lets define $\hat{\mathcal{C}}_{\mathcal{A}} := \mathcal{C} \oplus \mathcal{C}_{\mathcal{A}}$, $\hat{\mathcal{C}}_{\mathcal{B}} := \mathcal{C} \oplus \mathcal{C}_{\mathcal{B}}$ and $\hat{\mathcal{C}}_{AB} := \mathcal{C} \oplus \mathcal{C}_{\mathcal{A}} \oplus \mathcal{C}_{\mathcal{B}}$. Now we define a map $T_{AB,\delta} : ABC \rightarrow ABC\hat{\mathcal{C}}_{AB}$ as follows:

$$T_{AB,\delta}(|h\rangle^{ABC}) = \frac{1}{\sqrt{1+2\delta^2}} \{ |h\rangle^{ABC} \oplus \delta |h\rangle^{ABC\mathcal{A}} \oplus \delta |h\rangle^{ABC\mathcal{B}} \}$$

Similarly we define the maps $T_{A,\delta} : \mathcal{AC} \rightarrow \mathcal{AC}\hat{\mathcal{C}}_{\mathcal{A}}$ and $T_{B,\delta} : \mathcal{BC} \rightarrow \mathcal{BC}\hat{\mathcal{C}}_{\mathcal{B}}$ in an analogous way:

$$\begin{aligned}T_{A,\delta}(|h\rangle^{AC}) &= \frac{1}{\sqrt{1+\delta^2}} \{ |h\rangle^{AC} \oplus \delta |h\rangle^{AC\mathcal{A}} \} \\ T_{B,\delta}(|h\rangle^{BC}) &= \frac{1}{\sqrt{1+\delta^2}} \{ |h\rangle^{BC} \oplus \delta |h\rangle^{BC\mathcal{B}} \}\end{aligned}$$

Proposition 5.1.1. Consider we define $\rho_{ABC\hat{\mathcal{C}}_{AB}} := T_{AB,\delta}(\rho_{ABC})$ and $\rho_{AC\hat{\mathcal{C}}_{\mathcal{A}}} := T_{A,\delta}(\text{Tr}_{\mathcal{B}}[\rho_{ABC}])$ then we can show the following:

$$\| \text{Tr}_{\mathcal{B}}[\rho_{ABC\hat{\mathcal{C}}_{AB}}] - \rho_{AC\hat{\mathcal{C}}_{\mathcal{A}}} \|_{\infty} \leq 3\delta$$

Proof. We can write ρ_{ABC} in the following way:

$$\rho_{ABC} = \sum_h \lambda_h |h\rangle^{\mathcal{ABC}} \langle h|$$

Since for general quantum states system (simply put Hilbert Spaces) $\mathcal{A}$, $\mathcal{B}$ and $\mathcal{C}$ might not be well-distinguishable. Hence we can't perform augmentation for each individual Hilbert spaces and we only apply tilting to them. Thus we can write $\rho_{ABC\hat{\mathcal{A}}\mathcal{B}}$ as:

$$\begin{aligned} \rho_{ABC\hat{\mathcal{A}}\mathcal{B}} &= \sum_h \lambda_h T_{AB,\delta}(|h\rangle^{\mathcal{ABC}} \langle h|) \\ &= \frac{1}{1+2\delta^2} \sum_h \lambda_h (|h\rangle^{\mathcal{ABC}} + \delta|h\rangle^{\mathcal{ABC}\mathcal{A}} + \delta|h\rangle^{\mathcal{ABC}\mathcal{B}}) (\langle h|^{\mathcal{ABC}} + \delta\langle h|^{\mathcal{ABC}\mathcal{A}} + \delta\langle h|^{\mathcal{ABC}\mathcal{B}}) \\ &= \frac{1}{1+2\delta^2} \sum_h \lambda_h (|h\rangle^{\mathcal{ABC}} + \delta|h\rangle^{\mathcal{ABC}\mathcal{A}}) (\langle h|^{\mathcal{ABC}} + \delta\langle h|^{\mathcal{ABC}\mathcal{A}}) \\ &\quad + \frac{1}{1+2\delta^2} \sum_h \lambda_h \{ (|h\rangle^{\mathcal{ABC}} + \delta|h\rangle^{\mathcal{ABC}\mathcal{A}}) (\delta\langle h|^{\mathcal{ABC}\mathcal{B}}) + (\delta|h\rangle^{\mathcal{ABC}\mathcal{B}}) (\langle h|^{\mathcal{ABC}} + \delta\langle h|^{\mathcal{ABC}\mathcal{A}}) \\ &\quad + \delta^2 |h\rangle^{\mathcal{ABC}\mathcal{B}} \langle h| \} \end{aligned}$$

Now we can write,

$$\begin{aligned} &\text{Tr}_{\mathcal{B}}[\rho_{ABC\hat{\mathcal{A}}\mathcal{B}}] \\ &= \frac{1}{1+2\delta^2} \sum_h \lambda_h \text{Tr}_{\mathcal{B}} [(|h\rangle^{\mathcal{ABC}} + \delta|h\rangle^{\mathcal{ABC}\mathcal{A}}) (\langle h|^{\mathcal{ABC}} + \delta\langle h|^{\mathcal{ABC}\mathcal{A}})] \\ &\quad + \frac{1}{1+2\delta^2} \sum_h \lambda_h \text{Tr}_{\mathcal{B}} [(|h\rangle^{\mathcal{ABC}} + \delta|h\rangle^{\mathcal{ABC}\mathcal{A}}) (\delta\langle h|^{\mathcal{ABC}\mathcal{B}}) + (\delta|h\rangle^{\mathcal{ABC}\mathcal{B}}) (\langle h|^{\mathcal{ABC}} + \delta\langle h|^{\mathcal{ABC}\mathcal{A}}) \\ &\quad + \delta^2 |h\rangle^{\mathcal{ABC}\mathcal{B}} \langle h|] \\ &= \frac{1}{1+2\delta^2} \sum_h \lambda_h (|h\rangle^{\mathcal{AC}} + \delta|h\rangle^{\mathcal{AC}\mathcal{A}}) (\langle h|^{\mathcal{AC}} + \delta\langle h|^{\mathcal{AC}\mathcal{A}}) \\ &\quad + \frac{1}{1+2\delta^2} \sum_h \lambda_h \{ (|h\rangle^{\mathcal{AC}} + \delta|h\rangle^{\mathcal{AC}\mathcal{A}}) (\delta\langle h|^{\mathcal{AC}\mathcal{B}}) + (\delta|h\rangle^{\mathcal{AC}\mathcal{B}}) (\langle h|^{\mathcal{AC}} + \delta\langle h|^{\mathcal{AC}\mathcal{A}}) + \delta^2 |h\rangle^{\mathcal{AC}\mathcal{B}} \langle h| \} \\ &= \frac{1+\delta^2}{1+2\delta^2} \sum_h \lambda_h T_{\mathcal{A},\delta}(|h\rangle^{\mathcal{AC}} \langle h|) + \sum_h \lambda_h N_{\mathcal{A},\delta}(|h\rangle^{\mathcal{AC}} \langle h|) \\ &= \frac{1+\delta^2}{1+2\delta^2} T_{\mathcal{A},\delta}(\sum_h \lambda_h |h\rangle^{\mathcal{AC}} \langle h|) + N_{\mathcal{A},\delta}(\sum_h \lambda_h |h\rangle^{\mathcal{AC}} \langle h|) \\ &= \frac{1+\delta^2}{1+2\delta^2} T_{\mathcal{A},\delta}(\rho_{\mathcal{AC}}) + \frac{1}{1+2\delta^2} N_{\mathcal{A},\delta}(\rho_{\mathcal{AC}}) \end{aligned}$$

$$= \frac{1 + \delta^2}{1 + 2\delta^2} T_{\mathcal{A},\delta}(\text{Tr}_{\mathcal{B}}[\rho_{ABC}]) + N_{\mathcal{A},\delta}(\text{Tr}_{\mathcal{B}}[\rho_{ABC}]) \quad (5.1.1)$$

$$\begin{aligned} \Rightarrow \text{Tr}_{\mathcal{B}}[\rho_{ABC\hat{\mathcal{C}}_{AB}}] &= \frac{1 + \delta^2}{1 + 2\delta^2} \rho_{\mathcal{A}\hat{\mathcal{C}}_{\mathcal{A}}} + N_{\mathcal{A},\delta}(\text{Tr}_{\mathcal{B}}[\rho_{ABC}]) \\ \Rightarrow \text{Tr}_{\mathcal{B}}[\rho_{ABC\hat{\mathcal{C}}_{AB}}] &\leq \rho_{\mathcal{A}\hat{\mathcal{C}}_{\mathcal{A}}} + N_{\mathcal{A},\delta}(\text{Tr}_{\mathcal{B}}[\rho_{ABC}]) \\ \Rightarrow \text{Tr}_{\mathcal{B}}[\rho_{ABC\hat{\mathcal{C}}_{AB}}] - \rho_{\mathcal{A}\hat{\mathcal{C}}_{\mathcal{A}}} &\leq N_{\mathcal{A},\delta}(\text{Tr}_{\mathcal{B}}[\rho_{ABC}]) \end{aligned} \quad (5.1.2)$$

where,

$$N_{\mathcal{A},\delta}(|h\rangle\langle h|)^{\mathcal{AC}} := \frac{1}{1 + 2\delta^2} \{ (|h\rangle^{\mathcal{AC}} + \delta|h\rangle^{\mathcal{AC}_{\mathcal{A}}}) (\delta\langle h|^{\mathcal{AC}_{\mathcal{B}}}) + (\delta|h\rangle^{\mathcal{AC}_{\mathcal{B}}}) (\langle h|^{\mathcal{AC}} + \delta\langle h|^{\mathcal{AC}_{\mathcal{A}}}) + \delta^2|h\rangle^{\mathcal{AC}_{\mathcal{B}}}\langle h| \}$$

Now we have,

$$\begin{aligned} &\left\| N_{\mathcal{A},\delta}(|h\rangle\langle h|)^{\mathcal{AC}} \right\|_{\infty} \\ &= \frac{1}{1 + 2\delta^2} \left\| \{ (|h\rangle^{\mathcal{AC}} + \delta|h\rangle^{\mathcal{AC}_{\mathcal{A}}}) (\delta\langle h|^{\mathcal{AC}_{\mathcal{B}}}) + (\delta|h\rangle^{\mathcal{AC}_{\mathcal{B}}}) (\langle h|^{\mathcal{AC}} + \delta\langle h|^{\mathcal{AC}_{\mathcal{A}}}) + \delta^2|h\rangle^{\mathcal{AC}_{\mathcal{B}}}\langle h| \} \right\|_{\infty} \\ &\leq \frac{1}{1 + 2\delta^2} \left(2\delta \left\| (|h\rangle^{\mathcal{AC}} + |h\rangle^{\mathcal{AC}_{\mathcal{A}}}) (\delta\langle h|^{\mathcal{AC}_{\mathcal{B}}}) \right\|_{\infty} + \delta^2 \left\| |h\rangle^{\mathcal{AC}_{\mathcal{B}}}\langle h| \right\|_{\infty} \right) \\ &\leq 3\delta. \end{aligned}$$

Thus we have

$$\|N_{\mathcal{A},\delta}(\text{Tr}_{\mathcal{B}}[\rho_{ABC}])\|_{\infty} \leq 3\delta \quad (5.1.3)$$

So,

$$\|\text{Tr}_{\mathcal{B}}[\rho_{ABC\hat{\mathcal{C}}_{AB}}] - \rho_{\mathcal{A}\hat{\mathcal{C}}_{\mathcal{A}}}\|_{\infty} \leq 3\delta$$

□

Now similarly as in 5.1.1n we'll have the another two equations as well:

$$\text{Tr}_{\mathcal{A}}[\rho_{ABC\hat{\mathcal{C}}_{AB}}] = \frac{1 + \delta^2}{1 + 2\delta^2} T_{\mathcal{B},\delta}(\rho_{BC}) + N_{\mathcal{B},\delta}(\rho_{BC}) \quad (5.1.4)$$

$$\text{Tr}_{\mathcal{AB}}[\rho_{ABC\hat{\mathcal{C}}_{AB}}] = \frac{1 + \delta^2}{1 + 2\delta^2} (\rho_{\mathcal{C}}) + N_{\delta}(\rho_{\mathcal{C}}) \quad (5.1.5)$$

where,

$$\begin{aligned} N_{\mathcal{B},\delta}(|h\rangle\langle h|)^{\mathcal{BC}} &:= \frac{1}{1 + 2\delta^2} \{ (|h\rangle^{\mathcal{BC}} + \delta|h\rangle^{\mathcal{BC}_{\mathcal{B}}}) (\delta\langle h|^{\mathcal{BC}_{\mathcal{A}}}) + (\delta|h\rangle^{\mathcal{BC}_{\mathcal{A}}}) (\langle h|^{\mathcal{BC}} + \delta\langle h|^{\mathcal{BC}_{\mathcal{B}}}) \\ &\quad + \delta^2|h\rangle^{\mathcal{BC}_{\mathcal{A}}}\langle h| \} \end{aligned}$$

$$\begin{aligned} N_{\delta}(|h\rangle\langle h|)^{\mathcal{C}} &:= \frac{1}{1 + 2\delta^2} \{ (|h\rangle^{\mathcal{C}} + \delta|h\rangle^{\mathcal{C}_{\mathcal{A}}}) (\delta\langle h|^{\mathcal{C}_{\mathcal{B}}}) + (\delta|h\rangle^{\mathcal{C}_{\mathcal{B}}}) (\langle h|^{\mathcal{C}} + \delta\langle h|^{\mathcal{C}_{\mathcal{A}}}) \\ &\quad + \delta^2(|h\rangle^{\mathcal{C}_{\mathcal{A}}} + |h\rangle^{\mathcal{C}_{\mathcal{B}}}) (\langle h|^{\mathcal{C}_{\mathcal{A}}} + \langle h|^{\mathcal{C}_{\mathcal{B}}}) \} \end{aligned}$$

Similar to (5.1.3) we'll have

$$\begin{aligned}\|N_{\mathcal{B},\delta}(\rho_{\mathcal{BC}})\|_{\infty} &\leq 3\delta \\ \|N_{\delta}(\rho_{\mathcal{C}})\|_{\infty} &\leq 3\delta\end{aligned}$$

Since we assume δ is sufficiently small, we can claim the following:

$$\|\rho_{\mathcal{ABC}\hat{\mathcal{A}}\mathcal{B}} - \rho_{\mathcal{ABC}}\|_1 \leq 4\delta^2 \quad (5.1.6)$$

5.2 Intersection POVM Construction

Now we define 3 projectors $\hat{\Pi}_{\mathcal{A}}^{ABC}$, $\hat{\Pi}_{\mathcal{B}}^{ABC}$ and $\hat{\Pi}^{ABC}$ such that the following properties satisfies

$$\begin{aligned}\hat{\Pi}_{\mathcal{A}}^{ABC} : \quad & \text{Tr}[\hat{\Pi}_{\mathcal{A}}^{ABC} \rho_{\mathcal{ABC}}] \geq 1 - \epsilon \\ & \text{Tr}[\hat{\Pi}_{\mathcal{A}}^{ABC} (\rho_{\mathcal{B}} \otimes \rho_{\mathcal{AC}})] \leq 2^{-k_1} \\ \hat{\Pi}_{\mathcal{B}}^{ABC} : \quad & \text{Tr}[\hat{\Pi}_{\mathcal{B}}^{ABC} \rho_{\mathcal{ABC}}] \geq 1 - \epsilon \\ & \text{Tr}[\hat{\Pi}_{\mathcal{B}}^{ABC} (\rho_{\mathcal{A}} \otimes \rho_{\mathcal{BC}})] \leq 2^{-k_2} \\ \hat{\Pi}^{ABC} : \quad & \text{Tr}[\hat{\Pi}^{ABC} \rho_{\mathcal{ABC}}] \geq 1 - \epsilon \\ & \text{Tr}[\hat{\Pi}^{ABC} (\rho_{\mathcal{A}} \otimes \rho_{\mathcal{B}} \otimes \rho_{\mathcal{C}})] \leq 2^{-k_3}\end{aligned} \quad (5.2.1)$$

Let $W'_{\mathcal{A}}$ denotes the orthogonal complement of the support of $\hat{\Pi}_{\mathcal{A}}^{ABC}$ in $\mathcal{ABC}$. Define $W'_{\mathcal{B}}$, W' analogously for $\hat{\Pi}_{\mathcal{B}}^{ABC}$ and $\hat{\Pi}^{ABC}$ respectively. Define the tilted spaces

$$\begin{aligned}W'_{\mathcal{A};\delta} &:= T_{\mathcal{A},\delta}(W'_{\mathcal{A}}), \\ W'_{\mathcal{B};\delta} &:= T_{\mathcal{B},\delta}(W'_{\mathcal{B}}),\end{aligned}$$

Define the subspace

$$W'_{\delta} := W'_{\mathcal{A};\delta} + W'_{\mathcal{B};\delta} + W',$$

and $(\Pi')_{\text{Correct}}^{ABC\hat{\mathcal{A}}\mathcal{B}}$ to be the orthogonal projection in $\mathcal{ABC}\hat{\mathcal{A}}\mathcal{B}$ onto W'_{δ} . Let $\Pi_{\mathcal{ABC}}^{ABC\hat{\mathcal{A}}\mathcal{B}}$ be the orthogonal projection in $\mathcal{ABC}\hat{\mathcal{A}}\mathcal{B}$ onto $\mathcal{ABC}$. Then we define the projector $(\Pi')_{\text{Correct}}^{ABC\hat{\mathcal{A}}\mathcal{B}}$ in $\mathcal{ABC}\hat{\mathcal{A}}\mathcal{B}$ to be

$$(\Pi')_{\text{Correct}}^{ABC\hat{\mathcal{A}}\mathcal{B}} := \left(\mathbb{1}^{\mathcal{ABC}\hat{\mathcal{A}}\mathcal{B}} - (\Pi')_{W'_{\delta}}^{\mathcal{ABC}\hat{\mathcal{A}}\mathcal{B}} \right) \Pi_{\mathcal{ABC}}^{\mathcal{ABC}\hat{\mathcal{A}}\mathcal{B}} \left(\mathbb{1}^{\mathcal{ABC}\hat{\mathcal{A}}\mathcal{B}} - (\Pi')_{W'_{\delta}}^{\mathcal{ABC}\hat{\mathcal{A}}\mathcal{B}} \right)$$

5.3 Performance Analysis

So we try to find the error probability of not getting the actual state i.e. $\rho_{AB\hat{C}_{AB}}$,

$$\begin{aligned}
& \text{Tr} \left[\left(\mathbb{1}^{AB\hat{C}_{AB}} - (\Pi')_{Correct}^{AB\hat{C}_{AB}} \right) \rho_{AB\hat{C}_{AB}} \right] \\
& \leq_a \text{Tr} \left[\left(\mathbb{1}^{AB\hat{C}_{AB}} - (\Pi')_{Correct}^{AB\hat{C}_{AB}} \right) \rho_{ABC} \right] + 4\delta^2 \\
& = \text{Tr} [\rho_{ABC}] - \text{Tr} \left[\left((\Pi')_{Correct}^{AB\hat{C}_{AB}} \right) \rho_{ABC} \right] + 4\delta^2 \\
& = \text{Tr} [\rho_{ABC}] - \text{Tr} \left[\left(\mathbb{1}^{AB\hat{C}_{AB}} - (\Pi')_{W'\delta}^{AB\hat{C}_{AB}} \right) \Pi_{ABC}^{AB\hat{C}_{AB}} \left(\mathbb{1}^{AB\hat{C}_{AB}} - (\Pi')_{W'\delta}^{AB\hat{C}_{AB}} \right) \rho_{ABC} \right] + 4\delta^2 \\
& = 1 - \text{Tr} \left[\Pi_{ABC}^{AB\hat{C}_{AB}} \left(\mathbb{1}^{AB\hat{C}_{AB}} - (\Pi')_{W'\delta}^{AB\hat{C}_{AB}} \right) \rho_{ABC} \left(\mathbb{1}^{AB\hat{C}_{AB}} - (\Pi')_{W'\delta}^{AB\hat{C}_{AB}} \right) \Pi_{ABC}^{AB\hat{C}_{AB}} \right] + 4\delta^2 \\
& \leq_b 4\text{Tr} \left[\left(\mathbb{1}^{AB\hat{C}_{AB}} - \Pi_{ABC}^{AB\hat{C}_{AB}} \right) \rho_{ABC} \right] + 4\text{Tr} \left[(\Pi')_{W'\delta}^{AB\hat{C}_{AB}} \rho_{ABC} \right] + 4\delta^2 \\
& = 4\text{Tr} \left[(\Pi')_{W'\delta}^{AB\hat{C}_{AB}} \rho_{ABC} \right] + 4\delta^2 \\
& \leq_c \frac{4.6}{\delta^2} \left(\text{Tr} \left[(\Pi')_{W'A;\delta}^{AB\hat{C}_{AB}} \rho_{ABC} \right] + \text{Tr} \left[(\Pi')_{W'B;\delta}^{AB\hat{C}_{AB}} \rho_{ABC} \right] + \text{Tr} \left[(\Pi')_{W'}^{AB\hat{C}_{AB}} \rho_{ABC} \right] \right) + 4\delta^2 \\
& \leq \frac{24}{\delta^2} \left(\text{Tr} \left[(\Pi')_{W'A}^{ABC} \rho_{ABC} \right] + \text{Tr} \left[(\Pi')_{W'B}^{ABC} \rho_{ABC} \right] + \text{Tr} \left[(\Pi')_{W'}^{ABC} \rho_{ABC} \right] \right) + 4\delta^2 \\
& \leq \frac{24}{\delta^2} \left(3 - \text{Tr} \left[\hat{\Pi}_A^{ABC} \rho_{ABC} \right] - \text{Tr} \left[\hat{\Pi}_B^{ABC} \rho_{ABC} \right] - \text{Tr} \left[\hat{\Pi}^{ABC} \rho_{ABC} \right] \right) + 4\delta^2 \\
& \leq \frac{24.3\epsilon}{\delta^2} + 4\delta^2 \\
& = \frac{72\epsilon}{\delta^2} + 4\delta^2
\end{aligned}$$

We choose $\delta := \epsilon^{1/4}$, then

$$\text{Tr} \left[(\Pi')_{Correct}^{AB\hat{C}_{AB}} \rho_{AB\hat{C}_{AB}} \right] \geq 1 - 76\epsilon^{1/2}$$

Here, inequality **a** follows from (5.1.6) and Fact 10, **b** follows from Fact 8 and **c** follows from Corollary 2.2.3. Before going into the probability of error upper-bounds we can show the following inequality using 11

$$\begin{aligned}
\left\| (\Pi')_{Correct}^{AB\hat{C}_{AB}} \right\|_1 &= \left\| \left(\mathbb{1}^{AB\hat{C}_{AB}} - (\Pi')_{W'\delta}^{AB\hat{C}_{AB}} \right) \Pi_{ABC}^{AB\hat{C}_{AB}} \left(\mathbb{1}^{AB\hat{C}_{AB}} - (\Pi')_{W'\delta}^{AB\hat{C}_{AB}} \right) \right\|_1 \\
&\leq \left\| \left(\mathbb{1}^{AB\hat{C}_{AB}} - (\Pi')_{W'\delta}^{AB\hat{C}_{AB}} \right) \right\|_\infty^2 \left\| \Pi_{ABC}^{AB\hat{C}_{AB}} \right\|_1 = 2|ABC|
\end{aligned}$$

Now we'll try to get the upper-bound of error probabilities,

$$\begin{aligned}
& \text{Tr} \left[(\Pi')_{Correct}^{AB\hat{C}_{AB}} (\rho_{A\hat{C}_{AB}} \otimes \rho_B) \right] \\
& \leq \text{Tr} \left[(\Pi')_{Correct}^{AB\hat{C}_{AB}} (\rho_{A\hat{C}_{AB}} \otimes \rho_B) \right] \\
& \leq_a \text{Tr} \left[(\Pi')_{Correct}^{AB\hat{C}_{AB}} (\rho_{A\hat{C}_A} \otimes \rho_B) \right] + \text{Tr} \left[(\Pi')_{Correct}^{AB\hat{C}_{AB}} (N_{A,\delta}(\text{Tr}_B[\rho_{ABC}]) \otimes \rho_B) \right] \\
& \leq_b \text{Tr} \left[(\Pi')_{Correct}^{AB\hat{C}_{AB}} (\rho_{A\hat{C}_A} \otimes \rho_B) \right] + \left\| (\Pi')_{Correct}^{AB\hat{C}_{AB}} (N_{A,\delta}(\text{Tr}_B[\rho_{ABC}]) \otimes \rho_B) \right\|_1
\end{aligned}$$

$$\begin{aligned}
&\leq \text{Tr} \left[(\Pi')_{Correct}^{ABC\hat{C}_{AB}} (\rho_{\mathcal{A}\hat{C}_A} \otimes \rho_B) \right] + \left\| (\Pi')_{Correct}^{ABC\hat{C}_{AB}} \right\|_1 \| (N_{\mathcal{A},\delta}(\text{Tr}_B[\rho_{ABC}]) \otimes \rho_B) \|_\infty \\
&\leq \text{Tr} \left[(\Pi')_{Correct}^{ABC\hat{C}_{AB}} (T_{\mathcal{A},\delta}(\rho_{AC}) \otimes \rho_B) \right] + 6|ABC|\delta \\
&= \text{Tr} \left[\Pi_{ABC}^{ABC\hat{C}_{AB}} \left(\mathbb{1}^{ABC\hat{C}_{AB}} - (\Pi')_{W'_\delta}^{ABC\hat{C}_{AB}} \right) (T_{\mathcal{A},\delta}(\rho_{AC}) \otimes \rho_B) \left(\mathbb{1}^{ABC\hat{C}_{AB}} - (\Pi')_{W'_\delta}^{ABC\hat{C}_{AB}} \right) \right] + 6|ABC|\delta \\
&\leq \text{Tr} \left[\left(\mathbb{1}^{ABC\hat{C}_{AB}} - (\Pi')_{W'_\delta}^{ABC\hat{C}_{AB}} \right) (T_{\mathcal{A},\delta}(\rho_{AC}) \otimes \rho_B) \right] + 6|ABC|\delta \\
&\leq \text{Tr} \left[\left(\mathbb{1}^{ABC\hat{C}_{AB}} - (\Pi')_{W'_{\mathcal{A};\delta}}^{ABC\hat{C}_{AB}} \right) (T_{\mathcal{A},\delta}(\rho_{AC}) \otimes \rho_B) \right] + 6|ABC|\delta \\
&= \text{Tr} \left[\left(\mathbb{1}^{ABC\hat{C}_{AB}} - (\Pi')_{W'_{\mathcal{A};\delta}}^{ABC\hat{C}_{AB}} \right) \Pi_{B \otimes T_{\mathcal{A},\delta}(AC)}^{ABC\hat{C}_{AB}} (T_{\mathcal{A},\delta}(\rho_{AC}) \otimes \rho_B) \right] + 6|ABC|\delta \\
&\stackrel{d}{=} \text{Tr} \left[\left(\mathbb{1}^{B \otimes T_{\mathcal{A},\delta}(AC)} - (\Pi')_{T_{\mathcal{A},\delta}(W'_A)}^{B \otimes T_{\mathcal{A},\delta}(AC)} \right) (T_{\mathcal{A},\delta}(\rho_{AC}) \otimes \rho_B) \right] + 6|ABC|\delta \\
&\stackrel{e}{=} \text{Tr} \left[\left(\mathbb{1}^{ABC} - (\Pi')_{W'_A}^{ABC} \right) (\rho_{AC} \otimes \rho_B) \right] + 6|ABC|\delta \\
&= \text{Tr} \left[\hat{\Pi}_{\mathcal{A}}^{ABC} (\rho_{AC} \otimes \rho_B) \right] + 6|ABC|\delta \\
&\leq 2^{-k_1} + 6|ABC|\delta
\end{aligned}$$

Here, inequality **a** follows from (5.1.2), **b** follows from Fact 10, **c** follows from Fact 11 the equality **d** follows from that fact $T_{\mathcal{A},\delta}(\rho_{AC}) \leq \mathcal{A}\hat{C}_{AB}$, and equality **e** follows from the fact that $T_{\mathcal{A},\delta}(\rho_{AC})$ is an isometric map. Similarly we have two other upper-bounds as well,

$$\begin{aligned}
\text{Tr} \left[(\Pi')_{Correct}^{ABC\hat{C}_{AB}} (\rho_{B\hat{C}_{AB}} \otimes \rho_A) \right] &\leq 2^{-k_2} + 6|ABC|\delta \\
\text{tr} \left[(\Pi')_{Correct}^{ABC\hat{C}_{AB}} (\rho_{\hat{C}_{AB}} \otimes \rho_A \otimes \rho_B) \right] &\leq 2^{-k_3} + 6|ABC|\delta
\end{aligned}$$

In contrast to equation (4.3.2), the above equations lack a control variable such as $|\mathcal{L}|$. This means that there is a possibility that the space ABC is very large, rendering the term $6|ABC|\delta$ ineffective for quantum states that exist in a very large Hilbert space. Therefore, Sen's work [25] is not effective for general quantum states.

Chapter 6

Quantum Intersection Lower Bound

In subsection 1.2.1 we discussed about hypothesis testing and classical Stein's lemma and gave an intuition to extend it for quantum Stein's lemma using the approach similar to that we used to prove classical Stein's lemma but we noticed this would require a strong notion of intersection of measurements operators and in section 2.1 we realised notion of *intersection of measurement operators* is highly non-trivial.

To understand the need for a notion of intersection in the quantum scenario, consider the following statement about classical probability distributions. Consider three distributions P , Q_1 and Q_2 distributed over the set $\mathcal{X}$ such that $\text{supp}(P) \subseteq \text{supp}(Q_1)$ and $\text{supp}(P) \subseteq \text{supp}(Q_2)$.

Further, for $\alpha, \beta > 0$, let us define two sets $\mathcal{A}_1$ and $\mathcal{A}_2$ in the following way:

$$\begin{aligned}\mathcal{A}_1 &:= \{x : Q_1(x) \geq 2^{-k_1} P(x)\} \\ \mathcal{A}_2 &:= \{x : Q_2(x) \geq 2^{-k_2} P(x)\}\end{aligned}$$

with the following properties:

$$\Pr_P(\mathcal{A}_1) \geq 1 - \epsilon$$

$$\Pr_P(\mathcal{A}_2) \geq 1 - \epsilon.$$

Then, it is trivial to observe that for a set $\mathcal{A}^*$ which is the intersection of the two sets $\mathcal{A}_1$ and $\mathcal{A}_2$ i.e. $\mathcal{A}^* := \mathcal{A}_1 \cap \mathcal{A}_2$, we have the following properties

$$\Pr_P(\mathcal{A}^*) \geq 1 - 2\epsilon \tag{6.0.1}$$

$$\Pr_{Q_1}(\mathcal{A}^*) \geq 2^{-k_1}(1 - 2\epsilon) \tag{6.0.2}$$

$$\Pr_{Q_2}(\mathcal{A}^*) \geq 2^{-k_2}(1 - 2\epsilon) \tag{6.0.3}$$

where (6.0.1) follows from the union bound for probability and the other two inequalities (6.0.2), (6.0.3) follow from the fact that $\mathcal{A}^* \subseteq \mathcal{A}_1$ and $\mathcal{A}^* \subseteq \mathcal{A}_2$.

We aim to prove a similar statement in the quantum setting. To define the problem formally let us consider ρ, σ_1 and $\sigma_2 \in \mathcal{D}(\mathcal{H})$ such that $\text{supp}(\rho) \subseteq \text{supp}(\sigma_1)$ and $\text{supp}(\rho) \subseteq \text{supp}(\sigma_2)$. Further, for $\alpha, \beta > 0$, let Π_1, Π_2 be two measurement operators such that

$$\begin{aligned}\Pi_1 &:= \{\sigma_1 \succeq 2^{-k_1} \rho\} \\ \Pi_2 &:= \{\sigma_2 \succeq 2^{-k_2} \rho\}\end{aligned}$$

with the properties analogous to the classical scenario,

$$\begin{aligned}\text{Tr}[\Pi_1 \rho] &\geq 1 - \epsilon \\ \text{Tr}[\Pi_2 \rho] &\geq 1 - \epsilon\end{aligned}$$

Then, we want to find a projector Π^* such that it holds the following properties

$$\begin{aligned}\text{Tr}[\Pi^* \rho] &\geq 1 - f(\epsilon) \\ \text{Tr}[\Pi^* \sigma_1] &\geq 2^{-k_1} f(\epsilon) \\ \text{Tr}[\Pi^* \sigma_2] &\geq 2^{-k_2} f(\epsilon)\end{aligned}$$

where, $f(\epsilon)$ is some function of ϵ (in the classical case it was $(1 - 2\epsilon)$).

In this work we take care of the issues mentioned above and show it is indeed possible construct a Π^* such that it satisfies the properties analogous to the intersection subset as mentioned in the discussion above. An important ingredient which we use to construct this Π^* is the Jordan's lemma [20] which has found many applications in quantum information and computation, see for example, [8], [2] and also in post-quantum cryptography [9]. We now mention the main contribution of this work.

6.1 Objective

We aim to prove the below Theorem 6.1.1 with the help of certain presumptions we had in section 1.1.

Lemma 6.1.1. (Main Result) *Let ρ, σ_1 and $\sigma_2 \in \mathcal{D}(\mathcal{H})$ such that $\text{supp}(\rho) \subseteq \text{supp}(\sigma_1)$ and $\text{supp}(\rho) \subseteq \text{supp}(\sigma_2)$. Further, for $k_1, k_2 > 0$, let Π_1, Π_2 be such that $\Pi_1 := \{\sigma_1 \succeq 2^{-k_1} \rho\}$ and $\Pi_2 := \{\sigma_2 \succeq 2^{-k_2} \rho\}$ with the property that $\text{Tr}[\Pi_1 \rho] \geq 1 - \epsilon$ and $\text{Tr}[\Pi_2 \rho] \geq 1 - \epsilon$. Then, there exists a projector Π^* such that*

$$\text{Tr}[\Pi^* \rho] \geq (1 - 4\sqrt{\epsilon}), \quad (6.1.1)$$

$$\text{Tr}[\Pi^* \sigma_2] \geq 2^{-k_2} (1 - 20\sqrt{\epsilon}), \quad (6.1.2)$$

$$\text{Tr}[\Pi^* \sigma_1] \geq 2^{-k_1} (1 - 4\sqrt{\epsilon}). \quad (6.1.3)$$

A byproduct of the above lemma is the following corollary which has an application in proving a converse for the quantum Chernoff Stein's lemma which is similar in spirit to the converse provided in Tom and Cover book [26, Theorem 11.8.3].

Corollary 6.1.2. *Let $\rho^{\otimes n}$ and $\sigma^{\otimes n}$ be two quantum states over the Hilbert space $\mathcal{H}^{\otimes n}$ such that $D(\rho \parallel \sigma) < \infty$. Further, $\Pi_n^{(2)}$ is a projector such that $\text{Tr}[\Pi_n^{(2)} \rho^{\otimes n}] \geq 1 - \epsilon$. Then,*

$$\text{Tr}[\Pi_n^{(2)} \sigma^{\otimes n}] \geq 2^{-n(D(\rho \parallel \sigma) + \delta)} g(\epsilon),$$

where $g(\epsilon) = (1 - 19\sqrt{\epsilon})$ and $\delta > 0$ is arbitrary.

Corollary 6.1.2 can be viewed as the quantum generalisation of [26, Lemma 11.8.1].

6.2 Proof Construction

6.2.1 Construction of the Intersection Projector Π^*

Let Π_1 and Π_2 be as mentioned in Lemma 6.1.1. We will apply Fact 5 on the pair $\{\Pi_1, \Pi_2\}$. Consider $\{\Pi_\alpha\}_{\alpha=1}^k$ as the set of orthogonal projectors obtained by Fact 5 applied with respect to the pair $\{\Pi_1, \Pi_2\}$. Thus, $\sum_{\alpha=1}^k \Pi_\alpha = \mathbb{I}$. Furthermore, for every $\alpha \in [1 : k]$ we can represent $\Pi_\alpha \Pi_1 \Pi_\alpha := \Pi_{1,\alpha}$ as the following one dimensional projector:

$$\Pi_{1,\alpha} = |v_\alpha\rangle\langle v_\alpha|, \quad (6.2.1)$$

for some $|v_\alpha\rangle$ in the range of Π_α . Similarly, for every $\alpha \in [1 : k]$, we have

$$\Pi_{2,\alpha} = |w_\alpha\rangle\langle w_\alpha|, \quad (6.2.2)$$

for some $|w_\alpha\rangle$ in the range of Π_α . Thus, it is now follows from the property of $\{|v_\alpha\rangle\}_{\alpha=1}^k$ and $\{|w_\alpha\rangle\}_{\alpha=1}^k$ that

$$\Pi_1 := \sum_{\alpha=1}^k |v_\alpha\rangle\langle v_\alpha|, \quad (6.2.3)$$

$$\Pi_2 := \sum_{\alpha=1}^k |w_\alpha\rangle\langle w_\alpha|. \quad (6.2.4)$$

Further, for every $\alpha \in [1 : k]$, let

$$\rho_\alpha := \Pi_\alpha \rho \Pi_\alpha, \quad (6.2.5)$$

$$\sigma_{1,\alpha} := \Pi_\alpha \sigma_1 \Pi_\alpha, \quad (6.2.6)$$

$$\sigma_{2,\alpha} := \Pi_\alpha \sigma_2 \Pi_\alpha. \quad (6.2.7)$$

Thus, from the definitions of Π_1 and Π_2 , we have the following pair of inequalities for every $\alpha \in [1 : k]$,

$$\langle v_\alpha | \sigma_{1,\alpha} | v_\alpha \rangle \geq 2^{-k_1} \langle v_\alpha | \rho_\alpha | v_\alpha \rangle, \quad (6.2.8)$$

$$\langle w_\alpha | \sigma_{2,\alpha} | w_\alpha \rangle \geq 2^{-k_2} \langle w_\alpha | \rho_\alpha | w_\alpha \rangle, \quad (6.2.9)$$

For our future discussions it will be useful to make the following observations, for every $\alpha \in [1 : k]$, $\rho_\alpha \succeq 0$ and $\sum_{\alpha=1}^k \text{Tr}[\rho_\alpha] = 1$. Thus, $\{\text{Tr}[\rho_\alpha]\}_{\alpha=1}^k$ forms a valid probability distribution over

$\alpha \in [1 : k]$.

Let $\cos^2(\theta_\alpha) := |\langle v_\alpha | w_\alpha \rangle|^2$ and define the following sets:

$$\mathcal{E}_1 := \{ \alpha : \cos^2(\theta_\alpha) \geq 1 - 8\sqrt{\epsilon} \}, \quad (6.2.10)$$

$$\mathcal{E}_2 := \{ \alpha : \text{Tr}[|v_\alpha\rangle\langle v_\alpha| \rho_\alpha] \geq (1 - \sqrt{\epsilon}) \text{Tr}[\rho_\alpha] \}, \quad (6.2.11)$$

$$\mathcal{E}_3 := \{ \alpha : \text{Tr}[|w_\alpha\rangle\langle w_\alpha| \rho_\alpha] \geq (1 - \sqrt{\epsilon}) \text{Tr}[\rho_\alpha] \}, \quad (6.2.12)$$

$$\text{Good} := \mathcal{E}_1 \cap \mathcal{E}_2 \cap \mathcal{E}_3. \quad (6.2.13)$$

The intuition for defining the set Good is that we only want to consider those blocks where $|v_\alpha\rangle$ and $|w_\alpha\rangle$ are almost the same, i.e., the angle between them is very small along with the property that $|v_\alpha\rangle\langle v_\alpha|$ and $|w_\alpha\rangle\langle w_\alpha|$ behave almost like an identity operator over that block.

We now make the following claims about the sets $\mathcal{E}_1$, $\mathcal{E}_2$, $\mathcal{E}_3$ and Good.

Claim 1. The probability of the set $\mathcal{E}_1$ under the distribution $\text{Tr}[\rho_\alpha]$ satisfies the following lower bound: $\Pr\{\mathcal{E}_1\} \geq 1 - \sqrt{\epsilon}$.

Claim 2. The probability of the set $\mathcal{E}_2$ under the distribution $\text{Tr}[\rho_\alpha]$ satisfies the following lower bound: $\Pr\{\mathcal{E}_2\} \geq 1 - \sqrt{\epsilon}$.

Claim 3. The probability of the set $\mathcal{E}_3$ under the distribution $\text{Tr}[\rho_\alpha]$ satisfies the following lower bound: $\Pr\{\mathcal{E}_3\} \geq 1 - \sqrt{\epsilon}$.

Claim 4. The probability of the set Good under the distribution $\text{Tr}[\rho_\alpha]$ satisfies the following lower bound: $\Pr\{\text{Good}\} \geq 1 - 3\sqrt{\epsilon}$.

Claim 4 follows from Claims 1, 2 and 3 and union bound for probability. We will prove these claims later. To show the existence of Π^* which satisfies (6.1.1), (6.1.3) and (6.1.2) we construct Π^* as follows,

$$\Pi^* := \sum_{\alpha \in \text{Good}} |v_\alpha\rangle\langle v_\alpha|. \quad (6.2.14)$$

To show that Π^* satisfies (6.1.1), (6.1.3), (6.1.2) we first make few observations. From Jordan's lemma (Fact 5) we have that for every $\alpha \in [1 : k]$ the vectors $|w_\alpha\rangle$ and $|v_\alpha\rangle$ lie in either a two dimensional subspace or one-dimensional subspace. For the blocks where both $|w_\alpha\rangle$ and $|v_\alpha\rangle$ lie in one dimensional subspace then in that case $\cos(\theta(\alpha)) = 1$, i.e., $|w_\alpha\rangle = |v_\alpha\rangle$ for these cases. For the blocks where both $|w_\alpha\rangle$ and $|v_\alpha\rangle$ lie in a two dimensional subspace we have the following,

$$|v_\alpha\rangle = \cos(\theta_\alpha)|w_\alpha\rangle + \sin(\theta_\alpha)|w_\alpha^\perp\rangle. \quad (6.2.15)$$

6.2.2 Proof for the properties of Π^*

In this section we show that Π^* has properties similar to that of $\mathcal{A}^* = \mathcal{A}_1 \cap \mathcal{A}_2$ as discussed in the introduction. In particular we show that Π^* satisfies (6.1.1), (6.1.3), (6.1.2).

- **Proof of (6.1.1)**

Proof. Consider the following set of inequalities

$$\begin{aligned} \text{Tr}[\Pi^* \rho] &\stackrel{a}{=} \sum_{\alpha \in \text{Good}} \text{Tr}[|v_\alpha\rangle\langle v_\alpha| \rho_\alpha] \\ &\stackrel{b}{\geq} \sum_{\alpha \in \text{Good}} (1 - \sqrt{\epsilon}) \text{Tr}[\rho_\alpha] \\ &\stackrel{c}{\geq} 1 - 4\sqrt{\epsilon}, \end{aligned}$$

where a follows from the definition of (6.2.14), b follows from (6.2.13) and c follows from Claim 4.

- **Proof of (6.1.3)**

Proof. Consider the following set of inequalities

$$\begin{aligned} \text{Tr}[\Pi^* \sigma_1] &\stackrel{a}{=} \sum_{\alpha \in \text{Good}} \text{Tr}[|v_\alpha\rangle\langle v_\alpha| \sigma_{1,\alpha}] \\ &\stackrel{b}{\geq} 2^{-k_1} \sum_{\alpha \in \text{Good}} \text{Tr}[|v_\alpha\rangle\langle v_\alpha| \rho_\alpha] \\ &\stackrel{c}{\geq} 2^{-k_1} \sum_{\alpha \in \text{Good}} (1 - \sqrt{\epsilon}) \text{Tr}[\rho_\alpha] \\ &\stackrel{d}{\geq} 2^{-k_1} (1 - \sqrt{\epsilon}) (1 - 3\sqrt{\epsilon}) \\ &\geq 2^{-k_1} (1 - 4\sqrt{\epsilon}), \end{aligned}$$

where a follows from the definition of (6.2.14), b follows from (6.2.8), c follows from the definition of the set Good defined in (6.2.13) and d follows from Claim 4.

- **Proof of (6.1.2)**

Proof. Consider the following set of inequalities

$$\begin{aligned} \text{Tr}[\Pi^* \sigma_2] &\stackrel{a}{=} \sum_{\alpha \in \text{Good}} \text{Tr}[|v_\alpha\rangle\langle v_\alpha| \sigma_{2,\alpha}] \\ &\stackrel{b}{=} \sum_{\alpha \in \text{Good}} \left(\cos^2(\theta_\alpha) \text{Tr}[|w_\alpha\rangle\langle w_\alpha| \sigma_{2,\alpha}] + \sin^2(\theta_\alpha) \text{Tr}[|w_\alpha^\perp\rangle\langle w_\alpha^\perp| \sigma_{2,\alpha}] \right) \end{aligned}$$

$$\begin{aligned}
& + \cos(\theta_\alpha) \sin(\theta_\alpha) \text{Tr}[|w_\alpha\rangle\langle w_\alpha^\perp|\sigma_{2,\alpha}] + \cos(\theta_\alpha) \sin(\theta_\alpha) \text{Tr}[|w_\alpha^\perp\rangle\langle w_\alpha|\sigma_{2,\alpha}]) \\
& \stackrel{c}{=} \sum_{\alpha \in \text{Good}} \left(\cos(2\theta_\alpha) \text{Tr}[|w_\alpha\rangle\langle w_\alpha|\sigma_{2,\alpha}] + \cos(\theta_\alpha) \sin(\theta_\alpha) \text{Tr}[|w_\alpha\rangle\langle w_\alpha^\perp|\sigma_{2,\alpha}] \right. \\
& \quad \left. + \cos(\theta_\alpha) \sin(\theta_\alpha) \text{Tr}[|w_\alpha^\perp\rangle\langle w_\alpha|\sigma_{2,\alpha}] + \sin^2(\theta_\alpha) \text{Tr}[\sigma_{2,\alpha}] \right) \\
& \stackrel{d}{\geq} \sum_{\alpha \in \text{Good}} \left(\cos(2\theta_\alpha) \text{Tr}[|w_\alpha\rangle\langle w_\alpha|\sigma_{2,\alpha}] + \sin^2(\theta_\alpha) \text{Tr}[\sigma_{2,\alpha}] \right. \\
& \quad \left. + \sin^2(\theta_\alpha) \text{Tr}[|w_\alpha\rangle\langle w_\alpha^\perp|\sigma_{2,\alpha}] + \sin^2(\theta_\alpha) \text{Tr}[|w_\alpha^\perp\rangle\langle w_\alpha|\sigma_{2,\alpha}] \right) \\
& \stackrel{e}{\geq} \sum_{\alpha \in \text{Good}} \left(2^{-k_2} \cos(2\theta_\alpha) \text{Tr}[|w_\alpha\rangle\langle w_\alpha|\rho_\alpha] + \sin^2(\theta_\alpha) (\text{Tr}[\sigma_{2,\alpha}] + \text{Tr}[|w_\alpha\rangle\langle w_\alpha^\perp|\sigma_{2,\alpha}] \right. \\
& \quad \left. + \text{Tr}[|w_\alpha^\perp\rangle\langle w_\alpha|\sigma_{2,\alpha}]) \right) \\
& \stackrel{f}{\geq} \sum_{\alpha \in \text{Good}} (2^{-k_2} \cos(2\theta_\alpha) \text{Tr}[|w_\alpha\rangle\langle w_\alpha|\rho_\alpha]) \\
& \stackrel{g}{\geq} \sum_{\alpha \in \text{Good}} (2^{-k_2} \cos(2\theta_\alpha) (1 - \sqrt{\epsilon}) \text{Tr}[\rho_\alpha]) \\
& \stackrel{h}{\geq} 2^{-k_2} (1 - \sqrt{\epsilon}) (1 - 16\sqrt{\epsilon}) \sum_{\alpha \in \text{Good}} (\text{Tr}[\rho_\alpha]) \\
& \stackrel{i}{\geq} 2^{-k_2} (1 - \sqrt{\epsilon}) (1 - 16\sqrt{\epsilon}) (1 - 3\sqrt{\epsilon}) \\
& \geq 2^{-k_2} (1 - 20\sqrt{\epsilon}), \tag{6.2.16}
\end{aligned}$$

where a follows from the definition of $\Pi^\star$ (see (6.2.14)), b follows from (6.2.15), c follows because $\text{Tr}[\sigma_{2,\alpha}] = \text{Tr}[|w_\alpha\rangle\langle w_\alpha|\sigma_{2,\alpha}] + \text{Tr}[|w_\alpha^\perp\rangle\langle w_\alpha^\perp|\sigma_{2,\alpha}]$ and $\cos(2\theta_\alpha) = \cos^2(\theta_\alpha) - \sin^2(\theta_\alpha)$, d follows under the assumption that $\cos(\theta_\alpha) \geq \sin(\theta_\alpha)^1$, e follows from (6.2.9), f follows because $\sigma_{2,\alpha} \succeq 0$ and from Lemma (6.2.1), g follows from the definition of Good, defined in (6.2.13), h follows from the fact that $\cos(2\theta_\alpha) = 2\cos^2(\theta_\alpha) - 1$ and for $\alpha \in \text{Good}$, $\cos^2(\theta_\alpha) \geq 1 - 8\sqrt{\epsilon}$ and i follows from claim 4.

Lemma 6.2.1. *Let A be a 2×2 positive semi-definite matrix, i.e.,*

$$A = \begin{bmatrix} a & b \\ \bar{b} & d \end{bmatrix},$$

where $a, d \geq 0$, $\bar{b}$ is the conjugate of b and let $\text{Re}(b)$ represent the real part of b . Then, $\text{Tr}[A] + 2\text{Re}(b) \geq 0$.

Proof. From the positivity of the matrix A it follows that for any non-zero+ -vector $|x\rangle$,

¹For the case where $\sin(\theta_\alpha) \geq \cos(\theta_\alpha)$ we will bound d by lower bounding $\sin(\theta_\alpha)$ by $\cos(\theta_\alpha)$.

$\langle x|A|x \rangle \geq 0$. Now if we multiply A by the row vector $[1, 1]$ on the left and by the column vector $[1, 1]^T$ on the right, we'll get $\text{Tr}[A] + 2\text{Re}(b) \geq 0$ since A is positive. $\square$

6.3 Proof of Claims

6.3.1 Proof of claim 1

Proof. Using the basis obtained from Jordan's lemma we have the following representation of $\Pi_1 \Pi_2 \Pi_1$,

$$\Pi_1 \Pi_2 \Pi_1 := \sum_{\alpha=1}^k \cos^2(\theta_\alpha) |v_\alpha\rangle \langle v_\alpha|, \quad (6.3.1)$$

where $\cos^2(\theta_\alpha) := |\langle v_\alpha | w_\alpha \rangle|^2$. We now have the following set of inequalities:

$$\begin{aligned} 1 - 8\epsilon &\stackrel{a}{\leq} \text{Tr}[\Pi_1 \Pi_2 \Pi_1 \rho] \\ &\stackrel{b}{=} \sum_{\alpha=1}^k \cos^2(\theta_\alpha) \text{Tr}[|v_\alpha\rangle \langle v_\alpha| \rho_\alpha] \\ &\stackrel{c}{\leq} \sum_{\alpha=1}^k \cos^2(\theta_\alpha) \text{Tr}[\rho_\alpha], \end{aligned} \quad (6.3.2)$$

where a follows from the Gao's union bound (Fact 8), b follows from (6.3.1) and c follows from the property of the trace and the fact that $|v_\alpha\rangle \langle v_\alpha| \leq \Pi_\alpha$. Thus, it follows from (6.3.2) that

$$\mathbb{E}[\cos^2(\theta_\alpha)] \geq 1 - 8\epsilon, \quad (6.3.3)$$

where in the above the expectation is calculated with respect to the distribution $\text{Tr}[\rho_\alpha]$. We now prove the claim 1 as follows:

$$\begin{aligned} \Pr\{\sin^2(\theta_\alpha) > 8\sqrt{\epsilon}\} &\stackrel{a}{\leq} \frac{\mathbb{E}[\sin^2(\theta_\alpha)]}{8\sqrt{\epsilon}} \\ &\stackrel{b}{\leq} \sqrt{\epsilon}, \end{aligned} \quad (6.3.4)$$

where a follows from the Fact 6's inequality and b follows from (6.3.3). Claim 1 now easily follows from (6.3.4).

Proof of claim 2

Proof. The proof of claim 2 follows from the following set of inequalities

$$1 - \epsilon \stackrel{a}{\leq} \text{Tr}[\Pi_1 \rho]$$

$$\begin{aligned}
& \stackrel{b}{=} \sum_{\alpha=1}^k \text{Tr}[|v_\alpha\rangle\langle v_\alpha|\rho_\alpha] \\
& = \sum_{\alpha \in \mathcal{E}_2} \text{Tr}[|v_\alpha\rangle\langle v_\alpha|\rho_\alpha] + \sum_{\alpha \in \mathcal{E}_2^C} \text{Tr}[|v_\alpha\rangle\langle v_\alpha|\rho_\alpha] \\
& \stackrel{c}{\leq} \sum_{\alpha \in \mathcal{E}_2} \text{Tr}[\rho_\alpha] + (1 - \sqrt{\epsilon}) \sum_{\alpha \in \mathcal{E}_2^C} \text{Tr}[\rho_\alpha] \\
& = \Pr\{\mathcal{E}_2\} + (1 - \sqrt{\epsilon})(1 - \Pr\{\mathcal{E}_2\}) \\
& = 1 - \sqrt{\epsilon} + \sqrt{\epsilon} \Pr\{\mathcal{E}_2\} \\
& \Rightarrow \Pr\{\mathcal{E}_2\} \geq 1 - \sqrt{\epsilon},
\end{aligned} \tag{6.3.5}$$

where a follows from the property of the projector Π_1 given in Lemma 6.1.2, b follows from the Jordan's lemma and c from the definition of the set $\mathcal{E}_2$ and from monotonicity property of the trace and the fact that $|v_\alpha\rangle\langle v_\alpha| \leq \Pi_\alpha$. Claim 2 now easily follows from (6.3.6).

Proof of claim 3

Proof. The proof of claim 2 follows from the following set of inequalities

$$\begin{aligned}
1 - \epsilon & \stackrel{a}{\leq} \text{Tr}[\Pi_2\rho] \\
& \stackrel{b}{=} \sum_{\alpha=1}^k \text{Tr}[|w_\alpha\rangle\langle w_\alpha|\rho_\alpha] \\
& = \sum_{\alpha \in \mathcal{E}_2} \text{Tr}[|w_\alpha\rangle\langle w_\alpha|\rho_\alpha] + \sum_{\alpha \in \mathcal{E}_3^C} \text{Tr}[|w_\alpha\rangle\langle w_\alpha|\rho_\alpha] \\
& \stackrel{c}{\leq} \sum_{\alpha \in \mathcal{E}_2} \text{Tr}[\rho_\alpha] + (1 - \sqrt{\epsilon}) \sum_{\alpha \in \mathcal{E}_3^C} \text{Tr}[\rho_\alpha] \\
& = \Pr\{\mathcal{E}_3\} + (1 - \sqrt{\epsilon})(1 - \Pr\{\mathcal{E}_3\}) \\
& = 1 - \sqrt{\epsilon} + \sqrt{\epsilon} \Pr\{\mathcal{E}_3\} \\
& \Rightarrow \Pr\{\mathcal{E}_3\} \geq 1 - \sqrt{\epsilon},
\end{aligned} \tag{6.3.6}$$

where a follows from the property of the projector Π_1 given in Lemma 6.1.2, b follows from the Jordan's lemma and c from the definition of the set $\mathcal{E}_2$ and from monotonicity property of the trace and the fact that $|v_\alpha\rangle\langle v_\alpha| \leq \Pi_\alpha$. Claim 2 now easily follows from (6.3.6).

6.4 Quantum Stein's Lemma

Suppose we have a source which emits several i.i.d. copies of one of the two states ρ or σ and we should make a decision based on some POVMs. Let's say we have designed two POVMs $\{\Pi_n, (\mathbb{I} - \Pi_n)\}$ in such a way that given n realization of the unknown state, if someone obtains the measurement outcome associated to Π^n then the state was ρ and if the measurement outcome associated to $(\mathbb{I} - \Pi_n)$ then the unknown state was σ . Now here we can have two types of error.

- **Type-1:** if the measurement outcomes implies the state to be σ , but actually the state was ρ . Probability of such error is $\hat{\alpha}^n(\Pi_n) := \text{Tr}[(\mathbb{I} - \Pi_n)\rho^{\otimes n}]$
- **Type-2:** if the measurement outcomes implies the state to be ρ , but actually the state was σ . Probability of such error is $\hat{\beta}^n(\Pi_n) := \text{Tr}[\Pi_n\sigma^{\otimes n}]$

Quantum Chernoff Chernoff Stein's lemma studies the asymptotic behaviour of $\frac{1}{n} \log(\hat{\beta}_n^\epsilon)$, where,

$$\begin{aligned} \hat{\beta}_n^\epsilon &:= \min_{\Pi_n: \text{Tr}[\Pi_n \rho^{\otimes n}] \geq 1-\epsilon} \text{Tr}[\Pi_n \sigma^{\otimes n}] \\ \Rightarrow \hat{\beta}_n^\epsilon &:= \min_{\Pi_n: \hat{\alpha}^n(\Pi_n) \leq \epsilon} \hat{\beta}^n(\Pi_n). \end{aligned} \quad (6.4.1)$$

Notice that this definition of $\hat{\beta}_n^\epsilon$ is taken from the definition of *quantum hypothesis testing relative entropy* in equation (1.2.2). We now give a lower bound on $\frac{1}{n} \log(\hat{\beta}_n^\epsilon)$ (converse) which is similar in flavour to the proof given in Theorem 1.2.2. This may be seen as an application of our main result given in Theorem 6.1.1.

Theorem 6.4.1. (*Converse of Quantum Chernoff Stein's Lemma*) Let $\omega^{\otimes n} \in \mathcal{D}(\mathcal{H}^n)$ be n i.i.d. realisation of an unknown state. Consider the hypothesis test between two alternatives $H_1 : \omega \equiv \rho$ and $H_2 : \omega \equiv \sigma$, where $D(\rho \parallel \sigma) < \infty$. Let Π^n be an acceptance POVM of H_1 . Then for $0 < \epsilon < 1$,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log(\hat{\beta}_n^\epsilon) \geq -D(\rho \parallel \sigma), \quad (6.4.2)$$

where $\hat{\beta}_n^\epsilon$ is defined in equation (6.4.1) and f is a polynomial function on ϵ .

Before giving a proof of Theorem 6.4.1 let's prove Theorem 6.1.2 using the idea we discussed in above sections.

Proof of Corollary 6.1.2

The proof follows using techniques similar to the one used in the proof of Lemma 6.1.1. However, we give its proof for completeness. Let $\Pi_n^{(1)} := \{\sigma^{\otimes n} \succeq 2^{-n(D(\rho \parallel \sigma) + \delta)} \rho^{\otimes n}\}$. Then, from Fact 7, we have, $\text{Tr}[\Pi_n^{(1)} \rho^{\otimes n}] \geq 1 - \epsilon$. Fact 5 guarantees that for every $\alpha \in [1 : k']$ we can represent

$\Pi_{n,\alpha}\Pi_n^{(1)}\Pi_{n,\alpha} := \Pi_{n,\alpha}^{(1)}$ and $\Pi_{n,\alpha}\Pi_n^{(2)}\Pi_{n,\alpha} := \Pi_{n,\alpha}^{(2)}$ as the following one dimensional projectors:

$$\begin{aligned}\Pi_{n,\alpha}^{(1)} &= |v_{n,\alpha}\rangle\langle v_{n,\alpha}|, \\ \Pi_{n,\alpha}^{(2)} &= |w_{n,\alpha}\rangle\langle w_{n,\alpha}|,\end{aligned}$$

for some $|v_{n,\alpha}\rangle, |w_{n,\alpha}\rangle$ in the subspace spanned by the eigen vectors of $\Pi_{n,\alpha}$. Now for any $|w_{n,\alpha}\rangle$ it can be written in the following way:

$$|w_{n,\alpha}\rangle = \cos(\theta_{n,\alpha})|v_{n,\alpha}\rangle + \sin(\theta_{n,\alpha})|v_{n,\alpha}^\perp\rangle,$$

where $\cos^2(\theta_{n,\alpha}) := |\langle v_{n,\alpha}|w_{n,\alpha}\rangle|^2$. Similar to the proof of Lemma 6.1.1, we define $\text{Good} := \mathcal{E}_1 \cap \mathcal{E}_2$, where $\mathcal{E}_1$ and $\mathcal{E}_2$ is defined similarly as in 6.2.10 and 6.2.11. Observe here that we don't need the set $\mathcal{E}_3$ because $\Pi_n^{(2)}$ doesn't have any property as mentioned for Π_2 in Lemma 6.1.1. Using arguments similar to that the one used in the proof of claim 4, we can show that $\Pr\{\text{Good}\} \geq 1 - 2\sqrt{\epsilon}$. We now prove the corollary using the following set of inequalities:

$$\begin{aligned}& \text{Tr}[\Pi_n^{(2)}\sigma^{\otimes n}] \\& \stackrel{a}{=} \sum_{\alpha=1}^k \text{Tr}[|w_{n,\alpha}\rangle\langle w_{n,\alpha}|\sigma_\alpha^{\otimes n}] \\& \stackrel{b}{\geq} \sum_{\alpha \in \text{Good}} \text{Tr}[|w_{n,\alpha}\rangle\langle w_{n,\alpha}|\sigma_\alpha^{\otimes n}] \\& \stackrel{c}{\geq} \sum_{\alpha \in \text{Good}} \left(\cos(2\theta_{n,\alpha})\text{Tr}[|v_{n,\alpha}\rangle\langle v_{n,\alpha}|\sigma_\alpha^{\otimes n}] + \sin^2(\theta_{n,\alpha})\text{Tr}[\sigma_\alpha^{\otimes n}] \right. \\& \quad \left. + \sin^2(\theta_{n,\alpha})\text{Tr}[|v_{n,\alpha}\rangle\langle v_{n,\alpha}^\perp|\sigma_\alpha^{\otimes n}] + \sin^2(\theta_{n,\alpha})\text{Tr}[|v_{n,\alpha}^\perp\rangle\langle v_{n,\alpha}|\sigma_\alpha^{\otimes n}] \right) \\& \stackrel{d}{\geq} \sum_{\alpha \in \text{Good}} \left(2^{-n(D(\rho\|\sigma)+\delta)} \cos(2\theta_{n,\alpha})\text{Tr}[|v_{n,\alpha}\rangle\langle v_{n,\alpha}|\rho_\alpha^{\otimes n}] \right. \\& \quad \left. + \sin^2(\theta_{n,\alpha})(\text{Tr}[\sigma_\alpha^{\otimes n}] + \text{Tr}[|v_{n,\alpha}\rangle\langle v_{n,\alpha}^\perp|\sigma_\alpha^{\otimes n}] + \text{Tr}[|v_{n,\alpha}^\perp\rangle\langle v_{n,\alpha}|\sigma_\alpha^{\otimes n}]) \right) \\& \stackrel{e}{\geq} \sum_{\alpha \in \text{Good}} (2^{-n(D(\rho\|\sigma)+\delta)} \cos(2\theta_{n,\alpha})\text{Tr}[|v_{n,\alpha}\rangle\langle v_{n,\alpha}|\rho_\alpha^{\otimes n}]) \\& \stackrel{f}{\geq} \sum_{\alpha \in \text{Good}} (2^{-n(D(\rho\|\sigma)+\delta)} \cos(2\theta_{n,\alpha})(1 - \sqrt{\epsilon})\text{Tr}[\rho_\alpha^{\otimes n}]) \\& \stackrel{g}{\geq} 2^{-n(D(\rho\|\sigma)+\delta)}(1 - \sqrt{\epsilon})(1 - 16\sqrt{\epsilon}) \sum_{\alpha \in \text{Good}} (\text{Tr}[\rho_\alpha^{\otimes n}]) \\& \stackrel{h}{\geq} 2^{-n(D(\rho\|\sigma)+\delta)}(1 - \sqrt{\epsilon})(1 - 16\sqrt{\epsilon})(1 - 2\sqrt{\epsilon}) \\& = 2^{-n(D(\rho\|\sigma)+\delta)}(1 - 19\sqrt{\epsilon}),\end{aligned}$$

where a follows from the definition of (6.2.4), b follows because each term in the summation is

non-negative, c follows using arguments similar to the one used in the proof (6.2.16), d follows from the inequality (6.2.8), e follows from Lemma (6.2.1) and because $\sigma_\alpha^{\otimes n} \succeq 0$, f follows from the definition of $\mathcal{E}_2$ given in 6.2.11, g follows from the fact that $\cos(2\theta_{n,\alpha}) = \cos^2(\theta_{n,\alpha}) - \sin^2(\theta_{n,\alpha})$ and for $\alpha \in \text{Good}$, $\cos^2(\theta_{n,\alpha}) \geq 1 - 8\sqrt{\epsilon}$ and h follows because $\Pr\{\text{Good}\} \geq 1 - 2\sqrt{\epsilon}$ as discussed above. We now give a lower bound on $\frac{1}{n} \log(\hat{\beta}_n^\epsilon)$ (converse) using Corollary 6.1.2 which is similar in flavour to the proof given in the Tom and Cover Book [26]. This may be seen as an application of our main result 6.1.1.

6.4.1 Proof of Theorem 6.4.1

Proof. It follows Corollary 6.1.2 that for any arbitrary Π_n such that $\text{Tr}[\Pi_n \rho^{\otimes n}] \geq 1 - \epsilon$, we have $\text{Tr}[\Pi_n \sigma^{\otimes n}] \geq 2^{-n(D(\rho\|\sigma) + \delta)}(1 - 19\sqrt{\epsilon})$. Thus, we have

$$\begin{aligned} \frac{1}{n} \log \hat{\beta}_n^\epsilon &\geq \frac{1}{n} \log \hat{\beta}_n \\ &\stackrel{a}{=} \frac{1}{n} \log (\text{Tr}[\Pi_n \sigma^{\otimes n}]) \\ &\geq -(D(\rho\|\sigma) + \delta) + \frac{1}{n} \log(1 - 19\sqrt{\epsilon}), \end{aligned}$$

then

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \hat{\beta}_n^\epsilon \geq -(D(\rho\|\sigma) + \delta)$$

□

So we are finally done with all the discuss we can possible, now it's time to draw conclusion and discuss what can further be done in context with this dissertation.

Chapter 7

Conclusions and Future Work

In this work we have provided a notion of intersection of two projective measurements using Jordan's Lemma. Using this, we give a proof of the weak converse for the Quantum Chernoff Stein's Lemma. As a future work, it will be an interesting problem to prove a result similar to our main result for the reverse operator inequality. To elaborate more on this, consider ρ, σ_1 and $\sigma_2 \in \mathcal{D}(\mathcal{H})$ such that $\text{supp}(\rho) \subseteq \text{supp}(\sigma_1)$ and $\text{supp}(\rho) \subseteq \text{supp}(\sigma_2)$. Suppose for some $k_1, k_2 > 0$, let Π_1, Π_2 be such that $\Pi_1 := \{\sigma_1 \preceq 2^{-k_1} \rho\}$ and $\Pi_2 := \{\sigma_2 \preceq 2^{-k_2} \rho\}$ with the property that $\text{Tr}[\Pi_1 \rho] \geq 1 - \epsilon$ and $\text{Tr}[\Pi_2 \rho] \geq 1 - \epsilon$. Then, finding a projector Π^* such that $\text{Tr}[\Pi^* \rho] \geq 1 - f(\epsilon)$, $\text{Tr}[\Pi^* \sigma_1] \leq 2^{-k_1} f(\epsilon)$ and $\text{Tr}[\Pi^* \sigma_2] \leq 2^{-k_2} f(\epsilon)$, where, $f(\epsilon)$ is some function of ϵ . This will help in understanding many important problems in the area of quantum network information theory.

We have extended Sen's one-shot upper bound for CQ-MAC [25] by demonstrating that it applies even when the input alphabet spaces of CQ-MAC follow a joint probability distribution. This generalization can be useful for solving the Distributed Source Compression problem with quantum side information, making it a valuable application of Sen's work. It is worth noting that the Distributed Source Compression problem with classical side information shares a similar concept with classical MAC with two mutually dependent inputs. However, further research is required to fully explore these connections.

We attempted to apply Sen's method [25] to derive comparable upper bounds for general quantum states, which could aid in proving the Entanglement-Assisted MAC problem [17]. However, this approach proved to be ineffective for general quantum states as it is not well interpretable to augment each system of a general quantum state individually.

Sen's work [25] presents a statement that is weaker than the classical statement. The scope of the statement is limited to negative hypothesis states that are a tensor product of marginals and at most one arbitrary quantum state. However, it is crucial to address the open problem of proving a quantum lemma that can handle arbitrary negative hypothesis states, similar to the classical setting.

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